Noncommutative row-determinant pre-Pieri rules

Darij Grinberg

arxiv:2110.03108v2; CC0 1.0

Editorial note. Complete original fixed-version mathematical paper237–8195 retained with all native diagrams, original local supporting proofs and embedded bibliography. No active external TeX or image inputs occur.

Editorial note. Source0 only uses the author-provided detailed mode and established missing-font/class/unused-license-logo rendering interface at lines[37, 38, 49, 54, 137, 138]. Raw archive/source unchanged; no source hashing or cross-version byte comparisons.

Editorial note. One principal theorem-family/object. Other original definitions/results, known specializations, direct consequences and speculative statements remain original context, not extra counts. Exact original CC0 1.0 preserved; CC BY-SA remains share-alike and is not relabeled as CC BY or code MIT.

Editorial note. Authors retain mathematical responsibility. No proof repair, additional assumptions, independent mathematical review, counterexample search, numerical/symbolic replay or certification.

Allowed standard prerequisites (editorial).

- Complete original digraph/QSym, finite simplicial/discrete-Morse or row-determinant framework. All native standing assumptions, definitions, ring and parameter restrictions retained literally.. Full original author selected proof and source-local support supplied without reconstruction or certification.

Introduction

The Pieri rules in the theory of symmetric functions (see, e.g., [Macdon95, Chapter I, (5.16) and (5.17)] or [Stanle24, Theorem 7.15.7 and discussion before Corollary 7.15.9]) give simple formulas for multiplying a Schur function by a complete homogeneous or elementary symmetric function. One of the simplest ways to state them (sidestepping

the combinatorial background and the geometric motivation) is as follows: We define the ring Λ of symmetric functions as a polynomial ring in countably many indeterminates $h_1, h_2, h_3, \ldots$ over some commutative ring (say, over $\mathbb{Z}$); we furthermore set $h_0 := 1$ and $h_i := 0$ for all negative i. If $\lambda = (\lambda_1, \lambda_2, \lambda_3, \ldots)$ is an integer partition[1], then the corresponding *Schur function* $s_\lambda \in \Lambda$ can be defined by the formula

$$s_\lambda = \det\left(\left(h_{\lambda_i - i + j}\right)_{i,j \in [n]}\right), \tag{1}$$

where n is a nonnegative integer satisfying $\lambda_{n+1} = \lambda_{n+2} = \lambda_{n+3} = \cdots = 0$ (note that there are infinitely many possible values for n, but they all give the same s_λ). If the partition λ has the form $(1, 1, \ldots, 1, 0, 0, 0, \ldots)$ with k many 1's, then the corresponding Schur function s_λ is called e_k. (Of course, these are not the usual definitions of s_λ and e_k; see [Macdon95, Chapter I, (3.4) and (3.9)] for their equivalence to more standard definitions.) Now, the *first Pieri rule* ([Macdon95, Chapter I, (5.16)]) states that for each $p \in \mathbb{N}$ and each partition $\lambda = (\lambda_1, \lambda_2, \lambda_3, \ldots)$, we have

$$s_\lambda h_p = \sum_\mu s_\mu, \tag{2}$$

where the sum ranges over all partitions $\mu = (\mu_1, \mu_2, \mu_3, \ldots)$ such that μ/λ is a *horizontal p-strip*[2]. For example,

$$s_{(2,1)} h_2 = s_{(4,1)} + s_{(3,2)} + s_{(3,1,1)} + s_{(2,2,1)},$$

where we are using the standard convention of omitting zeroes from a partition (i.e., we identify a partition $\lambda = (\lambda_1, \lambda_2, \lambda_3, \ldots)$ with the n-tuple $(\lambda_1, \lambda_2, \ldots, \lambda_n)$ when $\lambda_{n+1} = \lambda_{n+2} = \lambda_{n+3} = \cdots = 0$). The *second Pieri rule* ([Macdon95, Chapter I, (5.17)]) states that for each $p \in \mathbb{N}$ and each partition $\lambda = (\lambda_1, \lambda_2, \lambda_3, \ldots)$, we have

$$s_\lambda e_p = \sum_\mu s_\mu, \tag{3}$$

where the sum ranges over all partitions $\mu = (\mu_1, \mu_2, \mu_3, \ldots)$ such that μ/λ is a *vertical p-strip*[3].

[1] An *integer partition* (or, for short, just *partition*) means a weakly decreasing sequence $(\lambda_1, \lambda_2, \lambda_3, \ldots)$ of nonnegative integers such that all but finitely many $i > 0$ satisfy $\lambda_i = 0$.

[2] The notion of a "horizontal p-strip" and the notation μ/λ are best explained in terms of Young diagrams. However, for our purposes, we can define them algebraically: We say that "μ/λ is a horizontal p-strip" if and only if

$$\mu_1 \geq \lambda_1 \geq \mu_2 \geq \lambda_2 \geq \mu_3 \geq \lambda_3 \geq \cdots \qquad \text{and} \quad \sum_{i \geq 1} (\mu_i - \lambda_i) = p.$$

[3] The notion of a "vertical p-strip" and the notation μ/λ are best explained in terms of Young diagrams. However, for our purposes, we can define them algebraically: We say that "μ/λ is a vertical p-strip" if and only if

$$(\mu_i - \lambda_i \in \{0, 1\} \text{ for each } i \geq 1) \qquad \text{and} \quad \sum_{i \geq 1} (\mu_i - \lambda_i) = p.$$

There is a less-known variant of the first Pieri rule (2), which sometimes appears as a stepping stone to its proof (e.g., in [Tamvak13, §2.4]). In order to state it, we agree to define a "Schur function" s_λ by the equality (1) not just whenever λ is a partition, but also whenever $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_n)$ is any n-tuple of integers (not necessarily nonnegative, not necessarily weakly decreasing). This does not significantly extend the notion of a "Schur function", since any such s_λ equals either 0 or $\pm s_\mu$ for an (honest) partition μ. (This follows easily from basic properties of determinants.) Now, if $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_n)$ is any integer partition with $\lambda_n = 0$, and if $p \in \mathbb{N}$ is arbitrary, then

$$s_\lambda h_p = \sum_\mu s_\mu, \tag{4}$$

where the sum now ranges over all n-tuples $\mu = (\mu_1, \mu_2, \ldots, \mu_n) \in \mathbb{N}^n$ such that

$$(\mu_i \geq \lambda_i \text{ for each } i) \text{ and } \sum_{i=1}^{n} (\mu_i - \lambda_i) = p.$$

In other words, if $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_n)$ is any integer partition with $\lambda_n = 0$, and if $p \in \mathbb{N}$ is arbitrary, then

$$s_\lambda h_p = \sum_{\substack{\beta = (\beta_1, \beta_2, \ldots, \beta_n) \in \mathbb{N}^n; \\ \beta_1 + \beta_2 + \cdots + \beta_n = p}} s_{\lambda + \beta}, \tag{5}$$

where $\lambda + \beta$ denotes the entrywise sum of the n-tuples λ and β.

Note that each addend in the sum in (2) is also contained in the sum in (4), but (usually) not the other way around: An n-tuple μ appearing on the right hand side of (4) might fail to be a partition, and even if it is one, it may violate the "horizontal p-strip" condition (by failing to satisfy $\lambda_i \geq \mu_{i+1}$ for some i). Some of these extraneous addends in (4) are 0, while others cancel each other out. This cancellation argument appears (in some slightly different contexts) in [Tamvak13, Lemma 2] and [LakTho07, (2.1) vs. (2.2)].

The "alternative first Pieri rule" (4) aka (5) is itself not hard to prove. In this note, we shall generalize it in multiple directions. The ultimate generalization – our Theorem 2.1 – we call the *first pre-Pieri rule*; it is an identity for row-determinants of matrices over noncommutative rings. We will give an elementary proof of Theorem 2.1 (using simple combinatorics and manipulation of sums[4]) and derive several corollaries, which include not only (5), but also a noncommutative "right-Pieri rule" for immaculate functions due to Berg, Bergeron, Saliola, Serrano and Zabrocki [BBSSZ13, Theorem 3.5] as well as a Pieri-like rule for Macdonald's 9th-variation Schur functions [Fun12, Proposition 3.9].[5]

[4] Our proof could be viewed as a sequence of sign-reversing involutions, although we do not state it in such a form.

[5] It should also be possible to derive the "uncancelled Pieri rule" [Grinbe19, Theorem 11.7] from our first pre-Pieri rule, but this will likely require more effort than it is worth. (Note that [Grinbe19, Theorem 11.7] is not about symmetric functions, but about symmetric polynomials in k variables; on the other hand, unlike (5), there is no $\lambda_n = 0$ requirement in [Grinbe19, Theorem 11.7]. These differences are fairly substantial, and it is not immediately obvious how to bridge them.)

We will also show a *second pre-Pieri rule*: an analogue of the first pre-Pieri rule with a parallel retinue of corollaries. One such corollary is an analogue of (5) for e_p instead of h_p; it says that if $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_n)$ is any partition, and if $p \in \{0, 1, \ldots, n\}$ satisfies $\lambda_{n-p+1} = \lambda_{n-p+2} = \cdots = \lambda_n = 0$, then

$$s_\lambda e_p = \sum_{\substack{\beta=(\beta_1,\beta_2,\ldots,\beta_n)\in\{0,1\}^n; \\ \beta_1+\beta_2+\cdots+\beta_n=p}} s_{\lambda+\beta}. \tag{6}$$

This can be viewed as an alternative version of the second Pieri rule (3), and indeed it is possible to obtain (3) from (6) by removing vanishing addends. (Unlike for the first Pieri rule, cancellations are not required.)

Finally, we shall speculate on the existence of a "pre-LR rule", which might include both pre-Pieri rules as particular cases.

1. Notations

Let us first introduce the notations that will be used throughout this note.

- Let R be a ring (unital and associative, but not necessarily commutative).
- Let $\mathbb{N} := \{0, 1, 2, \ldots\}$ and $\mathbb{P} := \{1, 2, 3, \ldots\}$.
- For any $n \in \mathbb{N}$, we let $[n]$ denote the n-element set $\{1, 2, \ldots, n\}$.
- If α is an n-tuple (for some $n \in \mathbb{N}$), and if $i \in [n]$, then we let α_i denote the i-th entry of α (so that $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_n)$).
- If α is an n-tuple of integers (for some $n \in \mathbb{N}$), then we define $|\alpha| := \alpha_1 + \alpha_2 + \cdots + \alpha_n$.
- For any $n \in \mathbb{N}$, we let S_n denote the n-th symmetric group (i.e., the group of all permutations of the set $[n]$).
- If $n \in \mathbb{N}$ and $\sigma \in S_n$, then we let $(-1)^\sigma$ denote the sign of the permutation σ.
- If $n \in \mathbb{N}$, and if we are given an element $a_{i,j} \in R$ for each pair $(i, j) \in [n] \times [n]$, then we let $(a_{i,j})_{i,j\in[n]}$ denote the $n \times n$-matrix whose (i, j)-th entry is $a_{i,j}$ for each $(i, j) \in [n] \times [n]$. That is, we let

$$(a_{i,j})_{i,j\in[n]} := \begin{pmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & \cdots & a_{n,n} \end{pmatrix} \in R^{n\times n}.$$

- If $A = (a_{i,j})_{i,j\in[n]}$ is any $n \times n$-matrix over R (for some $n \in \mathbb{N}$), then we define an element $\operatorname{rowdet} A \in R$ by

$$\operatorname{rowdet} A := \sum_{\sigma\in S_n} (-1)^\sigma a_{1,\sigma(1)} a_{2,\sigma(2)} \cdots a_{n,\sigma(n)}.$$

  This element is called the *row-determinant* of A. When the ring R is commutative, this row-determinant $\operatorname{rowdet} A$ is just the usual determinant of A.

- We regard the set $\mathbb{Z}^n$ as a $\mathbb{Z}$-module in the usual way: i.e., we have

$$\alpha + \beta = (\alpha_1 + \beta_1, \ \alpha_2 + \beta_2, \ \ldots, \ \alpha_n + \beta_n) \qquad \text{and}$$
$$\alpha - \beta = (\alpha_1 - \beta_1, \ \alpha_2 - \beta_2, \ \ldots, \ \alpha_n - \beta_n)$$

  for any $\alpha \in \mathbb{Z}^n$ and $\beta \in \mathbb{Z}^n$. Thus, $\alpha + \beta$ and $\alpha - \beta$ are defined for $\alpha \in \mathbb{N}^n$ and $\beta \in \mathbb{N}^n$ as well (since $\mathbb{N}^n$ is a subset of $\mathbb{Z}^n$).

2. The first pre-Pieri rule

2.1. The theorem

We can now state our "first pre-Pieri rule" in full generality:

Theorem 2.1 (first pre-Pieri rule)**.** Let $n \in \mathbb{P}$ and $p \in \mathbb{N}$. Let $h_{k,\ i}$ be an element of R for all $k \in \mathbb{Z}$ and $i \in [n]$.

For any $\alpha \in \mathbb{Z}^n$, we define

$$t_\alpha := \operatorname{rowdet}\left(\left(h_{\alpha_i+j,\ i}\right)_{i,j\in[n]}\right) \in R.$$

Let η be the n-tuple

$$(1, 2, \ldots, n) + \left(\underbrace{0, 0, \ldots, 0}_{n-1 \text{ zeroes}}, p\right) = (1, 2, \ldots, n-1, n+p) \in \mathbb{Z}^n.$$

Let $\alpha \in \mathbb{Z}^n$. Then,

$$\sum_{\substack{\beta\in\mathbb{N}^n;\\ |\beta|=p}} t_{\alpha+\beta} = \operatorname{rowdet}\left(\left(h_{\alpha_i+\eta_j,\ i}\right)_{i,j\in[n]}\right). \tag{7}$$

Example 2.2. For this example, set $n = 2$ and $p = 2$, and let $\alpha \in \mathbb{Z}^2$ be arbitrary. Fix arbitrary elements $h_{k,\ i} \in R$ for all $k \in \mathbb{Z}$ and $i \in [n]$. Then, the n-tuple η defined in Theorem 2.1 is $(1, 2) + (0, 2) = (1, 4)$. Hence, (7) says that

$$\sum_{\substack{\beta\in\mathbb{N}^2;\\ |\beta|=2}} t_{\alpha+\beta} = \operatorname{rowdet}\left(\left(h_{\alpha_i+\eta_j,\ i}\right)_{i,j\in[2]}\right) = \operatorname{rowdet}\begin{pmatrix} h_{\alpha_1+1,\ 1} & h_{\alpha_1+4,\ 1} \\ h_{\alpha_2+1,\ 2} & h_{\alpha_2+4,\ 2} \end{pmatrix}.$$

The left hand side of this equality can be rewritten as

$$\sum_{\substack{\beta\in\mathbb{N}^2;\\ |\beta|=2}} \underbrace{t_{\alpha+\beta}}_{\substack{=\operatorname{rowdet}\left(\left(h_{(\alpha+\beta)_i+j,\ i}\right)_{i,j\in[2]}\right)\\ \text{(by the definition of } t_{\alpha+\beta})}}$$

$$= \sum_{\substack{\beta\in\mathbb{N}^2;\\ |\beta|=2}} \operatorname{rowdet}\left(\left(h_{(\alpha+\beta)_i+j,\ i}\right)_{i,j\in[2]}\right)$$

$$= \sum_{\substack{\beta\in\mathbb{N}^2;\\ |\beta|=2}} \operatorname{rowdet}\left(\left(h_{\alpha_i+\beta_i+j,\ i}\right)_{i,j\in[2]}\right)$$

$$\left(\text{since } (\alpha+\beta)_i = \alpha_i+\beta_i \text{ for all } i\in[2]\right)$$

$$= \sum_{\substack{\beta\in\mathbb{N}^2;\\ |\beta|=2}} \operatorname{rowdet}\begin{pmatrix} h_{\alpha_1+\beta_1+1,\ 1} & h_{\alpha_1+\beta_1+2,\ 1} \\ h_{\alpha_2+\beta_2+1,\ 2} & h_{\alpha_2+\beta_2+2,\ 2} \end{pmatrix}$$

$$= \operatorname{rowdet}\begin{pmatrix} h_{\alpha_1+2+1,\ 1} & h_{\alpha_1+2+2,\ 1} \\ h_{\alpha_2+0+1,\ 2} & h_{\alpha_2+0+2,\ 2} \end{pmatrix} + \operatorname{rowdet}\begin{pmatrix} h_{\alpha_1+1+1,\ 1} & h_{\alpha_1+1+2,\ 1} \\ h_{\alpha_2+1+1,\ 2} & h_{\alpha_2+1+2,\ 2} \end{pmatrix}$$
$$+ \operatorname{rowdet}\begin{pmatrix} h_{\alpha_1+0+1,\ 1} & h_{\alpha_1+0+2,\ 1} \\ h_{\alpha_2+2+1,\ 2} & h_{\alpha_2+2+2,\ 2} \end{pmatrix}$$
$$\left(\begin{array}{c}\text{since there are exactly three 2-tuples } \beta\in\mathbb{N}^2\\ \text{satisfying } |\beta|=2, \text{ namely } (2,0),\ (1,1) \text{ and } (0,2)\end{array}\right)$$

$$= \operatorname{rowdet}\begin{pmatrix} h_{\alpha_1+3,\ 1} & h_{\alpha_1+4,\ 1} \\ h_{\alpha_2+1,\ 2} & h_{\alpha_2+2,\ 2} \end{pmatrix} + \operatorname{rowdet}\begin{pmatrix} h_{\alpha_1+2,\ 1} & h_{\alpha_1+3,\ 1} \\ h_{\alpha_2+2,\ 2} & h_{\alpha_2+3,\ 2} \end{pmatrix}$$
$$+ \operatorname{rowdet}\begin{pmatrix} h_{\alpha_1+1,\ 1} & h_{\alpha_1+2,\ 1} \\ h_{\alpha_2+3,\ 2} & h_{\alpha_2+4,\ 2} \end{pmatrix}.$$

Therefore, (7) rewrites as

$$\operatorname{rowdet}\begin{pmatrix} h_{\alpha_1+3,\ 1} & h_{\alpha_1+4,\ 1} \\ h_{\alpha_2+1,\ 2} & h_{\alpha_2+2,\ 2} \end{pmatrix} + \operatorname{rowdet}\begin{pmatrix} h_{\alpha_1+2,\ 1} & h_{\alpha_1+3,\ 1} \\ h_{\alpha_2+2,\ 2} & h_{\alpha_2+3,\ 2} \end{pmatrix}$$
$$+ \operatorname{rowdet}\begin{pmatrix} h_{\alpha_1+1,\ 1} & h_{\alpha_1+2,\ 1} \\ h_{\alpha_2+3,\ 2} & h_{\alpha_2+4,\ 2} \end{pmatrix}$$
$$= \operatorname{rowdet}\begin{pmatrix} h_{\alpha_1+1,\ 1} & h_{\alpha_1+4,\ 1} \\ h_{\alpha_2+1,\ 2} & h_{\alpha_2+4,\ 2} \end{pmatrix}.$$

This is easy to check directly by expanding all four row-determinants.

Example 2.3. For another example, set $n = 3$ and $p = 2$, and let $\alpha \in \mathbb{Z}^3$ be arbitrary. Fix arbitrary elements $h_{k,\ i} \in R$ for all $k \in \mathbb{Z}$ and $i \in [n]$. Then, the n-tuple η defined in Theorem 2.1 is $(1,2,3) + (0,0,2) = (1,2,5)$. Hence, (7) says

that

$$\sum_{\substack{\beta\in\mathbb{N}^3;\\ |\beta|=2}} t_{\alpha+\beta} = \operatorname{rowdet}\left(\left(h_{\alpha_i+\eta_j,\ i}\right)_{i,j\in[3]}\right)$$
$$= \operatorname{rowdet}\begin{pmatrix} h_{\alpha_1+1,\ 1} & h_{\alpha_1+2,\ 1} & h_{\alpha_1+5,\ 1} \\ h_{\alpha_2+1,\ 2} & h_{\alpha_2+2,\ 2} & h_{\alpha_2+5,\ 2} \\ h_{\alpha_3+1,\ 3} & h_{\alpha_3+2,\ 3} & h_{\alpha_3+5,\ 3} \end{pmatrix}.$$

The left hand side of this equality can be rewritten as

$$\sum_{\substack{\beta\in\mathbb{N}^3;\\ |\beta|=2}} \underbrace{t_{\alpha+\beta}}_{\substack{=\operatorname{rowdet}\left(\left(h_{(\alpha+\beta)_i+j,\ i}\right)_{i,j\in[3]}\right)\\ \text{(by the definition of } t_{\alpha+\beta})}}$$
$$= \sum_{\substack{\beta\in\mathbb{N}^3;\\ |\beta|=2}} \operatorname{rowdet}\left(\left(h_{(\alpha+\beta)_i+j,\ i}\right)_{i,j\in[3]}\right)$$
$$= \sum_{\substack{\beta\in\mathbb{N}^3;\\ |\beta|=2}} \operatorname{rowdet}\left(\left(h_{\alpha_i+\beta_i+j,\ i}\right)_{i,j\in[3]}\right)$$
$$\text{(since } (\alpha+\beta)_i = \alpha_i+\beta_i \text{ for all } i\in[3])$$
$$= \sum_{\substack{\beta\in\mathbb{N}^3;\\ |\beta|=2}} \operatorname{rowdet}\begin{pmatrix} h_{\alpha_1+\beta_1+1,\ 1} & h_{\alpha_1+\beta_1+2,\ 1} & h_{\alpha_1+\beta_1+3,\ 1} \\ h_{\alpha_2+\beta_2+1,\ 2} & h_{\alpha_2+\beta_2+2,\ 2} & h_{\alpha_2+\beta_2+3,\ 2} \\ h_{\alpha_3+\beta_3+1,\ 3} & h_{\alpha_3+\beta_3+2,\ 3} & h_{\alpha_3+\beta_3+3,\ 3} \end{pmatrix}.$$

This is a sum with 6 addends, since the 3-tuples $\beta\in\mathbb{N}^3$ satisfying $|\beta|=2$ are

$$(2,0,0), \quad (0,2,0), \quad (0,0,2), \quad (1,1,0), \quad (1,0,1), \quad (0,1,1).$$

The second addend (corresponding to the 3-tuple $\beta=(0,2,0)$) is

$$\operatorname{rowdet}\begin{pmatrix} h_{\alpha_1+0+1,\ 1} & h_{\alpha_1+0+2,\ 1} & h_{\alpha_1+0+3,\ 1} \\ h_{\alpha_2+2+1,\ 2} & h_{\alpha_2+2+2,\ 2} & h_{\alpha_2+2+3,\ 2} \\ h_{\alpha_3+0+1,\ 3} & h_{\alpha_3+0+2,\ 3} & h_{\alpha_3+0+3,\ 3} \end{pmatrix}$$
$$= \operatorname{rowdet}\begin{pmatrix} h_{\alpha_1+1,\ 1} & h_{\alpha_1+2,\ 1} & h_{\alpha_1+3,\ 1} \\ h_{\alpha_2+3,\ 2} & h_{\alpha_2+4,\ 2} & h_{\alpha_2+5,\ 2} \\ h_{\alpha_3+1,\ 3} & h_{\alpha_3+2,\ 3} & h_{\alpha_3+3,\ 3} \end{pmatrix}.$$

2.2. The proof

We shall now prepare for the proof of Theorem 2.1 by introducing some notations.

First, we introduce a right action of the symmetric group S_n on the set $\mathbb{Z}^n$ of all

n-tuples of integers:

Definition 2.4. Let $n \in \mathbb{N}$. Let $\eta \in \mathbb{Z}^n$ and $\sigma \in S_n$. Then, we define $\eta \circ \sigma$ to be the n-tuple $\left(\eta_{\sigma(1)}, \eta_{\sigma(2)}, \ldots, \eta_{\sigma(n)}\right) \in \mathbb{Z}^n$.

Thus, the n-tuple $\eta \circ \sigma$ is obtained from η by permuting the entries using the permutation σ.

The following two properties of this right action are near-obvious:

Proposition 2.5. Let $n \in \mathbb{N}$. Let $\eta \in \mathbb{Z}^n$ and $\sigma \in S_n$. Then,

$$|\eta \circ \sigma| = |\eta| .$$

Proof of Proposition 2.5. We have $\sigma \in S_n$. Thus, σ is a permutation of the set $[n]$ (since S_n was defined as the set of all permutations of the set $[n]$). In other words, σ is a bijection from $[n]$ to $[n]$. Therefore, $\eta_{\sigma(1)} + \eta_{\sigma(2)} + \cdots + \eta_{\sigma(n)} = \eta_1 + \eta_2 + \cdots + \eta_n$. However, Definition 2.4 yields $\eta \circ \sigma = \left(\eta_{\sigma(1)}, \eta_{\sigma(2)}, \ldots, \eta_{\sigma(n)}\right)$. Hence,

$$\begin{aligned} |\eta \circ \sigma| &= \left|\left(\eta_{\sigma(1)}, \eta_{\sigma(2)}, \ldots, \eta_{\sigma(n)}\right)\right| = \eta_{\sigma(1)} + \eta_{\sigma(2)} + \cdots + \eta_{\sigma(n)} \\ &\qquad \left(\text{by the definition of } \left|\left(\eta_{\sigma(1)}, \eta_{\sigma(2)}, \ldots, \eta_{\sigma(n)}\right)\right|\right) \\ &= \eta_1 + \eta_2 + \cdots + \eta_n = |\eta| \end{aligned}$$

(since the definition of $|\eta|$ yields $|\eta| = \eta_1 + \eta_2 + \cdots + \eta_n$). This proves Proposition 2.5. □

Proposition 2.6. Let $n \in \mathbb{N}$. Let $\eta \in \mathbb{Z}^n$. Assume that the n numbers $\eta_1, \eta_2, \ldots, \eta_n$ are distinct. Let $\sigma \in S_n$ and $\pi \in S_n$ be two distinct permutations. Then, $\eta \circ \sigma \neq \eta \circ \pi$.

Proof of Proposition 2.6. Both σ and π are elements of S_n, and thus are permutations of $[n]$ (since S_n was defined as the set of all permutations of $[n]$). We have assumed that σ and π are distinct. In other words, $\sigma \neq \pi$. In other words, there exists an $i \in [n]$ such that $\sigma(i) \neq \pi(i)$. Consider this i. The elements $\sigma(i)$ and $\pi(i)$ of $[n]$ are distinct (since $\sigma(i) \neq \pi(i)$).

We have assumed that the n numbers $\eta_1, \eta_2, \ldots, \eta_n$ are distinct. In other words, if p and q are two distinct elements of $[n]$, then $\eta_p \neq \eta_q$. We can apply this to $p = \sigma(i)$ and $q = \pi(i)$ (since $\sigma(i)$ and $\pi(i)$ are distinct), and thus obtain $\eta_{\sigma(i)} \neq \eta_{\pi(i)}$.

However, Definition 2.4 yields $\eta \circ \sigma = \left(\eta_{\sigma(1)}, \eta_{\sigma(2)}, \ldots, \eta_{\sigma(n)}\right)$; thus, $(\eta \circ \sigma)_i = \eta_{\sigma(i)}$. The same argument (applied to π instead of σ) yields $(\eta \circ \pi)_i = \eta_{\pi(i)}$. Thus, if we had $\eta \circ \sigma = \eta \circ \pi$, then we would have

$$\begin{aligned} \eta_{\sigma(i)} &= \left(\underbrace{\eta \circ \sigma}_{=\eta \circ \pi}\right)_i \qquad \left(\text{since } (\eta \circ \sigma)_i = \eta_{\sigma(i)}\right) \\ &= (\eta \circ \pi)_i = \eta_{\pi(i)}; \end{aligned}$$

but this would contradict $\eta_{\sigma(i)} \neq \eta_{\pi(i)}$. Hence, we cannot have $\eta \circ \sigma = \eta \circ \pi$. In other words, we have $\eta \circ \sigma \neq \eta \circ \pi$. This proves Proposition 2.6. $\square$

Finally, we will use the *Iverson bracket notation*:

Definition 2.7. If $\mathcal{A}$ is a logical statement, then $[\mathcal{A}]$ means the *truth value* of $\mathcal{A}$; this is the integer $\begin{cases} 1, & \text{if } \mathcal{A} \text{ is true;} \\ 0, & \text{if } \mathcal{A} \text{ is false.} \end{cases}$

For example, $[2+2=4] = 1$ and $[2+2=5] = 0$. The following easy property of truth values can serve as a warm-up:

Lemma 2.8. Let $u \in \mathbb{Z}$ and $k \in \mathbb{Z}$ satisfy $u \neq k$. Then, $[u \geq k] = [u \geq k+1]$.

Proof of Lemma 2.8. We are in one of the following two cases:
Case 1: We have $u < k$.
Case 2: We have $u \geq k$.

Let us first consider Case 1. In this case, we have $u < k$. Hence, we don't have $u \geq k$. Thus, we have $[u \geq k] = 0$. Furthermore, we don't have $u \geq k+1$ (since we have $u < k < k+1$). Hence, we have $[u \geq k+1] = 0$. Comparing this with $[u \geq k] = 0$, we obtain $[u \geq k] = [u \geq k+1]$. Thus, Lemma 2.8 is proven in Case 1.

Let us now consider Case 2. In this case, we have $u \geq k$. Combining this with $u \neq k$, we obtain $u > k$. Thus, $u \geq k+1$ (since u and k are integers). Therefore, $[u \geq k+1] = 1$. Moreover, $[u \geq k] = 1$ (since $u \geq k+1 \geq k$). Comparing these two equalities, we obtain $[u \geq k] = [u \geq k+1]$. Hence, Lemma 2.8 is proven in Case 2.

We have now proven Lemma 2.8 in both Cases 1 and 2. Thus, the proof of Lemma 2.8 is complete. $\square$

Next, we will use the following basic fact from the theory of finite sets:

Lemma 2.9. Let $m \in \mathbb{N}$. Let $u_1, u_2, \ldots, u_m$ be any m objects. Assume that $|\{u_1, u_2, \ldots, u_m\}| = m$. Then, the m objects $u_1, u_2, \ldots, u_m$ are distinct.

Proof of Lemma 2.9. The set $\{u_1, u_2, \ldots, u_m\}$ has size m (since $|\{u_1, u_2, \ldots, u_m\}| = m$). If the m objects $u_1, u_2, \ldots, u_m$ were not distinct, then there would be fewer than m distinct elements in the set $\{u_1, u_2, \ldots, u_m\}$; therefore, the set $\{u_1, u_2, \ldots, u_m\}$ would have size $< m$. But this would contradict the fact that this set has size m (which is not $< m$). Thus, the m objects $u_1, u_2, \ldots, u_m$ must be distinct. This proves Lemma 2.9. $\square$

Besides the above generalities, our proof of Theorem 2.1 will rely on some more specific lemmas. The first is an easy exercise on the pigeonhole principle:

Lemma 2.10. Let $n \in \mathbb{P}$. Let $\nu \in \mathbb{Z}^n$ and $\eta \in \mathbb{Z}^n$ satisfy $|\nu| = |\eta|$ and

$$\{\eta_1, \eta_2, \ldots, \eta_{n-1}\} \subseteq \{\nu_1, \nu_2, \ldots, \nu_n\} \qquad \text{and} \qquad |\{\eta_1, \eta_2, \ldots, \eta_{n-1}\}| = n-1.$$

Then, there exists some permutation $\pi \in S_n$ satisfying $\nu = \eta \circ \pi$.

Proof of Lemma 2.10. We know that n is a positive integer (since $n \in \mathbb{P}$). Hence, $n > 0$ and $n - 1 \in \mathbb{N}$.

The definition of $[n-1]$ yields $[n-1] = \{1, 2, \ldots, n-1\}$. The definition of $[n]$ yields $[n] = \{1, 2, \ldots, n\}$. Hence,

$$[n] \setminus \{n\} = \{1, 2, \ldots, n\} \setminus \{n\} = \{1, 2, \ldots, n-1\} = [n-1].$$

Thus, $[n-1] = [n] \setminus \{n\} \subseteq [n]$.

Now, recall that $n - 1 \in \mathbb{N}$ and $|\{\eta_1, \eta_2, \ldots, \eta_{n-1}\}| = n - 1$. Hence, Lemma 2.9 (applied to $m = n-1$ and $u_i = \eta_i$) shows that the $n-1$ objects $\eta_1, \eta_2, \ldots, \eta_{n-1}$ are distinct. In other words, if i and j are two distinct elements of $[n-1]$, then

$$\eta_i \neq \eta_j. \tag{8}$$

We define a map $f : [n-1] \to [n]$ as follows: For each $i \in [n-1]$, we choose some $j \in [n]$ satisfying $\eta_i = \nu_j$ (indeed, such a j exists, because

$$\begin{aligned} \eta_i &\in \{\eta_1, \eta_2, \ldots, \eta_{n-1}\} \qquad (\text{since } i \in [n-1] = \{1, 2, \ldots, n-1\}) \\ &\subseteq \{\nu_1, \nu_2, \ldots, \nu_n\} = \left\{ \nu_j \mid j \in \underbrace{\{1, 2, \ldots, n\}}_{=[n]} \right\} = \{\nu_j \mid j \in [n]\} \end{aligned}$$

), and we define $f(i)$ to be this j. Thus, we have defined a map $f : [n-1] \to [n]$.

For each $i \in [n-1]$, the image $f(i)$ of i is an element of $[n]$ and satisfies

$$\eta_i = \nu_{f(i)} \tag{9}$$

(since $f(i)$ is defined to be a $j \in [n]$ satisfying $\eta_i = \nu_j$).

The $n-1$ elements $f(1), f(2), \ldots, f(n-1)$ are distinct[6]. Hence,

$$|\{f(1), f(2), \ldots, f(n-1)\}| = n - 1.$$

[6] *Proof.* Let i and j be two distinct elements of $[n-1]$. We shall show that $f(i) \neq f(j)$.

Indeed, assume the contrary. Thus, $f(i) = f(j)$. However, (9) yields $\eta_i = \nu_{f(i)} = \nu_{f(j)}$ (since $f(i) = f(j)$). On the other hand, (9) (applied to j instead of i) yields $\eta_j = \nu_{f(j)}$. Comparing these two equalities, we obtain $\eta_i = \eta_j$. However, (8) yields $\eta_i \neq \eta_j$. This contradicts $\eta_i = \eta_j$. This contradiction shows that our assumption was false. Hence, $f(i) \neq f(j)$.

Forget that we fixed i and j. We thus have shown that $f(i) \neq f(j)$ whenever i and j are two distinct elements of $[n-1]$. In other words, the $n-1$ elements $f(1), f(2), \ldots, f(n-1)$ are distinct.

Clearly, $\{f(1), f(2), \ldots, f(n-1)\}$ is a subset of $[n]$ (since f is a map from $[n-1]$ to $[n]$). Thus,

$$|[n] \setminus \{f(1), f(2), \ldots, f(n-1)\}| = \underbrace{|[n]|}_{=n} - \underbrace{|\{f(1), f(2), \ldots, f(n-1)\}|}_{=n-1}$$
$$= n - (n-1) = 1.$$

In other words, $[n] \setminus \{f(1), f(2), \ldots, f(n-1)\}$ is a 1-element set. In other words,

$$[n] \setminus \{f(1), f(2), \ldots, f(n-1)\} = \{p\}$$

for some element p. Consider this p. We have $p \in \{p\} = [n] \setminus \{f(1), f(2), \ldots, f(n-1)\}$. In other words,

$$p \in [n] \qquad \text{and} \qquad p \notin \{f(1), f(2), \ldots, f(n-1)\}.$$

Now, for each $i \in [n]$, we have

$$\begin{cases} f(i), & \text{if } i \neq n; \\ p, & \text{if } i = n \end{cases} \in [n] \tag{10}$$

[7]. Hence, we can define a map $g : [n] \to [n]$ by setting

$$g(i) = \begin{cases} f(i), & \text{if } i \neq n; \\ p, & \text{if } i = n \end{cases} \qquad \text{for each } i \in [n].$$

Consider this map g.

The map g is injective[8]. However, a well-known fact from set theory (actually one of the versions of the pigeonhole principle) says that any injective map ϕ from a finite set

[7] *Proof of (10):* Let $i \in [n]$. We must prove (10). We are in one of the following two cases:

Case 1: We have $i \neq n$.

Case 2: We have $i = n$.

Let us first consider Case 1. In this case, we have $i \neq n$. Combining $i \in [n] = \{1, 2, \ldots, n\}$ with $i \neq n$, we obtain

$$i \in \{1, 2, \ldots, n\} \setminus \{n\} = \{1, 2, \ldots, n-1\} = [n-1]$$

(since $[n-1]$ is defined as $\{1, 2, \ldots, n-1\}$). Hence, $f(i) \in [n]$ (since f is a map from $[n-1]$ to $[n]$). Now, we have $i \neq n$ and thus $\begin{cases} f(i), & \text{if } i \neq n; \\ p, & \text{if } i = n \end{cases} = f(i) \in [n]$. Thus, (10) is proved in Case 1.

Let us now consider Case 2. In this case, we have $i = n$. Hence, $\begin{cases} f(i), & \text{if } i \neq n; \\ p, & \text{if } i = n \end{cases} = p \in [n]$. Thus, (10) is proved in Case 2.

We have now proved (10) in both Cases 1 and 2. Thus, the proof of (10) is complete.

[8] *Proof.* Let i and j be two distinct elements of $[n]$. We shall show that $g(i) \neq g(j)$.

Indeed, assume the contrary. Thus, $g(i) = g(j)$. We are looking to find a contradiction.

The situation is symmetric in i and j; thus, we WLOG assume that $i \leq j$ (since otherwise, we can just swap i with j). However, $i \neq j$ (since i and j are distinct). Thus, $i < j$ (since $i \leq j$). On

X to X must be a permutation of X. Applying this to $X = [n]$ and $\phi = g$, we conclude that g must be a permutation of $[n]$ (since g is an injective map from the finite set $[n]$ to $[n]$). Therefore, the inverse g^{-1} of g is well-defined and also is a permutation of $[n]$. In other words, $g^{-1} \in S_n$ (since S_n is the set of all permutations of $[n]$).

Next, we shall show that

$$\eta_i = \nu_{g(i)} \qquad \text{for each } i \in [n-1]\,. \tag{11}$$

[*Proof of (11):* Let $i \in [n-1]$. Thus, $i \leq n-1 < n$, so that $i \neq n$. However, $i \in [n-1] \subseteq [n]$. The definition of g yields $g(i) = \begin{cases} f(i), & \text{if } i \neq n; \\ p, & \text{if } i = n \end{cases} = f(i)$ (since $i \neq n$). On the other hand, (9) yields $\eta_i = \nu_{f(i)}$. This rewrites as $\eta_i = \nu_{g(i)}$ (since $g(i) = f(i)$). This proves (11).]

Now, recall that g is a permutation of $[n]$. In other words, g is a bijection from $[n]$

---

the other hand, $j \in [n] = \{1, 2, \ldots, n\}$, so that $j \leq n$. Thus, $i < j \leq n$, so that $i \neq n$. Now, the definition of g yields

$$g(i) = \begin{cases} f(i), & \text{if } i \neq n; \\ p, & \text{if } i = n \end{cases} = f(i) \qquad (\text{since } i \neq n)\,.$$

Combining $i \in [n]$ with $i \neq n$, we obtain $i \in [n] \setminus \{n\} = [n-1] = \{1, 2, \ldots, n-1\}$. Note that

$$g(i) = f(i) \in \{f(1), f(2), \ldots, f(n-1)\} \qquad (\text{since } i \in \{1, 2, \ldots, n-1\})\,.$$

If we had $j = n$, then we would have

$$\begin{aligned} g(i) = g(j) &= \begin{cases} f(j), & \text{if } j \neq n; \\ p, & \text{if } j = n \end{cases} \qquad (\text{by the definition of } g) \\ &= p \qquad (\text{since } j = n) \\ &\notin \{f(1), f(2), \ldots, f(n-1)\}\,, \end{aligned}$$

which would contradict $g(i) \in \{f(1), f(2), \ldots, f(n-1)\}$. Hence, we cannot have $j = n$. Thus, we have $j \neq n$. Combining $j \in [n]$ with $j \neq n$, we obtain $j \in [n] \setminus \{n\} = [n-1]$. Thus, both i and j are elements of $[n-1]$ (since $i \in [n-1]$ and $j \in [n-1]$). These elements i and j are furthermore distinct (as we know). Hence, we have $f(i) \neq f(j)$ (since the $n-1$ elements $f(1), f(2), \ldots, f(n-1)$ are distinct). However, from $g(i) = f(i)$, we obtain

$$\begin{aligned} f(i) = g(i) = g(j) &= \begin{cases} f(j), & \text{if } j \neq n; \\ p, & \text{if } j = n \end{cases} \qquad (\text{by the definition of } g) \\ &= f(j) \qquad (\text{since } j \neq n)\,, \end{aligned}$$

and this contradicts $f(i) \neq f(j)$. This contradiction shows that our assumption was false. Hence, $g(i) \neq g(j)$ is proven.

Now, forget that we fixed i and j. We thus have shown that if i and j are two distinct elements of $[n]$, then $g(i) \neq g(j)$. In other words, the map g is injective.

to $[n]$. Hence, we can substitute i for $g(i)$ in the sum $\sum\limits_{i\in[n]} \nu_{g(i)}$. We thus obtain

$$\sum_{i\in[n]} \nu_{g(i)} = \underbrace{\sum_{i\in[n]} \nu_i}_{=\sum\limits_{i=1}^{n}} = \sum_{i=1}^{n} \nu_i = \nu_1 + \nu_2 + \cdots + \nu_n = |\nu|$$

(since $|\nu|$ was defined to be $\nu_1 + \nu_2 + \cdots + \nu_n$). Hence,

$$\begin{aligned} |\nu| &= \underbrace{\sum_{i\in[n]} \nu_{g(i)}}_{=\sum\limits_{i=1}^{n}} = \sum_{i=1}^{n} \nu_{g(i)} = \nu_{g(n)} + \sum_{i=1}^{n-1} \underbrace{\nu_{g(i)}}_{\substack{=\eta_i \\ \text{(by (11))}}} \qquad \left(\begin{array}{c}\text{here, we have split off the}\\ \text{addend for } i=n \text{ from}\\ \text{the sum (since } n>0)\end{array}\right) \\ &= \nu_{g(n)} + \sum_{i=1}^{n-1} \eta_i. \end{aligned}$$

Comparing this with

$$\begin{aligned} |\nu| = |\eta| &= \eta_1 + \eta_2 + \cdots + \eta_n \qquad \text{(by the definition of } |\eta|) \\ &= \sum_{i=1}^{n} \eta_i = \eta_n + \sum_{i=1}^{n-1} \eta_i \qquad \left(\begin{array}{c}\text{here, we have split off}\\ \text{the addend for } i=n \text{ from}\\ \text{the sum (since } n>0)\end{array}\right), \end{aligned}$$

we obtain

$$\eta_n + \sum_{i=1}^{n-1} \eta_i = \nu_{g(n)} + \sum_{i=1}^{n-1} \eta_i.$$

Subtracting $\sum\limits_{i=1}^{n-1} \eta_i$ from both sides of this equality, we obtain

$$\eta_n = \nu_{g(n)}. \tag{12}$$

Now, we can easily conclude that

$$\eta_i = \nu_{g(i)} \qquad \text{for each } i \in [n]. \tag{13}$$

[*Proof of (13):* Let $i \in [n]$. We then must show that $\eta_i = \nu_{g(i)}$. If $i = n$, then this follows from (12). Thus, for the rest of this proof, we WLOG assume that $i \neq n$. Combining $i \in [n]$ and $i \neq n$, we obtain $i \in [n] \setminus \{n\} = [n-1]$. Hence, (11) yields $\eta_i = \nu_{g(i)}$. This proves (13).]

Consequently, we obtain that

$$\nu_j = \eta_{g^{-1}(j)} \qquad \text{for each } j \in [n]. \tag{14}$$

[*Proof of (14):* Let $j \in [n]$. Then, $g^{-1}(j) \in [n]$ (since g^{-1} is a map from $[n]$ to $[n]$). Hence, (13) (applied to $i = g^{-1}(j)$) yields $\eta_{g^{-1}(j)} = \nu_{g(g^{-1}(j))} = \nu_j$ (since $g(g^{-1}(j)) = j$). In other words, $\nu_j = \eta_{g^{-1}(j)}$. This proves (14).]

We can rewrite (14) as follows:

$$(\nu_1, \nu_2, \ldots, \nu_n) = \left(\eta_{g^{-1}(1)}, \eta_{g^{-1}(2)}, \ldots, \eta_{g^{-1}(n)}\right).$$

However, Definition 2.4 yields

$$\eta \circ g^{-1} = \left(\eta_{g^{-1}(1)}, \eta_{g^{-1}(2)}, \ldots, \eta_{g^{-1}(n)}\right).$$

Comparing these two equalities, we obtain

$$\eta \circ g^{-1} = (\nu_1, \nu_2, \ldots, \nu_n) = \nu.$$

In other words, $\nu = \eta \circ g^{-1}$. Hence, there exists some permutation $\pi \in S_n$ satisfying $\nu = \eta \circ \pi$ (namely, $\pi = g^{-1}$). This proves Lemma 2.10. □

Our second lemma is about an integer determinant:

Lemma 2.11. Let $n \in \mathbb{P}$ and $p \in \mathbb{N}$. Let η be the n-tuple

$$(1, 2, \ldots, n) + \left(\underbrace{0, 0, \ldots, 0}_{n-1 \text{ zeroes}}, p\right) = (1, 2, \ldots, n-1, n+p) \in \mathbb{Z}^n.$$

Let $\nu \in \mathbb{Z}^n$ be an n-tuple satisfying $|\nu| = |\eta|$. Then,

$$\det\left(([\nu_i \geq j])_{i,j \in [n]}\right) = \sum_{\substack{\sigma \in S_n; \\ \nu = \eta \circ \sigma}} (-1)^\sigma. \tag{15}$$

Note that the matrix $([\nu_i \geq j])_{i,j \in [n]}$ in (15) is a matrix with integer entries; thus, its determinant is a well-defined integer.

Before we prove Lemma 2.11, a remark is in order:

Remark 2.12. The sum $\sum_{\substack{\sigma \in S_n; \\ \nu = \eta \circ \sigma}} (-1)^\sigma$ on the right hand side of (15) always has either no addends or only one addend. (Indeed, it is easy to see that the n-tuples $\eta \circ \sigma$ for different $\sigma \in S_n$ are distinct; thus, no more than one of these n-tuples can equal ν.) Thus, this sum can be rewritten as

$$\begin{cases} (-1)^\sigma, & \text{if } \nu = \eta \circ \sigma \text{ for some } \sigma \in S_n; \\ 0, & \text{otherwise.} \end{cases}$$

Proof of Lemma 2.11. The definition of η yields

$$\eta = (1, 2, \ldots, n) + \left(\underbrace{0, 0, \ldots, 0}_{n-1 \text{ zeroes}}, p \right) = (1, 2, \ldots, n-1, n+p) .$$

Thus,

$$(\eta_1, \eta_2, \ldots, \eta_n) = \eta = (1, 2, \ldots, n-1, n+p) . \tag{16}$$

In other words, we have

$$\eta_i = i \qquad \text{for each } i \in [n-1] \tag{17}$$

and

$$\eta_n = n + p. \tag{18}$$

We have $p \geq 0$ (since $p \in \mathbb{N}$), so that $n + \underbrace{p}_{\geq 0} \geq n > n-1$. Thus, we have the following chain of inequalities:

$$1 < 2 < \cdots < n-1 < n+p$$

(indeed, $n-1 < n+p$ follows from $n+p > n-1$, whereas the rest of this chain of inequalities is obvious). In view of (16), we can rewrite this chain of inequalities as follows:

$$\eta_1 < \eta_2 < \cdots < \eta_n.$$

Hence, the n numbers $\eta_1, \eta_2, \ldots, \eta_n$ are distinct.

We have $n-1 \in \mathbb{N}$ (since $n \in \mathbb{P}$). Hence, $[n-1] = \{1, 2, \ldots, n-1\}$ (by the definition of $[n-1]$).

We are in one of the following two cases:
Case 1: We have $[n-1] \not\subseteq \{\nu_1, \nu_2, \ldots, \nu_n\}$.
Case 2: We have $[n-1] \subseteq \{\nu_1, \nu_2, \ldots, \nu_n\}$.

Let us first consider Case 1. In this case, we have $[n-1] \not\subseteq \{\nu_1, \nu_2, \ldots, \nu_n\}$. In other words, there exists some $k \in [n-1]$ such that $k \notin \{\nu_1, \nu_2, \ldots, \nu_n\}$. Consider this k. We have $k \in [n-1] = \{1, 2, \ldots, n-1\}$; thus, both k and $k+1$ are elements of $\{1, 2, \ldots, n\}$. Hence, the $n \times n$-matrix $([\nu_i \geq j])_{i,j \in [n]}$ has both a k-th column and a $(k+1)$-st column.

Now, it is easy to see that

$$[\nu_i \geq k] = [\nu_i \geq k+1] \qquad \text{for each } i \in [n] . \tag{19}$$

[*Proof of (19):* Let $i \in [n]$. Then, $\nu_i \in \mathbb{Z}$ (since $\nu \in \mathbb{Z}^n$). If we had $\nu_i = k$, then we would have $k = \nu_i \in \{\nu_1, \nu_2, \ldots, \nu_n\}$, which would contradict $k \notin \{\nu_1, \nu_2, \ldots, \nu_n\}$. Thus, we cannot have $\nu_i = k$. Hence, we have $\nu_i \neq k$. Therefore, Lemma 2.8 (applied to $u = \nu_i$) yields $[\nu_i \geq k] = [\nu_i \geq k+1]$. This proves (19).]

The equality (19) shows that each entry in the k-th column of the matrix $([\nu_i \geq j])_{i,j\in[n]}$ equals the corresponding entry in the $(k+1)$-st column of this matrix (because the i-th entry in the k-th column is $[\nu_i \geq k]$, whereas the corresponding entry in the $(k+1)$-st column is $[\nu_i \geq k+1]$). In other words, the k-th and the $(k+1)$-st columns of the matrix $([\nu_i \geq j])_{i,j\in[n]}$ are equal. Hence, this matrix $([\nu_i \geq j])_{i,j\in[n]}$ has two equal columns. However, it is well-known that if a square matrix[9] has two equal columns, then its determinant is 0. Hence, the determinant of the matrix $([\nu_i \geq j])_{i,j\in[n]}$ is 0 (since this matrix has two equal columns). In other words,

$$\det\left(([\nu_i \geq j])_{i,j\in[n]}\right) = 0. \tag{20}$$

On the other hand, there exists no $\sigma \in S_n$ satisfying $\nu = \eta \circ \sigma$ [10]. Hence, the sum $\sum\limits_{\substack{\sigma\in S_n;\\ \nu=\eta\circ\sigma}} (-1)^\sigma$ is empty. Thus,

$$\sum_{\substack{\sigma\in S_n;\\ \nu=\eta\circ\sigma}} (-1)^\sigma = (\text{empty sum}) = 0.$$

Comparing this with (20), we obtain $\det\left(([\nu_i \geq j])_{i,j\in[n]}\right) = \sum\limits_{\substack{\sigma\in S_n;\\ \nu=\eta\circ\sigma}} (-1)^\sigma$. Thus, Lemma 2.11 is proved in Case 1.

Let us now consider Case 2. In this case, we have $[n-1] \subseteq \{\nu_1, \nu_2, \ldots, \nu_n\}$. However, (17) shows that $(\eta_1, \eta_2, \ldots, \eta_{n-1}) = (1, 2, \ldots, n-1)$. Therefore,

$$\{\eta_1, \eta_2, \ldots, \eta_{n-1}\} = \{1, 2, \ldots, n-1\} = [n-1] \subseteq \{\nu_1, \nu_2, \ldots, \nu_n\}.$$

Moreover, from $\{\eta_1, \eta_2, \ldots, \eta_{n-1}\} = \{1, 2, \ldots, n-1\}$, we obtain $|\{\eta_1, \eta_2, \ldots, \eta_{n-1}\}| = |\{1, 2, \ldots, n-1\}| = n-1$ (since $n-1 \in \mathbb{N}$). Hence, Lemma 2.10 yields that there exists some permutation $\pi \in S_n$ satisfying $\nu = \eta \circ \pi$. Consider this π.

[9] In this proof, the word "matrix" always means a matrix with integer entries.

[10] *Proof.* Let $\sigma \in S_n$ satisfy $\nu = \eta \circ \sigma$. We shall obtain a contradiction.

Indeed, we have $\sigma \in S_n$. In other words, σ is a permutation of $[n]$ (since S_n was defined as the set of all permutations of $[n]$). Hence,

$$\begin{aligned}\left\{\eta_{\sigma(1)}, \eta_{\sigma(2)}, \ldots, \eta_{\sigma(n)}\right\} &= \{\eta_1, \eta_2, \ldots, \eta_n\} \\ &= \{1, 2, \ldots, n-1, n+p\} \qquad \text{(by (16))} \\ &\supseteq \{1, 2, \ldots, n-1\} = [n-1]\end{aligned}$$

(since $[n-1] = \{1, 2, \ldots, n-1\}$) and therefore

$$[n-1] \subseteq \left\{\eta_{\sigma(1)}, \eta_{\sigma(2)}, \ldots, \eta_{\sigma(n)}\right\}. \tag{21}$$

However, $(\nu_1, \nu_2, \ldots, \nu_n) = \nu = \eta \circ \sigma = \left(\eta_{\sigma(1)}, \eta_{\sigma(2)}, \ldots, \eta_{\sigma(n)}\right)$ (by Definition 2.4). Hence, $\{\nu_1, \nu_2, \ldots, \nu_n\} = \left\{\eta_{\sigma(1)}, \eta_{\sigma(2)}, \ldots, \eta_{\sigma(n)}\right\}$. In light of this, we can rewrite (21) as $[n-1] \subseteq \{\nu_1, \nu_2, \ldots, \nu_n\}$. But this contradicts $[n-1] \not\subseteq \{\nu_1, \nu_2, \ldots, \nu_n\}$.

Forget that we fixed σ. We thus have found a contradiction for each $\sigma \in S_n$ satisfying $\nu = \eta \circ \sigma$. Hence, there exists no such σ. Qed.

It is easy to see that π is the **only** permutation $\sigma \in S_n$ satisfying $\nu = \eta \circ \sigma$ [11]. Hence, the sum $\sum\limits_{\substack{\sigma \in S_n; \\ \nu = \eta \circ \sigma}} (-1)^\sigma$ has only one addend, namely the addend for $\sigma = \pi$. Thus,

$$\sum_{\substack{\sigma \in S_n; \\ \nu = \eta \circ \sigma}} (-1)^\sigma = (-1)^\pi . \tag{22}$$

We have $\nu = \eta \circ \pi = \left(\eta_{\pi(1)}, \eta_{\pi(2)}, \ldots, \eta_{\pi(n)}\right)$ (by Definition 2.4). Thus, for each $i \in [n]$, we have

$$\nu_i = \left(\eta_{\pi(1)}, \eta_{\pi(2)}, \ldots, \eta_{\pi(n)}\right)_i = \eta_{\pi(i)}. \tag{23}$$

On the other hand, it is well-known (see, e.g., [Grinbe21, Corollary 6.4.15] or [Grinbe15, Lemma 6.17 **(a)**]) that when the rows of a matrix are permuted, then the determinant of this matrix gets multiplied by $(-1)^\tau$, where τ is the permutation used to permute the rows. In other words: If $(a_{i,j})_{i,j \in [n]}$ is a square matrix (with integer entries), and if $\tau \in S_n$ is a permutation, then

$$\det\left(\left(a_{\tau(i),j}\right)_{i,j\in[n]}\right) = (-1)^\tau \cdot \det\left(\left(a_{i,j}\right)_{i,j\in[n]}\right).$$

Applying this to $a_{i,j} = [\eta_i \geq j]$ and $\tau = \pi$, we obtain

$$\det\left(\left(\left[\eta_{\pi(i)} \geq j\right]\right)_{i,j\in[n]}\right) = (-1)^\pi \cdot \det\left(\left([\eta_i \geq j]\right)_{i,j\in[n]}\right).$$

In view of (23), we can rewrite this as

$$\det\left(\left([\nu_i \geq j]\right)_{i,j\in[n]}\right) = (-1)^\pi \cdot \det\left(\left([\eta_i \geq j]\right)_{i,j\in[n]}\right). \tag{24}$$

However, we have $[\eta_i \geq j] = 0$ for any $i, j \in [n]$ satisfying $i < j$ [12]. In other words, the matrix $([\eta_i \geq j])_{i,j\in[n]}$ is lower-triangular. Therefore, its determinant is the product

[11] *Proof.* It is clear that π is a permutation $\sigma \in S_n$ satisfying $\nu = \eta \circ \sigma$ (because $\pi \in S_n$ is a permutation satisfying $\nu = \eta \circ \pi$). It thus remains to show that π is the **only** such permutation. In other words, it remains to show that if $\sigma \in S_n$ is a permutation satisfying $\nu = \eta \circ \sigma$, then $\sigma = \pi$. So let us show this.

Let $\sigma \in S_n$ be a permutation satisfying $\nu = \eta \circ \sigma$. Then, $\eta \circ \sigma = \nu = \eta \circ \pi$.

Recall that the n numbers $\eta_1, \eta_2, \ldots, \eta_n$ are distinct. Thus, Proposition 2.6 shows that if the permutations σ and π were distinct, then we would have $\eta \circ \sigma \neq \eta \circ \pi$; but this would contradict $\eta \circ \sigma = \eta \circ \pi$. Hence, σ and π cannot be distinct. In other words, we must have $\sigma = \pi$.

Forget that we fixed σ. We thus have shown that if $\sigma \in S_n$ is a permutation satisfying $\nu = \eta \circ \sigma$, then $\sigma = \pi$. In other words, π is the **only** permutation $\sigma \in S_n$ satisfying $\nu = \eta \circ \sigma$ (since we already know that π is such a permutation). Qed.

[12] *Proof.* Let $i, j \in [n]$ satisfy $i < j$. We must prove that $[\eta_i \geq j] = 0$.

We have $j \in [n] = \{1, 2, \ldots, n\}$, so that $j \leq n$. Thus, $i < j \leq n$, so that $i \neq n$. Combining $i \in [n] = \{1, 2, \ldots, n\}$ with $i \neq n$, we obtain $i \in \{1, 2, \ldots, n\} \setminus \{n\} = \{1, 2, \ldots, n-1\} = [n-1]$. Thus, (17) yields $\eta_i = i$. Hence, $\eta_i = i < j$. Thus, we don't have $\eta_i \geq j$. Hence, $[\eta_i \geq j] = 0$, qed.

$\prod_{i=1}^{n} [\eta_i \geq i]$ of its diagonal entries (because it is well-known that the determinant of any lower-triangular matrix is the product of its diagonal entries). In other words,

$$\det\left(([\eta_i \geq j])_{i,j\in[n]}\right) = \prod_{i=1}^{n} [\eta_i \geq i]\,. \tag{25}$$

However, each $i \in [n]$ satisfies $\eta_i \geq i$ [13] and therefore

$$[\eta_i \geq i] = 1. \tag{26}$$

Hence, (25) becomes

$$\det\left(([\eta_i \geq j])_{i,j\in[n]}\right) = \prod_{i=1}^{n} \underbrace{[\eta_i \geq i]}_{\substack{=1 \\ \text{(by (26))}}} = \prod_{i=1}^{n} 1 = 1.$$

Thus, (24) becomes

$$\det\left(([\nu_i \geq j])_{i,j\in[n]}\right) = (-1)^{\pi} \cdot \underbrace{\det\left(([\eta_i \geq j])_{i,j\in[n]}\right)}_{=1} = (-1)^{\pi} = \sum_{\substack{\sigma\in S_n; \\ \nu=\eta\circ\sigma}} (-1)^{\sigma}$$

(by (22)). Thus, Lemma 2.11 is proved in Case 2.

We have now proved Lemma 2.11 in both Cases 1 and 2. This completes the proof of Lemma 2.11. □

Our third lemma is another expression for the same determinant as in Lemma 2.11:

Lemma 2.13. Let $n \in \mathbb{N}$. For any permutation $\sigma \in S_n$, we let $\overline{\sigma}$ denote the n-tuple $(\sigma(1), \sigma(2), \ldots, \sigma(n)) \in \mathbb{Z}^n$.

Let $\nu \in \mathbb{Z}^n$ be an n-tuple. Then,

$$\det\left(([\nu_i \geq j])_{i,j\in[n]}\right) = \sum_{\substack{\sigma\in S_n; \\ \nu-\overline{\sigma}\in\mathbb{N}^n}} (-1)^{\sigma}\,.$$

[13] *Proof.* Let $i \in [n]$. We must prove that $\eta_i \geq i$. If $i \in [n-1]$, then this follows from (17) (in fact, in this case, (17) yields $\eta_i = i \geq i$). Hence, for the rest of this proof, we WLOG assume that we don't have $i \in [n-1]$. Thus, we have $i \notin [n-1]$. Combining $i \in [n]$ with $i \notin [n-1]$, we obtain

$$i \in \underbrace{[n]}_{=\{1,2,\ldots,n\}} \setminus \underbrace{[n-1]}_{=\{1,2,\ldots,n-1\}} = \{1,2,\ldots,n\} \setminus \{1,2,\ldots,n-1\} = \{n\}\,.$$

Thus, $i = n$. Hence, $\eta_i = \eta_n = n+p$ (by (18)). However, $p \geq 0$ (since $p \in \mathbb{N}$) and thus $n+p \geq n = i$ (since $i = n$). Thus, $\eta_i = n+p \geq i$, qed.

Proof of Lemma 2.13. We first observe that each $\sigma \in S_n$ satisfies

$$[\nu - \overline{\sigma} \in \mathbb{N}^n] = \prod_{i=1}^{n} [\nu_i \geq \sigma(i)]. \tag{27}$$

[*Proof of (27):* Let $\sigma \in S_n$ be arbitrary. We must prove the equality (27).

The definition of $\overline{\sigma}$ yields $\overline{\sigma} = (\sigma(1), \sigma(2), \ldots, \sigma(n))$. Hence, for each $i \in [n]$, we have

$$\overline{\sigma}_i = \sigma(i). \tag{28}$$

We are in one of the following two cases:

Case 1: We have $\nu - \overline{\sigma} \in \mathbb{N}^n$.

Case 2: We have $\nu - \overline{\sigma} \notin \mathbb{N}^n$.

Let us first consider Case 1. In this case, we have $\nu - \overline{\sigma} \in \mathbb{N}^n$. Now, let $i \in [n]$. Then, $(\nu - \overline{\sigma})_i \in \mathbb{N}$ (since $\nu - \overline{\sigma} \in \mathbb{N}^n$). In view of

$$(\nu - \overline{\sigma})_i = \nu_i - \underbrace{\overline{\sigma}_i}_{\substack{=\sigma(i) \\ \text{(by (28))}}} = \nu_i - \sigma(i),$$

we can rewrite this as $\nu_i - \sigma(i) \in \mathbb{N}$. Hence, $\nu_i - \sigma(i) \geq 0$, so that $\nu_i \geq \sigma(i)$. Thus, $[\nu_i \geq \sigma(i)] = 1$.

Forget that we fixed i. Thus, we have shown that $[\nu_i \geq \sigma(i)] = 1$ for each $i \in [n]$. Multiplying these equalities over all $i \in [n]$, we obtain $\prod_{i \in [n]} [\nu_i \geq \sigma(i)] = \prod_{i \in [n]} 1 = 1$. On the other hand, $[\nu - \overline{\sigma} \in \mathbb{N}^n] = 1$ (since $\nu - \overline{\sigma} \in \mathbb{N}^n$). Comparing these two equalities, we obtain

$$[\nu - \overline{\sigma} \in \mathbb{N}^n] = \underbrace{\prod_{i \in [n]} [\nu_i \geq \sigma(i)]}_{=\prod_{i=1}^{n}} = \prod_{i=1}^{n} [\nu_i \geq \sigma(i)].$$

Thus, (27) is proved in Case 1.

Let us now consider Case 2. In this case, we have $\nu - \overline{\sigma} \notin \mathbb{N}^n$. Hence, there exists some $j \in [n]$ such that $(\nu - \overline{\sigma})_j \notin \mathbb{N}$. Consider this j. We have $\overline{\sigma}_j = \sigma(j)$ (by (28), applied to $i = j$). Now,

$$(\nu - \overline{\sigma})_j = \nu_j - \underbrace{\overline{\sigma}_j}_{=\sigma(j)} = \nu_j - \sigma(j),$$

so that $\nu_j - \sigma(j) = (\nu - \overline{\sigma})_j \notin \mathbb{N}$. Combining this with $\nu_j - \sigma(j) \in \mathbb{Z}$ (which is obvious, since ν_j and $\sigma(j)$ belong to $\mathbb{Z}$), we obtain $\nu_j - \sigma(j) \in \mathbb{Z} \setminus \mathbb{N} = \{-1, -2, -3, \ldots\}$. In other words, $\nu_j - \sigma(j)$ is a negative integer. Hence $\nu_j - \sigma(j) < 0$, so that $\nu_j < \sigma(j)$. Thus, we don't have $\nu_j \geq \sigma(j)$. Hence, we have $[\nu_j \geq \sigma(j)] = 0$.

Now, consider the product $\prod_{i=1}^{n} [\nu_i \geq \sigma(i)]$. The term $[\nu_j \geq \sigma(j)]$ is one of the factors of this product (namely, the factor for $i = j$). Since this term is 0 (because we have just

shown that $[\nu_j \geq \sigma(j)] = 0$), we thus conclude that one of the factors of the product $\prod_{i=1}^{n} [\nu_i \geq \sigma(i)]$ is 0. Therefore, the entire product is 0. In other words,

$$\prod_{i=1}^{n} [\nu_i \geq \sigma(i)] = 0. \tag{29}$$

On the other hand, we do not have $\nu - \overline{\sigma} \in \mathbb{N}^n$ (since $\nu - \overline{\sigma} \notin \mathbb{N}^n$). Hence, $[\nu - \overline{\sigma} \in \mathbb{N}^n] = 0$. Comparing this with (29), we obtain

$$[\nu - \overline{\sigma} \in \mathbb{N}^n] = \prod_{i=1}^{n} [\nu_i \geq \sigma(i)].$$

Thus, (27) is proved in Case 2.

We have now proved (27) in both Cases 1 and 2. Thus, the proof of (27) is complete.]

Now, the definition of the determinant of a matrix yields

$$\begin{aligned}
&\det\left(([\nu_i \geq j])_{i,j\in[n]}\right) \\
&= \sum_{\sigma\in S_n} (-1)^\sigma \underbrace{\prod_{i=1}^{n} [\nu_i \geq \sigma(i)]}_{\substack{=[\nu-\overline{\sigma}\in\mathbb{N}^n] \\ \text{(by (27))}}} = \sum_{\sigma\in S_n} (-1)^\sigma [\nu - \overline{\sigma} \in \mathbb{N}^n] \\
&= \sum_{\substack{\sigma\in S_n; \\ \nu-\overline{\sigma}\in\mathbb{N}^n}} (-1)^\sigma \underbrace{[\nu - \overline{\sigma} \in \mathbb{N}^n]}_{\substack{=1 \\ \text{(since } \nu-\overline{\sigma}\in\mathbb{N}^n\text{)}}} + \sum_{\substack{\sigma\in S_n; \\ \nu-\overline{\sigma}\notin\mathbb{N}^n}} (-1)^\sigma \underbrace{[\nu - \overline{\sigma} \in \mathbb{N}^n]}_{\substack{=0 \\ \text{(since we don't have } \nu-\overline{\sigma}\in\mathbb{N}^n \\ \text{(because } \nu-\overline{\sigma}\notin\mathbb{N}^n\text{))}}} \\
&\qquad \left(\begin{array}{c}\text{since each } \sigma\in S_n \text{ satisfies either } \nu-\overline{\sigma}\in\mathbb{N}^n \\ \text{or } \nu-\overline{\sigma}\notin\mathbb{N}^n \text{ (but not both at the same time)}\end{array}\right) \\
&= \sum_{\substack{\sigma\in S_n; \\ \nu-\overline{\sigma}\in\mathbb{N}^n}} (-1)^\sigma + \underbrace{\sum_{\substack{\sigma\in S_n; \\ \nu-\overline{\sigma}\notin\mathbb{N}^n}} (-1)^\sigma 0}_{=0} = \sum_{\substack{\sigma\in S_n; \\ \nu-\overline{\sigma}\in\mathbb{N}^n}} (-1)^\sigma.
\end{aligned}$$

This proves Lemma 2.13. □

Our fourth and last lemma is a trivial property of the $\mathbb{Z}$-module $\mathbb{Z}^n$:

Lemma 2.14. Let $\alpha \in \mathbb{Z}^n$ and $\beta \in \mathbb{Z}^n$. Then, $|\alpha + \beta| = |\alpha| + |\beta|$.

Proof of Lemma 2.14. The definition of $|\alpha|$ yields $|\alpha| = \alpha_1 + \alpha_2 + \cdots + \alpha_n$. Similarly, $|\beta| = \beta_1 + \beta_2 + \cdots + \beta_n$. Adding these two equalities together, we obtain

$$|\alpha| + |\beta| = (\alpha_1 + \alpha_2 + \cdots + \alpha_n) + (\beta_1 + \beta_2 + \cdots + \beta_n). \tag{30}$$

The definition of $\alpha+\beta$ yields $\alpha+\beta=(\alpha_1+\beta_1,\alpha_2+\beta_2,\ldots,\alpha_n+\beta_n)$. Hence,

$$\begin{aligned}|\alpha+\beta| &= |(\alpha_1+\beta_1,\alpha_2+\beta_2,\ldots,\alpha_n+\beta_n)| \\ &= (\alpha_1+\beta_1)+(\alpha_2+\beta_2)+\cdots+(\alpha_n+\beta_n) \\ &\qquad\qquad (\text{by the definition of } |(\alpha_1+\beta_1,\alpha_2+\beta_2,\ldots,\alpha_n+\beta_n)|) \\ &= (\alpha_1+\alpha_2+\cdots+\alpha_n)+(\beta_1+\beta_2+\cdots+\beta_n) \\ &= |\alpha|+|\beta| \qquad (\text{by (30)}).\end{aligned}$$

This proves Lemma 2.14. □

We are now ready to prove Theorem 2.1:

Proof of Theorem 2.1. We have $n\in\mathbb{P}$. In other words, n is a positive integer. Thus, $n-1\in\mathbb{N}$ and $n>0$. The definition of η yields

$$\eta=(1,2,\ldots,n)+\left(\underbrace{0,0,\ldots,0}_{n-1 \text{ zeroes}},p\right)=(1,2,\ldots,n-1,n+p).$$

Thus,

$$\begin{aligned}|\eta| &= |(1,2,\ldots,n-1,n+p)| = 1+2+\cdots+(n-1)+(n+p) \\ &\qquad\qquad (\text{by the definition of } |(1,2,\ldots,n-1,n+p)|) \\ &= \underbrace{(1+2+\cdots+(n-1))+n}_{=1+2+\cdots+n}+p \\ &= (1+2+\cdots+n)+p. \end{aligned} \tag{31}$$

For any permutation $\sigma\in S_n$, we let $\overline{\sigma}$ denote the n-tuple $(\sigma(1),\sigma(2),\ldots,\sigma(n))\in\mathbb{Z}^n$. This n-tuple $\overline{\sigma}$ satisfies $|\overline{\sigma}|=1+2+\cdots+n$ [14] and therefore

$$\underbrace{|\overline{\sigma}|}_{=1+2+\cdots+n}+p=(1+2+\cdots+n)+p=|\eta| \tag{32}$$

(by (31)). Moreover, for each $\sigma\in S_n$ and each $i\in[n]$, we have

$$\overline{\sigma}_i=\sigma(i) \tag{33}$$

(since $\overline{\sigma}=(\sigma(1),\sigma(2),\ldots,\sigma(n))$).

Now, we claim the following:

[14] *Proof.* Let $\sigma\in S_n$. Thus, σ is a permutation of $[n]$ (since S_n was defined as the set of all permutations of $[n]$). In other words, σ is a bijection from $[n]$ to $[n]$. However, the definition of $\overline{\sigma}$ yields $\overline{\sigma}=(\sigma(1),\sigma(2),\ldots,\sigma(n))$. Thus,

$$\begin{aligned}|\overline{\sigma}| &= |(\sigma(1),\sigma(2),\ldots,\sigma(n))| = \sigma(1)+\sigma(2)+\cdots+\sigma(n) \\ &\qquad\qquad (\text{by the definition of } |(\sigma(1),\sigma(2),\ldots,\sigma(n))|) \\ &= \sum_{i\in[n]}\sigma(i)=\sum_{i\in[n]} i \qquad \left(\begin{array}{c}\text{here, we have substituted } i \text{ for } \sigma(i) \text{ in the sum,}\\ \text{since the map } \sigma:[n]\to[n] \text{ is a bijection}\end{array}\right) \\ &= 1+2+\cdots+n,\end{aligned}$$

qed.

Claim 1: Let $\beta \in \mathbb{N}^n$ and $\sigma \in S_n$ be arbitrary. Then, the statement "$|\beta| = p$" is equivalent to "$|\beta + \overline{\sigma}| = |\eta|$".

[*Proof of Claim 1:* We have $\beta + \overline{\sigma} = \overline{\sigma} + \beta$ (since $\mathbb{Z}^n$ is an abelian group). Thus, $|\beta + \overline{\sigma}| = |\overline{\sigma} + \beta| = |\overline{\sigma}| + |\beta|$ (by Lemma 2.14, applied to $\alpha = \overline{\sigma}$). Therefore,

$$\underbrace{|\beta + \overline{\sigma}|}_{=|\overline{\sigma}|+|\beta|} - \underbrace{|\eta|}_{\substack{=|\overline{\sigma}|+p \\ \text{(by (32))}}} = (|\overline{\sigma}| + |\beta|) - (|\overline{\sigma}| + p)$$

$$= |\beta| - p. \tag{34}$$

Hence, we have the following chain of logical equivalences:

$$(|\beta| = p) \iff \left(\underbrace{|\beta| - p}_{\substack{=|\beta+\overline{\sigma}|-|\eta| \\ \text{(by (34))}}} = 0 \right) \iff (|\beta + \overline{\sigma}| - |\eta| = 0)$$

$$\iff (|\beta + \overline{\sigma}| = |\eta|).$$

In other words, the statement "$|\beta| = p$" is equivalent to "$|\beta + \overline{\sigma}| = |\eta|$". This proves Claim 1.]

Let us now set

$$\prod_{i=1}^{n} a_i := a_1 a_2 \cdots a_n \tag{35}$$

for any $a_1, a_2, \ldots, a_n \in R$. (This is, of course, the usual meaning of the notation $\prod_{i=1}^{n} a_i$ when the ring R is commutative; however, we are now extending it to the case of arbitrary R by using the formula (35).)

Thus, if $(a_{i,j})_{i,j\in[n]}$ is any $n \times n$-matrix over R, then the definition of rowdet $\left((a_{i,j})_{i,j\in[n]}\right)$ yields

$$\operatorname{rowdet}\left((a_{i,j})_{i,j\in[n]}\right) = \sum_{\sigma\in S_n} (-1)^\sigma \underbrace{a_{1,\sigma(1)} a_{2,\sigma(2)} \cdots a_{n,\sigma(n)}}_{=\prod_{i=1}^{n} a_{i,\sigma(i)}}$$

$$= \sum_{\sigma\in S_n} (-1)^\sigma \prod_{i=1}^{n} a_{i,\sigma(i)}. \tag{36}$$

For each $\beta \in \mathbb{N}^n$, we have

$$t_{\alpha+\beta} = \operatorname{rowdet}\left(\left(h_{(\alpha+\beta)_i+j,\ i}\right)_{i,j\in[n]}\right) \qquad \text{(by the definition of } t_{\alpha+\beta}\text{)}$$

$$= \operatorname{rowdet}\left(\left(h_{\alpha_i+\beta_i+j,\ i}\right)_{i,j\in[n]}\right)$$

$$\text{(since } (\alpha+\beta)_i = \alpha_i + \beta_i \text{ for each } i \in [n]\text{)}$$

$$= \sum_{\sigma\in S_n} (-1)^\sigma \prod_{i=1}^{n} h_{\alpha_i+\beta_i+\sigma(i),\ i} \tag{37}$$

$$\text{(by (36), applied to } a_{i,j} = h_{\alpha_i+\beta_i+j,\ i}\text{)}.$$

However, for each $\beta \in \mathbb{N}^n$ and each $\sigma \in S_n$, we have

$$\beta_i + \underbrace{\sigma(i)}_{\substack{=\overline{\sigma}_i \\ \text{(by (33))}}} = \beta_i + \overline{\sigma}_i = (\beta + \overline{\sigma})_i$$

and therefore

$$h_{\alpha_i+\beta_i+\sigma(i),\ i} = h_{\alpha_i+(\beta+\overline{\sigma})_i,\ i}. \tag{38}$$

Hence, for each $\beta \in \mathbb{N}^n$, we have the equality

$$\begin{aligned} t_{\alpha+\beta} &= \sum_{\sigma \in S_n} (-1)^\sigma \prod_{i=1}^{n} \underbrace{h_{\alpha_i+\beta_i+\sigma(i),\ i}}_{\substack{=h_{\alpha_i+(\beta+\overline{\sigma})_i,\ i} \\ \text{(by (38))}}} \qquad \text{(by (37))} \\ &= \sum_{\sigma \in S_n} (-1)^\sigma \prod_{i=1}^{n} h_{\alpha_i+(\beta+\overline{\sigma})_i,\ i}. \end{aligned}$$

Summing these equalities over all $\beta \in \mathbb{N}^n$ satisfying $|\beta| = p$, we obtain

$$\begin{aligned} \sum_{\substack{\beta \in \mathbb{N}^n; \\ |\beta|=p}} t_{\alpha+\beta} &= \underbrace{\sum_{\substack{\beta \in \mathbb{N}^n; \\ |\beta|=p}} \sum_{\sigma \in S_n}}_{= \sum_{\sigma \in S_n} \sum_{\substack{\beta \in \mathbb{N}^n; \\ |\beta|=p}}} (-1)^\sigma \prod_{i=1}^{n} h_{\alpha_i+(\beta+\overline{\sigma})_i,\ i} \\ &= \sum_{\sigma \in S_n} \sum_{\substack{\beta \in \mathbb{N}^n; \\ |\beta|=p}} (-1)^\sigma \prod_{i=1}^{n} h_{\alpha_i+(\beta+\overline{\sigma})_i,\ i} \\ &= \sum_{\sigma \in S_n} (-1)^\sigma \sum_{\substack{\beta \in \mathbb{N}^n; \\ |\beta|=p}} \prod_{i=1}^{n} h_{\alpha_i+(\beta+\overline{\sigma})_i,\ i}. \end{aligned} \tag{39}$$

Now, fix a permutation $\sigma \in S_n$. We shall rewrite the sum $\sum\limits_{\substack{\beta \in \mathbb{N}^n; \\ |\beta|=p}} \prod\limits_{i=1}^{n} h_{\alpha_i+(\beta+\overline{\sigma})_i,\ i}$ in terms of $\beta + \overline{\sigma}$. Indeed, Claim 1 shows that for any n-tuple $\beta \in \mathbb{N}^n$, the statement "$|\beta| = p$" is equivalent to "$|\beta + \overline{\sigma}| = |\eta|$". Hence, we have the following equality of summation signs:

$$\begin{aligned} \sum_{\substack{\beta \in \mathbb{N}^n; \\ |\beta|=p}} &= \sum_{\substack{\beta \in \mathbb{N}^n; \\ |\beta+\overline{\sigma}|=|\eta|}} = \sum_{\substack{\beta \in \mathbb{Z}^n; \\ \beta \in \mathbb{N}^n; \\ |\beta+\overline{\sigma}|=|\eta|}} \qquad \text{(since } \mathbb{N}^n \subseteq \mathbb{Z}^n\text{)} \\ &= \sum_{\substack{\beta \in \mathbb{Z}^n; \\ (\beta+\overline{\sigma})-\overline{\sigma} \in \mathbb{N}^n; \\ |\beta+\overline{\sigma}|=|\eta|}} \qquad \text{(since } \beta = (\beta+\overline{\sigma})-\overline{\sigma}\text{)}. \end{aligned}$$

Thus,

$$\underbrace{\sum_{\substack{\beta\in\mathbb{N}^n;\\ |\beta|=p}}}_{=\sum\limits_{\substack{\beta\in\mathbb{Z}^n;\\ (\beta+\overline{\sigma})-\overline{\sigma}\in\mathbb{N}^n;\\ |\beta+\overline{\sigma}|=|\eta|}}} \prod_{i=1}^{n} h_{\alpha_i+(\beta+\overline{\sigma})_i,\ i} = \sum_{\substack{\beta\in\mathbb{Z}^n;\\ (\beta+\overline{\sigma})-\overline{\sigma}\in\mathbb{N}^n;\\ |\beta+\overline{\sigma}|=|\eta|}} \prod_{i=1}^{n} h_{\alpha_i+(\beta+\overline{\sigma})_i,\ i}.$$

However, $\mathbb{Z}^n$ is a group (under addition). Hence, for any $\rho \in \mathbb{Z}^n$, the map

$$\begin{aligned} \mathbb{Z}^n &\to \mathbb{Z}^n, \\ \beta &\mapsto \beta + \rho \end{aligned}$$

is a bijection (whose inverse is the map $\mathbb{Z}^n \to \mathbb{Z}^n,\ \nu \mapsto \nu - \rho$). Applying this to $\rho = \overline{\sigma}$, we conclude that the map

$$\begin{aligned} \mathbb{Z}^n &\to \mathbb{Z}^n, \\ \beta &\mapsto \beta + \overline{\sigma} \end{aligned}$$

is a bijection (whose inverse is the map $\mathbb{Z}^n \to \mathbb{Z}^n,\ \nu \mapsto \nu - \overline{\sigma}$). Therefore, we can substitute ν for $\beta + \overline{\sigma}$ in the sum $\sum\limits_{\substack{\beta\in\mathbb{Z}^n;\\ (\beta+\overline{\sigma})-\overline{\sigma}\in\mathbb{N}^n;\\ |\beta+\overline{\sigma}|=|\eta|}} \prod\limits_{i=1}^{n} h_{\alpha_i+(\beta+\overline{\sigma})_i,\ i}$. We thus obtain

$$\sum_{\substack{\beta\in\mathbb{Z}^n;\\ (\beta+\overline{\sigma})-\overline{\sigma}\in\mathbb{N}^n;\\ |\beta+\overline{\sigma}|=|\eta|}} \prod_{i=1}^{n} h_{\alpha_i+(\beta+\overline{\sigma})_i,\ i} = \sum_{\substack{\nu\in\mathbb{Z}^n;\\ \nu-\overline{\sigma}\in\mathbb{N}^n;\\ |\nu|=|\eta|}} \prod_{i=1}^{n} h_{\alpha_i+\nu_i,\ i}.$$

Hence, our above computation becomes

$$\begin{aligned} \sum_{\substack{\beta\in\mathbb{N}^n;\\ |\beta|=p}} \prod_{i=1}^{n} h_{\alpha_i+(\beta+\overline{\sigma})_i,\ i} &= \sum_{\substack{\beta\in\mathbb{Z}^n;\\ (\beta+\overline{\sigma})-\overline{\sigma}\in\mathbb{N}^n;\\ |\beta+\overline{\sigma}|=|\eta|}} \prod_{i=1}^{n} h_{\alpha_i+(\beta+\overline{\sigma})_i,\ i} \\ &= \sum_{\substack{\nu\in\mathbb{Z}^n;\\ \nu-\overline{\sigma}\in\mathbb{N}^n;\\ |\nu|=|\eta|}} \prod_{i=1}^{n} h_{\alpha_i+\nu_i,\ i}. \end{aligned} \tag{40}$$

Forget that we fixed σ. We thus have proved (40) for each permutation $\sigma \in S_n$.

Now, (39) becomes

$$\sum_{\substack{\beta\in\mathbb{N}^n;\\ |\beta|=p}} t_{\alpha+\beta} = \sum_{\sigma\in S_n} (-1)^\sigma \underbrace{\sum_{\substack{\beta\in\mathbb{N}^n;\\ |\beta|=p}} \prod_{i=1}^{n} h_{\alpha_i+(\beta+\overline{\sigma})_i,\ i}}_{\substack{= \sum\limits_{\substack{\nu\in\mathbb{Z}^n;\\ \nu-\overline{\sigma}\in\mathbb{N}^n;\\ |\nu|=|\eta|}} \prod\limits_{i=1}^{n} h_{\alpha_i+\nu_i,\ i} \\ \text{(by (40))}}}$$

$$= \sum_{\sigma\in S_n} (-1)^\sigma \sum_{\substack{\nu\in\mathbb{Z}^n;\\ \nu-\overline{\sigma}\in\mathbb{N}^n;\\ |\nu|=|\eta|}} \prod_{i=1}^{n} h_{\alpha_i+\nu_i,\ i}$$

$$= \sum_{\sigma\in S_n} \underbrace{\sum_{\substack{\nu\in\mathbb{Z}^n;\\ \nu-\overline{\sigma}\in\mathbb{N}^n;\\ |\nu|=|\eta|}} (-1)^\sigma \prod_{i=1}^{n} h_{\alpha_i+\nu_i,\ i}}_{= \sum\limits_{\substack{\nu\in\mathbb{Z}^n;\\ |\nu|=|\eta|}} \sum\limits_{\substack{\sigma\in S_n;\\ \nu-\overline{\sigma}\in\mathbb{N}^n}}}$$

$$= \sum_{\substack{\nu\in\mathbb{Z}^n;\\ |\nu|=|\eta|}} \sum_{\substack{\sigma\in S_n;\\ \nu-\overline{\sigma}\in\mathbb{N}^n}} (-1)^\sigma \prod_{i=1}^{n} h_{\alpha_i+\nu_i,\ i}$$

$$= \sum_{\substack{\nu\in\mathbb{Z}^n;\\ |\nu|=|\eta|}} \left(\sum_{\substack{\sigma\in S_n;\\ \nu-\overline{\sigma}\in\mathbb{N}^n}} (-1)^\sigma \right) \prod_{i=1}^{n} h_{\alpha_i+\nu_i,\ i}. \tag{41}$$

Now, if $\nu \in \mathbb{Z}^n$ is an n-tuple satisfying $|\nu| = |\eta|$, then

$$\sum_{\substack{\sigma\in S_n;\\ \nu-\overline{\sigma}\in\mathbb{N}^n}} (-1)^\sigma = \det\left(([\nu_i \geq j])_{i,j\in[n]}\right) \qquad \text{(by Lemma 2.13)}$$

$$= \sum_{\substack{\sigma\in S_n;\\ \nu=\eta\circ\sigma}} (-1)^\sigma \qquad \text{(by Lemma 2.11)}\,. \tag{42}$$

Thus, (41) becomes

$$\sum_{\substack{\beta\in\mathbb{N}^n;\\ |\beta|=p}} t_{\alpha+\beta} = \sum_{\substack{\nu\in\mathbb{Z}^n;\\ |\nu|=|\eta|}} \underbrace{\left(\sum_{\substack{\sigma\in S_n;\\ \nu-\overline{\sigma}\in\mathbb{N}^n}} (-1)^\sigma\right)}_{\substack{=\sum\limits_{\substack{\sigma\in S_n;\\ \nu=\eta\circ\sigma}} (-1)^\sigma \\ \text{(by (42))}}} \prod_{i=1}^{n} h_{\alpha_i+\nu_i,\ i}$$

$$= \sum_{\substack{\nu\in\mathbb{Z}^n;\\ |\nu|=|\eta|}} \left(\sum_{\substack{\sigma\in S_n;\\ \nu=\eta\circ\sigma}} (-1)^\sigma\right) \prod_{i=1}^{n} h_{\alpha_i+\nu_i,\ i}$$

$$= \sum_{\substack{\nu\in\mathbb{Z}^n;\\ |\nu|=|\eta|}} \sum_{\substack{\sigma\in S_n;\\ \nu=\eta\circ\sigma}} (-1)^\sigma \prod_{i=1}^{n} h_{\alpha_i+\nu_i,\ i}. \tag{43}$$

On the other hand,

$$\operatorname{rowdet}\left(\left(h_{\alpha_i+\eta_j,\ i}\right)_{i,j\in[n]}\right) = \sum_{\sigma\in S_n} (-1)^\sigma \prod_{i=1}^{n} h_{\alpha_i+\eta_{\sigma(i)},\ i} \tag{44}$$

(by (36), applied to $a_{i,j} = h_{\alpha_i+\eta_j,\ i}$). However, for each $\sigma \in S_n$, the n-tuple $\eta\circ\sigma$ belongs to $\mathbb{Z}^n$ and satisfies $|\eta\circ\sigma| = |\eta|$ (by Proposition 2.5). In other words, for each $\sigma \in S_n$, the n-tuple $\eta\circ\sigma$ is a $\nu \in \mathbb{Z}^n$ satisfying $|\nu| = |\eta|$. Hence, any sum ranging over all $\sigma \in S_n$ can be split according to the value of $\eta\circ\sigma$. In other words, we have the following equality of summation signs:

$$\sum_{\sigma\in S_n} = \sum_{\substack{\nu\in\mathbb{Z}^n;\\ |\nu|=|\eta|}} \underbrace{\sum_{\substack{\sigma\in S_n;\\ \eta\circ\sigma=\nu}}}_{=\sum\limits_{\substack{\sigma\in S_n;\\ \nu=\eta\circ\sigma}}} = \sum_{\substack{\nu\in\mathbb{Z}^n;\\ |\nu|=|\eta|}} \sum_{\substack{\sigma\in S_n;\\ \nu=\eta\circ\sigma}}.$$

Thus, (44) becomes

$$\operatorname{rowdet}\left(\left(h_{\alpha_i+\eta_j,\ i}\right)_{i,j\in[n]}\right) = \underbrace{\sum_{\sigma\in S_n}}_{=\sum\limits_{\substack{\nu\in\mathbb{Z}^n;\\ |\nu|=|\eta|}} \sum\limits_{\substack{\sigma\in S_n;\\ \nu=\eta\circ\sigma}}} (-1)^\sigma \prod_{i=1}^{n} h_{\alpha_i+\eta_{\sigma(i)},\ i}$$

$$= \sum_{\substack{\nu\in\mathbb{Z}^n;\\ |\nu|=|\eta|}} \sum_{\substack{\sigma\in S_n;\\ \nu=\eta\circ\sigma}} (-1)^\sigma \prod_{i=1}^{n} \underbrace{h_{\alpha_i+\eta_{\sigma(i)},\ i}}_{\substack{=h_{\alpha_i+\nu_i,\ i} \\ \text{(since } \eta_{\sigma(i)}=\nu_i \\ \text{(because } \nu=\eta\circ\sigma=\left(\eta_{\sigma(1)},\eta_{\sigma(2)},\ldots,\eta_{\sigma(n)}\right) \\ \text{(by Definition 2.4) and therefore } \nu_i=\eta_{\sigma(i)}))}}$$

$$= \sum_{\substack{\nu\in\mathbb{Z}^n;\\ |\nu|=|\eta|}} \sum_{\substack{\sigma\in S_n;\\ \nu=\eta\circ\sigma}} (-1)^\sigma \prod_{i=1}^{n} h_{\alpha_i+\nu_i,\ i}.$$

Comparing this with (43), we obtain

$$\sum_{\substack{\beta\in\mathbb{N}^n;\\ |\beta|=p}} t_{\alpha+\beta} = \operatorname{rowdet}\left(\left(h_{\alpha_i+\eta_j,\ i}\right)_{i,j\in[n]}\right).$$

This proves Theorem 2.1. □

3. Corollaries

3.1. The $h_{k,\ n} = 0$ case

We shall now derive some corollaries from Theorem 2.1 by imposing some conditions on R or on the $h_{k,\ i}$. We begin with the most basic one, in which we force $h_{k,\ n}$ to be 0 for all negative k:

Corollary 3.1. Let $n \in \mathbb{P}$ and $p \in \mathbb{Z}$. Let $h_{k,\ i}$ be an element of R for all $k \in \mathbb{Z}$ and $i \in [n]$. Assume that

$$h_{k,\ n} = 0 \qquad \text{for all } k < 0. \tag{45}$$

For any $\alpha \in \mathbb{Z}^n$, we define

$$t_\alpha := \operatorname{rowdet}\left(\left(h_{\alpha_i+j,\ i}\right)_{i,j\in[n]}\right) \in R.$$

Let $\alpha \in \mathbb{Z}^n$ be such that $\alpha_n \leq -n$. Then,

$$\sum_{\substack{\beta\in\mathbb{N}^n;\\ |\beta|=p}} t_{\alpha+\beta} = \operatorname{rowdet}\left(\left(h_{\alpha_i+j,\ i}\right)_{i,j\in[n-1]}\right) \cdot h_{\alpha_n+n+p,\ n}.$$

In order to derive this from Theorem 2.1, we shall need the following formula for the row-determinant of a matrix whose last row is 0 except for its rightmost entry:

Lemma 3.2. Let n be a positive integer. Let $A = \left(a_{i,j}\right)_{i,j\in[n]} \in R^{n\times n}$ be an $n \times n$-matrix. Assume that

$$a_{n,j} = 0 \qquad \text{for every } j \in \{1, 2, \ldots, n-1\}. \tag{46}$$

Then,

$$\operatorname{rowdet} A = \operatorname{rowdet}\left(\left(a_{i,j}\right)_{i,j\in[n-1]}\right) \cdot a_{n,n}.$$

Example 3.3. For $n = 3$ and $A = \begin{pmatrix} a & b & c \\ d & e & f \\ 0 & 0 & g \end{pmatrix}$, the claim of Lemma 3.2 says that

$$\operatorname{rowdet}\begin{pmatrix} a & b & c \\ d & e & f \\ 0 & 0 & g \end{pmatrix} = \operatorname{rowdet}\begin{pmatrix} a & b \\ d & e \end{pmatrix} \cdot g.$$

(The two zeroes in the third row of A are necessary for Lemma 3.2 to be applicable, as they guarantee that the condition (46) is satisfied.)

Proof of Lemma 3.2. This is a generalization of [Grinbe15, Theorem 6.43] to the case of arbitrary R (not necessarily commutative). The proof given in [Grinbe15] still works for this generalization, as long as we keep in mind that the products are noncommutative.

For the sake of completeness, here is this argument in detail, except for the parts that can be copied from [Grinbe15, proof of Theorem 6.43] verbatim.

We shall first prove the following auxiliary claim (an analogue of [Grinbe15, Lemma 6.44]):

Claim 1: We have

$$\sum_{\substack{\sigma\in S_n;\\ \sigma(n)=n}} (-1)^{\sigma} a_{1,\sigma(1)} a_{2,\sigma(2)} \cdots a_{n-1,\sigma(n-1)} = \operatorname{rowdet}\left(\left(a_{i,j}\right)_{i,j\in[n-1]}\right).$$

[*Proof of Claim 1:* We define a subset T of S_n by

$$T = \{\tau \in S_n \mid \tau(n) = n\}.$$

For every $\sigma \in S_{n-1}$, we define a map $\widehat{\sigma} : [n] \to [n]$ by setting

$$\widehat{\sigma}(i) = \begin{cases} \sigma(i), & \text{if } i < n; \\ n, & \text{if } i = n \end{cases} \qquad \text{for every } i \in [n].$$

It is easy to see (and has been proved in [Grinbe15, proof of Lemma 6.44][15]) that this map $\widehat{\sigma}$ is well-defined and belongs to T. Thus, we can define a map $\Phi : S_{n-1} \to T$ by setting

$$\Phi(\sigma) = \widehat{\sigma} \qquad \text{for every } \sigma \in S_{n-1}.$$

Consider this map Φ. It is not hard to see (and has been proved in [Grinbe15, proof of Lemma 6.44]) that the map Φ is a bijection. Furthermore, every $\sigma \in S_{n-1}$ satisfies

$$(-1)^{\widehat{\sigma}} = (-1)^{\sigma} \tag{47}$$

(again, this has been proved in [Grinbe15, proof of Lemma 6.44]). Finally, every $\sigma \in S_{n-1}$ satisfies

$$\begin{aligned} &a_{1,\widehat{\sigma}(1)} a_{2,\widehat{\sigma}(2)} \cdots a_{n-1,\widehat{\sigma}(n-1)} \\ &= a_{1,\sigma(1)} a_{2,\sigma(2)} \cdots a_{n-1,\sigma(n-1)} \end{aligned} \tag{48}$$

[15] A reader looking these arguments up in [Grinbe15, proof of Lemma 6.44] should keep in mind that we are here using the notation $[n]$ for what has been called $\{1, 2, \ldots, n\}$ in [Grinbe15, proof of Lemma 6.44].

[16].

Now,

$$\begin{aligned}
&\underbrace{\sum_{\substack{\sigma\in S_n;\\ \sigma(n)=n}}}_{=\sum\limits_{\sigma\in\{\tau\in S_n \mid \tau(n)=n\}}} (-1)^{\sigma}\, a_{1,\sigma(1)}a_{2,\sigma(2)}\cdots a_{n-1,\sigma(n-1)} \\
&\qquad\qquad \left(\text{since } \{\tau\in S_n \mid \tau(n)=n\}=T\right) \\
&= \sum_{\sigma\in T} (-1)^{\sigma}\, a_{1,\sigma(1)}a_{2,\sigma(2)}\cdots a_{n-1,\sigma(n-1)} \\
&= \sum_{\sigma\in S_{n-1}} \underbrace{(-1)^{\Phi(\sigma)}\, a_{1,(\Phi(\sigma))(1)}a_{2,(\Phi(\sigma))(2)}\cdots a_{n-1,(\Phi(\sigma))(n-1)}}_{\substack{=(-1)^{\widehat{\sigma}} a_{1,\widehat{\sigma}(1)}a_{2,\widehat{\sigma}(2)}\cdots a_{n-1,\widehat{\sigma}(n-1)} \\ \left(\text{since } \Phi(\sigma)=\widehat{\sigma}\right)}} \\
&\qquad\qquad \left(\begin{array}{c}\text{here, we have substituted } \Phi(\sigma) \text{ for } \sigma \text{ in the sum,}\\ \text{since the map } \Phi : S_{n-1}\to T \text{ is a bijection}\end{array}\right) \\
&= \sum_{\sigma\in S_{n-1}} \underbrace{(-1)^{\widehat{\sigma}}}_{\substack{=(-1)^{\sigma}\\ \text{(by (47))}}} \underbrace{a_{1,\widehat{\sigma}(1)}a_{2,\widehat{\sigma}(2)}\cdots a_{n-1,\widehat{\sigma}(n-1)}}_{\substack{=a_{1,\sigma(1)}a_{2,\sigma(2)}\cdots a_{n-1,\sigma(n-1)}\\ \text{(by (48))}}} \\
&= \sum_{\sigma\in S_{n-1}} (-1)^{\sigma}\, a_{1,\sigma(1)}a_{2,\sigma(2)}\cdots a_{n-1,\sigma(n-1)}.
\end{aligned}$$

But the definition of the row-determinant of a matrix yields

$$\operatorname{rowdet}\left(\left(a_{i,j}\right)_{i,j\in[n-1]}\right) = \sum_{\sigma\in S_{n-1}} (-1)^{\sigma}\, a_{1,\sigma(1)}a_{2,\sigma(2)}\cdots a_{n-1,\sigma(n-1)}.$$

Comparing these two equalities, we obtain

$$\sum_{\substack{\sigma\in S_n;\\ \sigma(n)=n}} (-1)^{\sigma}\, a_{1,\sigma(1)}a_{2,\sigma(2)}\cdots a_{n-1,\sigma(n-1)} = \operatorname{rowdet}\left(\left(a_{i,j}\right)_{i,j\in[n-1]}\right).$$

This proves Claim 1.]

Now, every permutation $\sigma\in S_n$ satisfying $\sigma(n)\neq n$ satisfies

$$a_{n,\sigma(n)} = 0 \tag{49}$$

[16] *Proof of (48):* Let $\sigma\in S_{n-1}$. Let $i\in[n-1]$. Then, $i\in[n-1]\subseteq[n]$. Moreover, from $i\in[n-1]=\{1,2,\ldots,n-1\}$, we obtain $i\leq n-1<n$. Now, the definition of $\widehat{\sigma}$ yields

$$\widehat{\sigma}(i) = \begin{cases}\sigma(i), & \text{if } i<n;\\ n, & \text{if } i=n\end{cases} = \sigma(i) \qquad (\text{since } i<n).$$

Hence, $a_{i,\widehat{\sigma}(i)} = a_{i,\sigma(i)}$.

Forget that we fixed i. We thus have proved that the equality $a_{i,\widehat{\sigma}(i)} = a_{i,\sigma(i)}$ holds for each $i\in[n-1]$. Multiplying these equalities for all $i\in[n-1]$, we obtain $a_{1,\widehat{\sigma}(1)}a_{2,\widehat{\sigma}(2)}\cdots a_{n-1,\widehat{\sigma}(n-1)} = a_{1,\sigma(1)}a_{2,\sigma(2)}\cdots a_{n-1,\sigma(n-1)}$. This proves (48).

[17].

Recall that $A = (a_{i,j})_{i,j\in[n]}$. Hence, the definition of the row-determinant of a matrix yields

$$\begin{aligned}
\operatorname{rowdet} A &= \sum_{\sigma\in S_n} (-1)^\sigma \underbrace{a_{1,\sigma(1)}a_{2,\sigma(2)}\cdots a_{n,\sigma(n)}}_{=\left(a_{1,\sigma(1)}a_{2,\sigma(2)}\cdots a_{n-1,\sigma(n-1)}\right)a_{n,\sigma(n)}} \\
&= \sum_{\sigma\in S_n} (-1)^\sigma \left(a_{1,\sigma(1)}a_{2,\sigma(2)}\cdots a_{n-1,\sigma(n-1)}\right) a_{n,\sigma(n)} \\
&= \sum_{\substack{\sigma\in S_n;\\ \sigma(n)=n}} (-1)^\sigma \left(a_{1,\sigma(1)}a_{2,\sigma(2)}\cdots a_{n-1,\sigma(n-1)}\right) \underbrace{a_{n,\sigma(n)}}_{\substack{=a_{n,n}\\ \text{(since } \sigma(n)=n\text{)}}} \\
&\quad + \sum_{\substack{\sigma\in S_n;\\ \sigma(n)\neq n}} (-1)^\sigma \left(a_{1,\sigma(1)}a_{2,\sigma(2)}\cdots a_{n-1,\sigma(n-1)}\right) \underbrace{a_{n,\sigma(n)}}_{\substack{=0\\ \text{(by (49))}}} \\
&\qquad \left(\begin{array}{c} \text{since every } \sigma\in S_n \text{ satisfies} \\ \text{either } \sigma(n)=n \text{ or } \sigma(n)\neq n \text{ (but not both)} \end{array} \right) \\
&= \sum_{\substack{\sigma\in S_n;\\ \sigma(n)=n}} (-1)^\sigma \left(a_{1,\sigma(1)}a_{2,\sigma(2)}\cdots a_{n-1,\sigma(n-1)}\right) a_{n,n} \\
&\quad + \underbrace{\sum_{\substack{\sigma\in S_n;\\ \sigma(n)\neq n}} (-1)^\sigma \left(a_{1,\sigma(1)}a_{2,\sigma(2)}\cdots a_{n-1,\sigma(n-1)}\right) 0}_{=0} \\
&= \sum_{\substack{\sigma\in S_n;\\ \sigma(n)=n}} (-1)^\sigma \left(a_{1,\sigma(1)}a_{2,\sigma(2)}\cdots a_{n-1,\sigma(n-1)}\right) a_{n,n} \\
&= \underbrace{\left(\sum_{\substack{\sigma\in S_n;\\ \sigma(n)=n}} (-1)^\sigma a_{1,\sigma(1)}a_{2,\sigma(2)}\cdots a_{n-1,\sigma(n-1)} \right)}_{\substack{=\operatorname{rowdet}\left((a_{i,j})_{i,j\in[n-1]}\right)\\ \text{(by Claim 1)}}} \cdot a_{n,n} \\
&= \operatorname{rowdet}\left((a_{i,j})_{i,j\in[n-1]}\right)\cdot a_{n,n}.
\end{aligned}$$

This proves Lemma 3.2. □

Proof of Corollary 3.1. If $p < 0$, then Corollary 3.1 is easily seen to hold[18]. Thus, for the rest of this proof, we WLOG assume that we don't have $p < 0$. Hence, we have $p \geq 0$. Hence, $p \in \mathbb{N}$ (since $p \in \mathbb{Z}$).

[17] *Proof of (49):* Let $\sigma \in S_n$ be a permutation satisfying $\sigma(n) \neq n$. Since $\sigma(n) \in [n] = \{1,2,\ldots,n\}$ and $\sigma(n) \neq n$, we have $\sigma(n) \in \{1,2,\ldots,n\} \setminus \{n\} = \{1,2,\ldots,n-1\}$. Hence, (46) (applied to $j = \sigma(n)$) shows that $a_{n,\sigma(n)} = 0$. This proves (49).

[18] *Proof.* Assume that $p < 0$. Then, $\underbrace{\alpha_n}_{\leq -n} + n + p \leq (-n) + n + p = p < 0$. Thus, (45) (applied to

Define an n-tuple $\eta \in \mathbb{Z}^n$ as in Theorem 2.1. Thus,

$$(\eta_1, \eta_2, \ldots, \eta_n) = \eta = (1, 2, \ldots, n-1, n+p)\,.$$

In other words, we have

$$\eta_j = j \qquad \text{for each } j \in [n-1] \tag{50}$$

and

$$\eta_n = n + p. \tag{51}$$

Now, Theorem 2.1 yields

$$\sum_{\substack{\beta \in \mathbb{N}^n; \\ |\beta| = p}} t_{\alpha+\beta} = \operatorname{rowdet}\left(\left(h_{\alpha_i+\eta_j,\ i}\right)_{i,j\in[n]}\right). \tag{52}$$

However, it is easy to see that $h_{\alpha_n+\eta_j,\ n} = 0$ for every $j \in \{1, 2, \ldots, n-1\}$ [19]. Hence, Lemma 3.2 (applied to $\left(h_{\alpha_i+\eta_j,\ i}\right)_{i,j\in[n]}$ and $h_{\alpha_i+\eta_j,\ i}$ instead of A and $a_{i,j}$) yields

$$\operatorname{rowdet}\left(\left(h_{\alpha_i+\eta_j,\ i}\right)_{i,j\in[n]}\right) = \operatorname{rowdet}\left(\left(h_{\alpha_i+\eta_j,\ i}\right)_{i,j\in[n-1]}\right) \cdot h_{\alpha_n+\eta_n,\ n}.$$

$k = \alpha_n + n + p$) yields $h_{\alpha_n+n+p,\ n} = 0$.

However, if $\beta \in \mathbb{N}^n$, then

$$\begin{aligned} |\beta| &= \beta_1 + \beta_2 + \cdots + \beta_n \qquad \text{(by the definition of } |\beta|) \\ &= \sum_{i=1}^{n} \underbrace{\beta_i}_{\substack{\geq 0 \\ \text{(since } \beta_i \in \mathbb{N} \\ \text{(because } \beta \in \mathbb{N}^n))}} \geq \sum_{i=1}^{n} 0 = 0 > p \qquad \text{(since } p < 0) \end{aligned}$$

and therefore $|\beta| \neq p$. In other words, there exists no $\beta \in \mathbb{N}^n$ satisfying $|\beta| = p$. Hence, the sum $\sum_{\substack{\beta \in \mathbb{N}^n; \\ |\beta| = p}} t_{\alpha+\beta}$ is empty. Thus,

$$\sum_{\substack{\beta \in \mathbb{N}^n; \\ |\beta| = p}} t_{\alpha+\beta} = (\text{empty sum}) = 0.$$

Comparing this with

$$\operatorname{rowdet}\left(\left(h_{\alpha_i+j,\ i}\right)_{i,j\in[n-1]}\right) \cdot \underbrace{h_{\alpha_n+n+p,\ n}}_{=0} = 0,$$

we obtain

$$\sum_{\substack{\beta \in \mathbb{N}^n; \\ |\beta| = p}} t_{\alpha+\beta} = \operatorname{rowdet}\left(\left(h_{\alpha_i+j,\ i}\right)_{i,j\in[n-1]}\right) \cdot h_{\alpha_n+n+p,\ n}.$$

Thus, we have shown that Corollary 3.1 holds under the assumption that $p < 0$.

[19] *Proof.* Let $j \in \{1, 2, \ldots, n-1\}$. We must show that $h_{\alpha_n+\eta_j,\ n} = 0$.

We have $j \in \{1, 2, \ldots, n-1\} = [n-1]$ and therefore $\eta_j = j$ (by (50)). From $j \in \{1, 2, \ldots, n-1\}$, we also obtain $j \leq n-1 < n$, so that $\underbrace{\alpha_n}_{\leq -n} + \underbrace{\eta_j}_{=j<n} < (-n) + n = 0$. Therefore, (45) (applied to $k = \alpha_n + \eta_j$) yields $h_{\alpha_n+\eta_j,\ n} = 0$, qed.

Hence, (52) becomes

$$\sum_{\substack{\beta\in\mathbb{N}^n;\\ |\beta|=p}} t_{\alpha+\beta} = \operatorname{rowdet}\left(\left(h_{\alpha_i+\eta_j,\ i}\right)_{i,j\in[n]}\right)$$
$$= \operatorname{rowdet}\left(\left(h_{\alpha_i+\eta_j,\ i}\right)_{i,j\in[n-1]}\right)\cdot h_{\alpha_n+\eta_n,\ n}. \tag{53}$$

However, each $j\in[n-1]$ satisfies $\eta_j = j$ (by (50)). Thus, $h_{\alpha_i+\eta_j,\ i} = h_{\alpha_i+j,\ i}$ for every $i,j\in[n-1]$. In other words,

$$\left(h_{\alpha_i+\eta_j,\ i}\right)_{i,j\in[n-1]} = \left(h_{\alpha_i+j,\ i}\right)_{i,j\in[n-1]}. \tag{54}$$

Using (54) and (51), we can rewrite (53) as

$$\sum_{\substack{\beta\in\mathbb{N}^n;\\ |\beta|=p}} t_{\alpha+\beta} = \operatorname{rowdet}\left(\left(h_{\alpha_i+j,\ i}\right)_{i,j\in[n-1]}\right)\cdot h_{\alpha_n+n+p,\ n}.$$

Thus, Corollary 3.1 is proved. □

3.2. The commutative case

Corollary 3.4. Assume that R is commutative. Let $n\in\mathbb{P}$ and $p\in\mathbb{Z}$. Let $h_{k,\ i}$ be an element of R for all $k\in\mathbb{Z}$ and $i\in[n]$. Assume that

$$h_{k,\ n} = 0 \qquad \text{for all } k<0.$$

For any $\alpha\in\mathbb{Z}^n$, we define

$$t_\alpha := \det\left(\left(h_{\alpha_i+j,\ i}\right)_{i,j\in[n]}\right)\in R.$$

Let $\alpha\in\mathbb{Z}^n$ be such that $\alpha_n\le -n$. Then,

$$\sum_{\substack{\beta\in\mathbb{N}^n;\\ |\beta|=p}} t_{\alpha+\beta} = \det\left(\left(h_{\alpha_i+j,\ i}\right)_{i,j\in[n-1]}\right)\cdot h_{\alpha_n+n+p,\ n}.$$

Proof of Corollary 3.4. We have assumed that the ring R is commutative. Thus, every square matrix over R has a well-defined determinant. Furthermore, the row-determinant of any square matrix over R is the same as the determinant of this matrix (since our above definition of the row-determinant clearly generalizes the standard definition of a determinant). In other words, if A is any square matrix over R, then

$$\operatorname{rowdet} A = \det A. \tag{55}$$

Now, for any $\alpha\in\mathbb{Z}^n$, we have

$$\operatorname{rowdet}\left(\left(h_{\alpha_i+j,\ i}\right)_{i,j\in[n]}\right) = \det\left(\left(h_{\alpha_i+j,\ i}\right)_{i,j\in[n]}\right) \tag{56}$$

(by (55), applied to $A = (h_{\alpha_i+j,\ i})_{i,j\in[n]}$). Hence, for any $\alpha \in \mathbb{Z}^n$, we have

$$\begin{aligned} t_\alpha &= \det\left((h_{\alpha_i+j,\ i})_{i,j\in[n]}\right) && \text{(by our definition of } t_\alpha\text{)} \\ &= \operatorname{rowdet}\left((h_{\alpha_i+j,\ i})_{i,j\in[n]}\right) && \text{(by (56))}. \end{aligned}$$

But this is precisely how t_α was defined in Corollary 3.1. Thus, our notations match the notations of Corollary 3.1. Hence, we can apply Corollary 3.1. As a result, we obtain

$$\begin{aligned} \sum_{\substack{\beta\in\mathbb{N}^n;\\ |\beta|=p}} t_{\alpha+\beta} &= \underbrace{\operatorname{rowdet}\left((h_{\alpha_i+j,\ i})_{i,j\in[n-1]}\right)}_{\substack{=\det\left((h_{\alpha_i+j,\ i})_{i,j\in[n-1]}\right) \\ \text{(by (55), applied to } A=\left(h_{\alpha_i+j,\ i}\right)_{i,j\in[n-1]})}} \cdot h_{\alpha_n+n+p,\ n} \\ &= \det\left((h_{\alpha_i+j,\ i})_{i,j\in[n-1]}\right) \cdot h_{\alpha_n+n+p,\ n}. \end{aligned}$$

This proves Corollary 3.4. □

3.3. A Schur-like reindexing

Here are some more consequences of the first pre-Pieri rule:

Corollary 3.5. Let $n \in \mathbb{P}$ and $p \in \mathbb{Z}$. Let $h_{k,\ i}$ be an element of R for all $k \in \mathbb{Z}$ and $i \in [n]$. Assume that

$$h_{k,\ n} = 0 \qquad \text{for all } k < 0.$$

For any $m \in \{0, 1, \ldots, n\}$ and any $\lambda \in \mathbb{Z}^m$, we define

$$s_\lambda := \operatorname{rowdet}\left((h_{\lambda_i+j-i,\ i})_{i,j\in[m]}\right) \in R.$$

Fix an n-tuple $\mu \in \mathbb{Z}^n$ with $\mu_n = 0$. Let $\overline{\mu} = (\mu_1, \mu_2, \ldots, \mu_{n-1})$. Then,

$$s_{\overline{\mu}} \cdot h_{p,\ n} = \sum_{\substack{\beta\in\mathbb{N}^n;\\ |\beta|=p}} s_{\mu+\beta}.$$

Proof of Corollary 3.5. Define $t_\alpha \in R$ for each $\alpha \in \mathbb{Z}^n$ as in Corollary 3.1.

Define an n-tuple $\alpha \in \mathbb{Z}^n$ by

$$\alpha = (\mu_1 - 1,\ \mu_2 - 2,\ \ldots,\ \mu_n - n).$$

Thus,

$$\alpha_i = \mu_i - i \qquad \text{for each } i \in [n]. \tag{57}$$

Applying this to $i = n$, we obtain $\alpha_n = \underbrace{\mu_n}_{=0} - n = -n \leq -n$. Hence, Corollary 3.1 yields

$$\sum_{\substack{\beta\in\mathbb{N}^n;\\ |\beta|=p}} t_{\alpha+\beta} = \operatorname{rowdet}\left((h_{\alpha_i+j,\ i})_{i,j\in[n-1]}\right) \cdot h_{\alpha_n+n+p,\ n}. \tag{58}$$

However, each $\beta \in \mathbb{N}^n$ satisfies

$$t_{\alpha+\beta} = s_{\mu+\beta} \tag{59}$$

[20] Furthermore, we have

$$\operatorname{rowdet}\left(\left(h_{\alpha_i+j,\ i}\right)_{i,j\in[n-1]}\right) = s_{\overline{\mu}} \tag{60}$$

[20] *Proof.* Let $\beta \in \mathbb{N}^n$. For each $i, j \in [n]$, we have

$$\underbrace{(\alpha+\beta)_i}_{=\alpha_i+\beta_i} + j = \underbrace{\alpha_i}_{\substack{=\mu_i - i \\ \text{(by (57))}}} + \beta_i + j = \mu_i - i + \beta_i + j = \underbrace{\mu_i + \beta_i}_{=(\mu+\beta)_i} + j - i = (\mu+\beta)_i + j - i$$

and therefore

$$h_{(\alpha+\beta)_i+j,\ i} = h_{(\mu+\beta)_i+j-i,\ i}.$$

In other words, we have

$$\left(h_{(\alpha+\beta)_i+j,\ i}\right)_{i,j\in[n]} = \left(h_{(\mu+\beta)_i+j-i,\ i}\right)_{i,j\in[n]}.$$

The definition of $t_{\alpha+\beta}$ yields

$$t_{\alpha+\beta} = \operatorname{rowdet}\left(\underbrace{\left(h_{(\alpha+\beta)_i+j,\ i}\right)_{i,j\in[n]}}_{=\left(h_{(\mu+\beta)_i+j-i,\ i}\right)_{i,j\in[n]}}\right) = \operatorname{rowdet}\left(\left(h_{(\mu+\beta)_i+j-i,\ i}\right)_{i,j\in[n]}\right) = s_{\mu+\beta}$$

(since the definition of $s_{\mu+\beta}$ yields $s_{\mu+\beta} = \operatorname{rowdet}\left(\left(h_{(\mu+\beta)_i+j-i,\ i}\right)_{i,j\in[n]}\right)$). Qed.

[21]. Finally, from $\alpha_n = -n$, we obtain

$$\alpha_n + n + p = (-n) + n + p = p. \tag{63}$$

Using (59), (60) and (63), we can rewrite (58) as

$$\sum_{\substack{\beta \in \mathbb{N}^n; \\ |\beta| = p}} s_{\mu+\beta} = s_{\overline{\mu}} \cdot h_{p,\ n}.$$

This proves Corollary 3.5. □

Corollary 3.6. Assume that R is commutative. Let $n \in \mathbb{P}$ and $p \in \mathbb{Z}$. Let $h_{k,\ i}$ be an element of R for all $k \in \mathbb{Z}$ and $i \in [n]$. Assume that

$$h_{k,\ n} = 0 \qquad \text{for all } k < 0.$$

For any $m \in \{0, 1, \ldots, n\}$ and any $\lambda \in \mathbb{Z}^m$, we define

$$s_\lambda := \det\left(\left(h_{\lambda_i + j - i,\ i}\right)_{i,j \in [m]}\right) \in R.$$

Fix an n-tuple $\mu \in \mathbb{Z}^n$ with $\mu_n = 0$. Let $\overline{\mu} = (\mu_1, \mu_2, \ldots, \mu_{n-1})$. Then,

$$h_{p,\ n} \cdot s_{\overline{\mu}} = \sum_{\substack{\beta \in \mathbb{N}^n; \\ |\beta| = p}} s_{\mu+\beta}. \tag{64}$$

[21] *Proof.* We have $\overline{\mu} = (\mu_1, \mu_2, \ldots, \mu_{n-1})$; thus,

$$\overline{\mu}_i = \mu_i \qquad \text{for each } i \in [n-1]. \tag{61}$$

The definition of $s_{\overline{\mu}}$ yields

$$s_{\overline{\mu}} = \operatorname{rowdet}\left(\left(h_{\overline{\mu}_i + j - i,\ i}\right)_{i,j \in [n-1]}\right). \tag{62}$$

For each $i, j \in [n-1]$, we have

$$\underbrace{\alpha_i}_{\substack{=\mu_i - i \\ \text{(by (57))}}} + j = \underbrace{\mu_i}_{\substack{=\overline{\mu}_i \\ \text{(by (61))}}} - i + j = \overline{\mu}_i - i + j = \overline{\mu}_i + j - i$$

and thus $h_{\alpha_i + j,\ i} = h_{\overline{\mu}_i + j - i,\ i}$. In other words, we have

$$\left(h_{\alpha_i + j,\ i}\right)_{i,j \in [n-1]} = \left(h_{\overline{\mu}_i + j - i,\ i}\right)_{i,j \in [n-1]}.$$

Hence,

$$\operatorname{rowdet}\left(\underbrace{\left(h_{\alpha_i + j,\ i}\right)_{i,j \in [n-1]}}_{= \left(h_{\overline{\mu}_i + j - i,\ i}\right)_{i,j \in [n-1]}}\right) = \operatorname{rowdet}\left(\left(h_{\overline{\mu}_i + j - i,\ i}\right)_{i,j \in [n-1]}\right) = s_{\overline{\mu}}$$

(by (62)).

Proof of Corollary 3.6. If A is any square matrix over R, then

$$\operatorname{rowdet} A = \det A. \tag{65}$$

(Indeed, we have already shown this in our proof of Corollary 3.4.) Now, for any $m \in \{0, 1, \ldots, n\}$ and any $\lambda \in \mathbb{Z}^m$, we have

$$\operatorname{rowdet}\left(\left(h_{\lambda_i+j-i,\ i}\right)_{i,j\in[m]}\right) = \det\left(\left(h_{\lambda_i+j-i,\ i}\right)_{i,j\in[m]}\right) \tag{66}$$

(by (65), applied to $A = \left(h_{\lambda_i+j-i,\ i}\right)_{i,j\in[m]}$). Hence, for any $m \in \{0, 1, \ldots, n\}$ and any $\lambda \in \mathbb{Z}^m$, we have

$$\begin{aligned} s_\lambda &= \det\left(\left(h_{\lambda_i+j-i,\ i}\right)_{i,j\in[m]}\right) && \text{(by our definition of } s_\lambda\text{)} \\ &= \operatorname{rowdet}\left(\left(h_{\lambda_i+j-i,\ i}\right)_{i,j\in[m]}\right) && \text{(by (66))}. \end{aligned}$$

But this is precisely how s_λ was defined in Corollary 3.5. Thus, our notations match the notations of Corollary 3.5. Hence, we can apply Corollary 3.5. As a result, we obtain

$$s_{\overline{\mu}} \cdot h_{p,\ n} = \sum_{\substack{\beta\in\mathbb{N}^n; \\ |\beta|=p}} s_{\mu+\beta}.$$

Hence,

$$\sum_{\substack{\beta\in\mathbb{N}^n; \\ |\beta|=p}} s_{\mu+\beta} = s_{\overline{\mu}} \cdot h_{p,\ n} = h_{p,\ n} \cdot s_{\overline{\mu}}.$$

This proves Corollary 3.6. □

We note that (5) is the particular case of Corollary 3.6 for $R = \Lambda$, $h_{k,\ i} = h_k$ and $\mu = \lambda$.

3.4. Recovering Fun's rule

Let us now explain how [Fun12, Proposition 3.9] follows from Corollary 3.6. Here is the claim of [Fun12, Proposition 3.9] with slightly modified notations:

Proposition 3.7. Assume that R is commutative. Let $\ell \in \mathbb{N}$, $p \in \mathbb{Z}$ and $q \in \mathbb{Z}$. Let $g_{k,\ j}$ be an element of R for all $k, j \in \mathbb{Z}$. Assume that

$$g_{k,\ j} = 0 \qquad \text{for all } k < 0 \text{ and } j \in \mathbb{Z}. \tag{67}$$

For any $m \in \mathbb{N}$ and any $\mu \in \mathbb{Z}^m$ and $\beta \in \mathbb{Z}^m$, we define

$$s_{\mu,\ \beta} := \det\left(\left(g_{\mu_i+j-i,\ \beta_i+j-i}\right)_{i,j\in[m]}\right) \in R.$$

Let $\mu \in \mathbb{Z}^{\ell+1}$ be an $(\ell+1)$-tuple satisfying $\mu_{\ell+1} = 0$. Let $\overline{\mu}$ be the ℓ-tuple $(\mu_1, \mu_2, \ldots, \mu_\ell) \in \mathbb{Z}^\ell$.

Let $\beta \in \mathbb{Z}^\ell$ be an ℓ-tuple, and let $\beta' \in \mathbb{Z}^{\ell+1}$ be the $(\ell+1)$-tuple $(\beta_1, \beta_2, \ldots, \beta_\ell, q-p)$.

Then,

$$g_{p,\ q} \cdot s_{\overline{\mu},\ \beta} = \sum_{\substack{\delta \in \mathbb{N}^{\ell+1}; \\ |\delta| = p}} s_{\mu+\delta,\ \beta'+\delta}.$$

To be precise, Proposition 3.7 does not only differ from [Fun12, Proposition 3.9] in the notations (what we call q and $g_{k,\ j}$ corresponds to β_e and $h_{k,\ j}$ in [Fun12, Proposition 3.9]; furthermore, the summation index δ in our sum is called σ in [Fun12, Proposition 3.9]), but is also slightly more general (our μ can be any $(\ell+1)$-tuple in $\mathbb{Z}^{\ell+1}$ satisfying $\mu_{\ell+1} = 0$ rather than just a partition of length ℓ; our $g_{k,\ j}$ are not required to satisfy $g_{0,\ j} = 1$; our p is not assumed to be positive). Note that our definition of $s_{\mu,\ \beta}$ in Proposition 3.7 is equivalent to the one in [Fun12], because of [Fun12, (3.3)].

As promised, we can now easily derive Proposition 3.7 from Corollary 3.6:

Proof of Proposition 3.7. We have $\overline{\mu} = (\mu_1, \mu_2, \ldots, \mu_\ell)$. Thus,

$$\overline{\mu}_i = \mu_i \qquad \text{for each } i \in [\ell]\,. \tag{68}$$

From $\beta' = (\beta_1, \beta_2, \ldots, \beta_\ell, q-p)$, we obtain

$$\beta'_i = \beta_i \qquad \text{for each } i \in [\ell] \tag{69}$$

and $\beta'_{\ell+1} = q - p$.

Set

$$n := \ell + 1. \tag{70}$$

Set

$$h_{k,\ i} := g_{k,\ \beta'_i - \mu_i + k} \qquad \text{for every } k \in \mathbb{Z} \text{ and } i \in [n]\,.$$

Then, we have $h_{k,\ n} = 0$ for all $k < 0$ [22]. Furthermore, from $n = \ell + 1$, we obtain $\mu_n = \mu_{\ell+1} = 0$ and $\beta'_n = \beta'_{\ell+1} = q - p$; therefore, $\underbrace{\beta'_n}_{=q-p} - \underbrace{\mu_n}_{=0} + p = (q-p) - 0 + p = q$.

Now, the definition of $h_{p,\ n}$ yields

$$h_{p,\ n} = g_{p,\ \beta'_n - \mu_n + p} = g_{p,\ q} \tag{71}$$

(since $\beta'_n - \mu_n + p = q$).

For any $m \in \{0, 1, \ldots, n\}$ and any $\lambda \in \mathbb{Z}^m$, we define $s_\lambda \in R$ as in Corollary 3.6. Thus, (64) (with the summation index β renamed as δ) yields

$$h_{p,\ n} \cdot s_{\overline{\mu}} = \sum_{\substack{\delta \in \mathbb{N}^n; \\ |\delta| = p}} s_{\mu+\delta}. \tag{72}$$

[22] *Proof.* Let $k < 0$ be an integer. Then, the definition of $h_{k,\ n}$ yields $h_{k,\ n} = g_{k,\ \beta'_n - \mu_n + k} = 0$ (by (67), applied to $j = \beta'_n - \mu_n + k$). Qed.

However, it is easy to see that

$$s_{\overline{\mu}} = s_{\overline{\mu},\ \beta} \tag{73}$$

[23]. Furthermore,

$$s_{\mu+\delta} = s_{\mu+\delta,\ \beta'+\delta} \qquad \text{for each } \delta \in \mathbb{N}^{\ell+1} \tag{75}$$

[24].

Using (70), (71), (73) and (75), we can rewrite (72) as

$$g_{p,\ q} \cdot s_{\overline{\mu},\ \beta} = \sum_{\substack{\delta \in \mathbb{N}^{\ell+1}; \\ |\delta| = p}} s_{\mu+\delta,\ \beta'+\delta}.$$

[23] *Proof of (73):* For each $i, j \in [\ell]$, we have

$$\begin{aligned} h_{\overline{\mu}_i+j-i,\ i} &= h_{\mu_i+j-i,\ i} \qquad \text{(since (68) yields } \overline{\mu}_i = \mu_i\text{)} \\ &= g_{\mu_i+j-i,\ \beta'_i-\mu_i+(\mu_i+j-i)} \qquad \text{(by the definition of } h_{\mu_i+j-i,\ i}\text{)} \\ &= g_{\mu_i+j-i,\ \beta_i+j-i} \\ &\qquad \left(\text{since } \beta'_i - \mu_i + (\mu_i + j - i) = \underbrace{\beta'_i}_{\substack{=\beta_i \\ \text{(by (69))}}} + j - i = \beta_i + j - i \right) \\ &= g_{\overline{\mu}_i+j-i,\ \beta_i+j-i} \end{aligned} \tag{74}$$

(since (68) yields $\mu_i = \overline{\mu}_i$).

The definition of $s_{\overline{\mu}}$ yields

$$s_{\overline{\mu}} = \det\left(\left(\underbrace{h_{\overline{\mu}_i+j-i,\ i}}_{\substack{=g_{\overline{\mu}_i+j-i,\ \beta_i+j-i} \\ \text{(by (74))}}} \right)_{i,j\in[\ell]} \right) = \det\left(\left(g_{\overline{\mu}_i+j-i,\ \beta_i+j-i} \right)_{i,j\in[\ell]} \right).$$

On the other hand, the definition of $s_{\overline{\mu},\ \beta}$ yields

$$s_{\overline{\mu},\ \beta} = \det\left(\left(g_{\overline{\mu}_i+j-i,\ \beta_i+j-i} \right)_{i,j\in[\ell]} \right).$$

Comparing these two equalities, we obtain $s_{\overline{\mu}} = s_{\overline{\mu},\ \beta}$, qed.

[24] *Proof of (75):* Let $\delta \in \mathbb{N}^{\ell+1}$.

For each $i, j \in [\ell+1]$, we have

$$\begin{aligned} h_{(\mu+\delta)_i+j-i,\ i} &= h_{\mu_i+\delta_i+j-i,\ i} \qquad \text{(since } (\mu+\delta)_i = \mu_i + \delta_i\text{)} \\ &= g_{\mu_i+\delta_i+j-i,\ \beta'_i-\mu_i+(\mu_i+\delta_i+j-i)} \qquad \text{(by the definition of } h_{\mu_i+\delta_i+j-i,\ i}\text{)} \\ &= g_{\mu_i+\delta_i+j-i,\ (\beta'+\delta)_i+j-i} \\ &\qquad \left(\text{since } \beta'_i - \mu_i + (\mu_i + \delta_i + j - i) = \underbrace{\beta'_i + \delta_i}_{=(\beta'+\delta)_i} + j - i = (\beta'+\delta)_i + j - i \right) \\ &= g_{(\mu+\delta)_i+j-i,\ (\beta'+\delta)_i+j-i} \end{aligned} \tag{76}$$

(since $\mu_i + \delta_i = (\mu+\delta)_i$).

This proves Proposition 3.7. □

3.5. Recovering the immaculate Pieri rule

Next, we shall exhibit [BBSSZ13, Theorem 3.5] as a consequence of Corollary 3.5. To this aim, we will briefly introduce the relevant parts of the scene of [BBSSZ13, Theorem 3.5]. We fix a commutative ring $\mathbf{k}$, and we let NSym be the algebra of noncommutative polynomials in countably many variables $H_1, H_2, H_3, \ldots$ over $\mathbf{k}$. We also set $H_0 := 1$ and $H_k := 0$ for all $k < 0$. For every $m \in \mathbb{N}$ and every $\alpha \in \mathbb{Z}^m$, we set

$$H_\alpha := H_{\alpha_1} H_{\alpha_2} \cdots H_{\alpha_m} \in \mathrm{NSym}\,.$$

For every $m \in \mathbb{N}$ and every $\alpha \in \mathbb{Z}^m$, we set

$$\mathfrak{S}_\alpha := \sum_{\sigma \in S_m} (-1)^\sigma \, H_{(\alpha_1+\sigma(1)-1,\ \alpha_2+\sigma(2)-2,\ \ldots,\ \alpha_m+\sigma(m)-m)} \in \mathrm{NSym}\,.$$

(This is not how $\mathfrak{S}_\alpha$ is defined in [BBSSZ13], but it is an equivalent definition, because [BBSSZ13, Theorem 3.27] shows that the $\mathfrak{S}_\alpha$ from [BBSSZ13] is given by the same formula.) Now, [BBSSZ13, Theorem 3.5] (the "right-Pieri rule for multiplication by H_s" in the terminology of [BBSSZ13]) states the following:

Proposition 3.8. Let $n \in \mathbb{N}$ and $\alpha \in \mathbb{Z}^n$. Let $s \in \mathbb{Z}$. Then,

$$\mathfrak{S}_\alpha H_s = \sum_{\substack{\beta \in \mathbb{Z}^{n+1}; \\ |\beta| = |\alpha| + s; \\ \alpha_i \leq \beta_i \text{ for all } i \in [n]; \\ 0 \leq \beta_{n+1}}} \mathfrak{S}_\beta .$$

(The conditions under the summation sign here are essentially what is denoted by "$\alpha \subset_s \beta$" in [BBSSZ13].)

To be fully precise, Proposition 3.8 is a bit more general than [BBSSZ13, Theorem 3.5], since α is assumed to be a composition (i.e., a tuple of positive integers) in [BBSSZ13, Theorem 3.5], while we are only assuming that $\alpha \in \mathbb{Z}^n$.

The definition of $s_{\mu+\delta}$ yields

$$s_{\mu+\delta} = \det\left(\left(\underbrace{h_{(\mu+\delta)_i+j-i,\ i}}_{=g_{(\mu+\delta)_i+j-i,\ (\beta'+\delta)_i+j-i} \text{ (by (76))}}\right)_{i,j\in[\ell+1]}\right) = \det\left(\left(g_{(\mu+\delta)_i+j-i,\ (\beta'+\delta)_i+j-i}\right)_{i,j\in[\ell+1]}\right).$$

On the other hand, the definition of $s_{\mu+\delta,\ \beta'+\delta}$ yields

$$s_{\mu+\delta,\ \beta'+\delta} = \det\left(\left(g_{(\mu+\delta)_i+j-i,\ (\beta'+\delta)_i+j-i}\right)_{i,j\in[\ell+1]}\right).$$

Comparing these two equalities, we obtain $s_{\mu+\delta} = s_{\mu+\delta,\ \beta'+\delta}$. This proves (75).

Proof of Proposition 3.8. Let R be the ring NSym. Set

$$h_{k,\ i} := H_k \in R \qquad \text{for each } k \in \mathbb{Z} \text{ and } i \in \mathbb{Z}.$$

Thus, $h_{s,\ n+1} = H_s$.

For any $m \in \{0, 1, \ldots, n\}$ and any $\lambda \in \mathbb{Z}^m$, we define $s_\lambda \in R$ as in Corollary 3.5. (This s_λ has nothing to do with the integer s.)

Let $\mu \in \mathbb{Z}^{n+1}$ be the $(n+1)$-tuple $(\alpha_1, \alpha_2, \ldots, \alpha_n, 0)$. Thus, $\mu_{n+1} = 0$ and $|\mu| = |\alpha|$ [25]. Moreover, we have

$$(\mu_1, \mu_2, \ldots, \mu_{n+1}) = \mu = (\alpha_1, \alpha_2, \ldots, \alpha_n, 0)\,.$$

Thus,

$$\mu_i = \alpha_i \qquad \text{for each } i \in [n]\,. \tag{77}$$

Let $\overline{\mu} = (\mu_1, \mu_2, \ldots, \mu_n) \in \mathbb{Z}^n$. Thus, $\overline{\mu} = \alpha$ [26].

For each $k < 0$, we have

$$\begin{aligned} h_{k,\ n+1} &= H_k \qquad \text{(by the definition of } h_{k,\ n+1}\text{)} \\ &= 0 \qquad \text{(since } k < 0\text{)}\,. \end{aligned}$$

Thus, we have shown that $h_{k,\ n+1} = 0$ for all $k < 0$. Hence, Corollary 3.5 (applied to $n+1$ and s instead of n and p) yields

$$s_{\overline{\mu}} \cdot h_{s,\ n+1} = \sum_{\substack{\beta \in \mathbb{N}^{n+1}; \\ |\beta| = s}} s_{\mu+\beta}.$$

In view of $h_{s,\ n+1} = H_s$ and $\overline{\mu} = \alpha$, we can rewrite this as

$$s_\alpha \cdot H_s = \sum_{\substack{\beta \in \mathbb{N}^{n+1}; \\ |\beta| = s}} s_{\mu+\beta}. \tag{78}$$

It is easy to see that

$$\mathfrak{S}_\lambda = s_\lambda \qquad \text{for any } m \in \mathbb{N} \text{ and any } \lambda \in \mathbb{Z}^m. \tag{79}$$

[25] *Proof.* From $\mu = (\alpha_1, \alpha_2, \ldots, \alpha_n, 0)$, we obtain

$$\begin{aligned} |\mu| &= |(\alpha_1, \alpha_2, \ldots, \alpha_n, 0)| = \alpha_1 + \alpha_2 + \cdots + \alpha_n + 0 \qquad \text{(by the definition of } |(\alpha_1, \alpha_2, \ldots, \alpha_n, 0)|\text{)} \\ &= \alpha_1 + \alpha_2 + \cdots + \alpha_n = |\alpha| \qquad \text{(since } |\alpha| \text{ is defined to be } \alpha_1 + \alpha_2 + \cdots + \alpha_n\text{)}\,. \end{aligned}$$

[26] *Proof.* We just have seen that

$$\mu_i = \alpha_i \qquad \text{for each } i \in [n]\,.$$

In other words, $(\mu_1, \mu_2, \ldots, \mu_n) = (\alpha_1, \alpha_2, \ldots, \alpha_n)$. Hence, $\overline{\mu} = (\mu_1, \mu_2, \ldots, \mu_n) = (\alpha_1, \alpha_2, \ldots, \alpha_n) = \alpha$ (since $\alpha \in \mathbb{Z}^n$).

[*Proof of (79):* Let $m \in \mathbb{N}$ and $\lambda \in \mathbb{Z}^m$. The definition of $\mathfrak{S}_\lambda$ yields

$$\begin{aligned}\mathfrak{S}_\lambda &= \sum_{\sigma \in S_m} (-1)^\sigma \underbrace{H_{(\lambda_1+\sigma(1)-1,\ \lambda_2+\sigma(2)-2,\ \ldots,\ \lambda_m+\sigma(m)-m)}}_{\substack{= H_{\lambda_1+\sigma(1)-1} H_{\lambda_2+\sigma(2)-2} \cdots H_{\lambda_m+\sigma(m)-m} \\ \text{(by the definition of } H_{(\lambda_1+\sigma(1)-1,\ \lambda_2+\sigma(2)-2,\ \ldots,\ \lambda_m+\sigma(m)-m)})}} \\ &= \sum_{\sigma \in S_m} (-1)^\sigma H_{\lambda_1+\sigma(1)-1} H_{\lambda_2+\sigma(2)-2} \cdots H_{\lambda_m+\sigma(m)-m}.\end{aligned}$$

On the other hand, the definition of s_λ yields

$$\begin{aligned}s_\lambda &= \operatorname{rowdet}\left(\left(\underbrace{h_{\lambda_i+j-i,\ i}}_{\substack{= H_{\lambda_i+j-i} \\ \text{(by the definition of } h_{\lambda_i+j-i,\ i})}}\right)_{i,j\in[m]}\right) = \operatorname{rowdet}\left(\left(H_{\lambda_i+j-i}\right)_{i,j\in[m]}\right) \\ &= \sum_{\sigma \in S_m} (-1)^\sigma H_{\lambda_1+\sigma(1)-1} H_{\lambda_2+\sigma(2)-2} \cdots H_{\lambda_m+\sigma(m)-m}\end{aligned}$$

(by the definition of a row-determinant). Comparing these two equalities, we obtain $\mathfrak{S}_\lambda = s_\lambda$. This proves (79).]

Now, (79) (applied to $m = n$ and $\lambda = \alpha$) yields $\mathfrak{S}_\alpha = s_\alpha$. Moreover, for each $\beta \in \mathbb{N}^{n+1}$, we have $\mathfrak{S}_{\mu+\beta} = s_{\mu+\beta}$ (by (79), applied to $m = n+1$ and $\lambda = \mu + \beta$). In view of these two equalities, we can rewrite (78) as

$$\mathfrak{S}_\alpha \cdot H_s = \sum_{\substack{\beta \in \mathbb{N}^{n+1}; \\ |\beta| = s}} \mathfrak{S}_{\mu+\beta}. \tag{80}$$

Now, let W be the set of all $(n+1)$-tuples $\gamma \in \mathbb{Z}^{n+1}$ satisfying $|\gamma| = |\alpha| + s$ and $(\alpha_i \leq \gamma_i$ for all $i \in [n])$ and $0 \leq \gamma_{n+1}$. Thus,

$$\sum_{\gamma \in W} \mathfrak{S}_\gamma = \sum_{\substack{\gamma \in \mathbb{Z}^{n+1}; \\ |\gamma| = |\alpha| + s; \\ \alpha_i \leq \gamma_i \text{ for all } i \in [n]; \\ 0 \leq \gamma_{n+1}}} \mathfrak{S}_\gamma. \tag{81}$$

Now, for each $\beta \in \mathbb{N}^{n+1}$ satisfying $|\beta| = s$, we have $\mu + \beta \in W$ [27]. Hence, we can define a map

$$\begin{aligned}f : \left\{\beta \in \mathbb{N}^{n+1} \ \mid\ |\beta| = s\right\} &\to W, \\ \beta &\mapsto \mu + \beta.\end{aligned}$$

[27] *Proof.* Let $\beta \in \mathbb{N}^{n+1}$ satisfy $|\beta| = s$. We must prove that $\mu + \beta \in W$.

We have $\mu + \beta \in \mathbb{Z}^{n+1}$ for obvious reasons. Moreover, Lemma 2.14 (applied to $n+1$ and μ instead of n and α) yields

$$|\mu + \beta| = \underbrace{|\mu|}_{=|\alpha|} + \underbrace{|\beta|}_{=s} = |\alpha| + s.$$

Furthermore, for each $i \in [n]$, we have

$$\begin{aligned}(\mu+\beta)_i &= \mu_i + \beta_i \geq \mu_i \qquad \left(\text{since } \beta_i \geq 0 \text{ (because } \beta_i \in \mathbb{N} \text{ (since } \beta \in \mathbb{N}^{n+1})\right)\right) \\ &= \alpha_i \qquad \text{(by (77))},\end{aligned}$$

Consider this map f.

On the other hand, for each $\delta \in W$, we have $\delta - \mu \in \{\beta \in \mathbb{N}^{n+1} \mid |\beta| = s\}$ [28]. Thus, we can define a map

$$g : W \to \left\{\beta \in \mathbb{N}^{n+1} \mid |\beta| = s\right\},$$
$$\delta \mapsto \delta - \mu.$$

Consider this map g.

---

so that $\alpha_i \leq (\mu+\beta)_i$. In other words, we have $\alpha_i \leq (\mu+\beta)_i$ for all $i \in [n]$. Finally, we have

$$(\mu+\beta)_{n+1} = \underbrace{\mu_{n+1}}_{=0} + \beta_{n+1} = \beta_{n+1} \geq 0 \qquad \left(\text{since } \beta_{n+1} \in \mathbb{N} \text{ (because } \beta \in \mathbb{N}^{n+1}\text{)}\right),$$

so that $0 \leq (\mu+\beta)_{n+1}$.

We now have proved that $\mu+\beta$ is an $(n+1)$-tuple in $\mathbb{Z}^{n+1}$ that satisfies $|\mu+\beta| = |\alpha| + s$ and $(\alpha_i \leq (\mu+\beta)_i$ for all $i \in [n])$ and $0 \leq (\mu+\beta)_{n+1}$. In other words, $\mu+\beta$ is an $(n+1)$-tuple $\gamma \in \mathbb{Z}^{n+1}$ satisfying $|\gamma| = |\alpha| + s$ and $(\alpha_i \leq \gamma_i$ for all $i \in [n])$ and $0 \leq \gamma_{n+1}$. In other words, $\mu+\beta \in W$ (since W is the set of all such $(n+1)$-tuples γ). Qed.

[28] *Proof.* Let $\delta \in W$. In other words, δ is an $(n+1)$-tuple $\gamma \in \mathbb{Z}^{n+1}$ satisfying $|\gamma| = |\alpha| + s$ and $(\alpha_i \leq \gamma_i$ for all $i \in [n])$ and $0 \leq \gamma_{n+1}$ (since W is the set of all such $(n+1)$-tuples γ). In other words, δ is an $(n+1)$-tuple in $\mathbb{Z}^{n+1}$ that satisfies $|\delta| = |\alpha| + s$ and $(\alpha_i \leq \delta_i$ for all $i \in [n])$ and $0 \leq \delta_{n+1}$. In particular, we therefore have

$$\alpha_i \leq \delta_i \qquad \text{for all } i \in [n]. \tag{82}$$

Now, $\delta = (\delta - \mu) + \mu$, so that

$$|\delta| = |(\delta - \mu) + \mu| = |\delta - \mu| + |\mu|$$

(by Lemma 2.14, applied to $n+1$, $\delta - \mu$ and μ instead of n, α and β). Thus,

$$|\delta - \mu| = \underbrace{|\delta|}_{=|\alpha|+s} - \underbrace{|\mu|}_{=|\alpha|} = |\alpha| + s - |\alpha| = s.$$

Let $i \in [n+1]$. We shall now prove that $(\delta-\mu)_i \geq 0$. Indeed, if $i = n+1$, then we have

$$(\delta-\mu)_i = (\delta-\mu)_{n+1} = \delta_{n+1} - \underbrace{\mu_{n+1}}_{=0} = \delta_{n+1} \geq 0 \qquad (\text{since } 0 \leq \delta_{n+1}).$$

Hence, $(\delta-\mu)_i \geq 0$ is proved in the case when $i = n+1$. Thus, for the rest of this proof of $(\delta-\mu)_i \geq 0$, we WLOG assume that $i \neq n+1$. Combining $i \in [n+1]$ with $i \neq n+1$, we obtain $i \in [n+1] \setminus \{n+1\} = [n]$. Hence,

$$(\delta-\mu)_i = \delta_i - \underbrace{\mu_i}_{\substack{=\alpha_i \\ \text{(by (77))}}} = \delta_i - \alpha_i \geq 0$$

(by (82)). Thus, the proof of $(\delta-\mu)_i \geq 0$ is complete.

Now, from $(\delta-\mu)_i \geq 0$, we obtain $(\delta-\mu)_i \in \mathbb{N}$ (since $(\delta-\mu)_i$ is an integer).

Forget that we fixed i. We thus have shown that $(\delta-\mu)_i \in \mathbb{N}$ for each $i \in [n+1]$. In other words, $\delta - \mu \in \mathbb{N}^{n+1}$.

We now know that $\delta - \mu \in \mathbb{N}^{n+1}$ and $|\delta - \mu| = s$. In other words, $\delta - \mu$ is a $\beta \in \mathbb{N}^{n+1}$ satisfying $|\beta| = s$. In other words, $\delta - \mu \in \left\{\beta \in \mathbb{N}^{n+1} \mid |\beta| = s\right\}$. Qed.

We have $f \circ g = \mathrm{id}$ [29] and $g \circ f = \mathrm{id}$ [30]. Thus, the maps f and g are mutually inverse. Hence, the map f is invertible, i.e., is a bijection. In other words, the map

$$\left\{\beta \in \mathbb{N}^{n+1} \mid |\beta| = s\right\} \to W,$$
$$\beta \mapsto \mu + \beta$$

is a bijection (since the map f is precisely this map). Hence, we can substitute $\mu + \beta$ for γ in the sum $\sum_{\gamma \in W} \mathfrak{S}_\gamma$. Thus, we obtain

$$\sum_{\gamma \in W} \mathfrak{S}_\gamma = \sum_{\substack{\beta \in \mathbb{N}^{n+1}; \\ |\beta| = s}} \mathfrak{S}_{\mu+\beta}.$$

Comparing this with (80), we obtain

$$\mathfrak{S}_\alpha \cdot H_s = \sum_{\gamma \in W} \mathfrak{S}_\gamma = \sum_{\substack{\gamma \in \mathbb{Z}^{n+1}; \\ |\gamma| = |\alpha| + s; \\ \alpha_i \leq \gamma_i \text{ for all } i \in [n]; \\ 0 \leq \gamma_{n+1}}} \mathfrak{S}_\gamma \qquad \text{(by (81))}$$
$$= \sum_{\substack{\beta \in \mathbb{Z}^{n+1}; \\ |\beta| = |\alpha| + s; \\ \alpha_i \leq \beta_i \text{ for all } i \in [n]; \\ 0 \leq \beta_{n+1}}} \mathfrak{S}_\beta$$

(here, we have renamed the summation index γ as β). This proves Proposition 3.8. □

4. The second pre-Pieri rule

The "second pre-Pieri rule" structurally resembles the first, but involves a sum over (a subset of) $\{0,1\}^n$ instead of $\mathbb{N}^n$. This is, of course, analogous to the relationship between the elementary symmetric functions and the complete homogeneous symmetric functions, or the relationship between sets and multisets, or various other "combinatorial reciprocities".

[29] *Proof.* Let $\delta \in W$. Then, the definition of g yields $g(\delta) = \delta - \mu$. However,

$$(f \circ g)(\delta) = f(g(\delta)) = \mu + \underbrace{g(\delta)}_{=\delta-\mu} \qquad \text{(by the definition of } f\text{)}$$
$$= \mu + (\delta - \mu) = \delta = \mathrm{id}(\delta).$$

Forget that we fixed δ. We thus have shown that $(f \circ g)(\delta) = \mathrm{id}(\delta)$ for each $\delta \in W$. In other words, $f \circ g = \mathrm{id}$.

[30] *Proof.* Let $\gamma \in \left\{\beta \in \mathbb{N}^{n+1} \mid |\beta| = s\right\}$. Then, the definition of f yields $f(\gamma) = \mu + \gamma$. However,

$$(g \circ f)(\gamma) = g(f(\gamma)) = \underbrace{f(\gamma)}_{=\mu+\gamma} - \mu \qquad \text{(by the definition of } g\text{)}$$
$$= (\mu + \gamma) - \mu = \gamma = \mathrm{id}(\gamma).$$

Forget that we fixed γ. We thus have shown that $(g \circ f)(\gamma) = \mathrm{id}(\gamma)$ for each $\gamma \in \left\{\beta \in \mathbb{N}^{n+1} \mid |\beta| = s\right\}$. In other words, $g \circ f = \mathrm{id}$.

4.1. The theorem

Before we state the second pre-Pieri rule, we observe that $\{0,1\}^n \subseteq \mathbb{Z}^n$ for each $n \in \mathbb{N}$.

Theorem 4.1 (second pre-Pieri rule)**.** Let $n \in \mathbb{N}$ and $p \in \{0,1,\ldots,n\}$. Let $h_{k,\ i}$ be an element of R for all $k \in \mathbb{Z}$ and $i \in [n]$.

For any $\alpha \in \mathbb{Z}^n$, we define

$$t_\alpha := \operatorname{rowdet}\left(\left(h_{\alpha_i+j,\ i}\right)_{i,j\in[n]}\right) \in R.$$

Let ξ be the n-tuple

$$\begin{aligned}&(1,2,\ldots,n) + \left(\underbrace{0,0,\ldots,0}_{n-p \text{ zeroes}}, \underbrace{1,1,\ldots,1}_{p \text{ ones}}\right)\\ &= (1,2,\ldots,n-p,n-p+2,n-p+3,\ldots,n+1) \in \mathbb{Z}^n.\end{aligned}$$

Let $\alpha \in \mathbb{Z}^n$. Then,

$$\sum_{\substack{\beta\in\{0,1\}^n;\\ |\beta|=p}} t_{\alpha+\beta} = \operatorname{rowdet}\left(\left(h_{\alpha_i+\xi_j,\ i}\right)_{i,j\in[n]}\right). \tag{83}$$

Example 4.2. For this example, set $n = 3$ and $p = 2$, and let $\alpha \in \mathbb{Z}^3$ be arbitrary. Fix arbitrary elements $h_{k,\ i} \in R$ for all $k \in \mathbb{Z}$ and $i \in [n]$. Then, the n-tuple ξ defined in Theorem 4.1 is $(1,2,3)+(0,1,1) = (1,3,4)$. Hence, (83) says that

$$\sum_{\substack{\beta\in\{0,1\}^3;\\ |\beta|=2}} t_{\alpha+\beta} = \operatorname{rowdet}\left(\left(h_{\alpha_i+\xi_j,\ i}\right)_{i,j\in[3]}\right) = \operatorname{rowdet}\begin{pmatrix} h_{\alpha_1+1,\ 1} & h_{\alpha_1+3,\ 1} & h_{\alpha_1+4,\ 1} \\ h_{\alpha_2+1,\ 2} & h_{\alpha_2+3,\ 2} & h_{\alpha_2+4,\ 2} \\ h_{\alpha_3+1,\ 3} & h_{\alpha_3+3,\ 3} & h_{\alpha_3+4,\ 3} \end{pmatrix}.$$

The left hand side of this equality can be rewritten as

$$\sum_{\substack{\beta\in\{0,1\}^3;\\ |\beta|=2}} \underbrace{t_{\alpha+\beta}}_{\substack{=\operatorname{rowdet}\left(\left(h_{(\alpha+\beta)_i+j,\ i}\right)_{i,j\in[3]}\right)\\ \text{(by the definition of } t_{\alpha+\beta})}}$$

$$= \sum_{\substack{\beta\in\{0,1\}^3;\\ |\beta|=2}} \operatorname{rowdet}\left(\left(h_{(\alpha+\beta)_i+j,\ i}\right)_{i,j\in[3]}\right)$$

$$= \sum_{\substack{\beta\in\{0,1\}^3;\\ |\beta|=2}} \operatorname{rowdet}\left(\left(h_{\alpha_i+\beta_i+j,\ i}\right)_{i,j\in[3]}\right)$$

$$\left(\text{since } (\alpha+\beta)_i = \alpha_i+\beta_i \text{ for all } i\in[3]\right)$$

$$= \sum_{\substack{\beta\in\{0,1\}^3;\\ |\beta|=2}} \operatorname{rowdet}\begin{pmatrix} h_{\alpha_1+\beta_1+1,\ 1} & h_{\alpha_1+\beta_1+2,\ 1} & h_{\alpha_1+\beta_1+3,\ 1} \\ h_{\alpha_2+\beta_2+1,\ 2} & h_{\alpha_2+\beta_2+2,\ 2} & h_{\alpha_2+\beta_2+3,\ 2} \\ h_{\alpha_3+\beta_3+1,\ 3} & h_{\alpha_3+\beta_3+2,\ 3} & h_{\alpha_3+\beta_3+3,\ 3} \end{pmatrix}$$

$$= \operatorname{rowdet}\begin{pmatrix} h_{\alpha_1+1+1,\ 1} & h_{\alpha_1+1+2,\ 1} & h_{\alpha_1+1+3,\ 1} \\ h_{\alpha_2+1+1,\ 2} & h_{\alpha_2+1+2,\ 2} & h_{\alpha_2+1+3,\ 2} \\ h_{\alpha_3+0+1,\ 3} & h_{\alpha_3+0+2,\ 3} & h_{\alpha_3+0+3,\ 3} \end{pmatrix}$$

$$+ \operatorname{rowdet}\begin{pmatrix} h_{\alpha_1+1+1,\ 1} & h_{\alpha_1+1+2,\ 1} & h_{\alpha_1+1+3,\ 1} \\ h_{\alpha_2+0+1,\ 2} & h_{\alpha_2+0+2,\ 2} & h_{\alpha_2+0+3,\ 2} \\ h_{\alpha_3+1+1,\ 3} & h_{\alpha_3+1+2,\ 3} & h_{\alpha_3+1+3,\ 3} \end{pmatrix}$$

$$+ \operatorname{rowdet}\begin{pmatrix} h_{\alpha_1+0+1,\ 1} & h_{\alpha_1+0+2,\ 1} & h_{\alpha_1+0+3,\ 1} \\ h_{\alpha_2+1+1,\ 2} & h_{\alpha_2+1+2,\ 2} & h_{\alpha_2+1+3,\ 2} \\ h_{\alpha_3+1+1,\ 3} & h_{\alpha_3+1+2,\ 3} & h_{\alpha_3+1+3,\ 3} \end{pmatrix}$$

$$\left(\begin{array}{c}\text{since there are exactly three 2-tuples } \beta\in\{0,1\}^3\\ \text{satisfying } |\beta|=2, \text{ namely } (1,1,0),\ (1,0,1) \text{ and } (0,1,1)\end{array}\right)$$

$$= \operatorname{rowdet}\begin{pmatrix} h_{\alpha_1+2,\ 1} & h_{\alpha_1+3,\ 1} & h_{\alpha_1+4,\ 1} \\ h_{\alpha_2+2,\ 2} & h_{\alpha_2+3,\ 2} & h_{\alpha_2+4,\ 2} \\ h_{\alpha_3+1,\ 3} & h_{\alpha_3+2,\ 3} & h_{\alpha_3+3,\ 3} \end{pmatrix}$$

$$+ \operatorname{rowdet}\begin{pmatrix} h_{\alpha_1+2,\ 1} & h_{\alpha_1+3,\ 1} & h_{\alpha_1+4,\ 1} \\ h_{\alpha_2+1,\ 2} & h_{\alpha_2+2,\ 2} & h_{\alpha_2+3,\ 2} \\ h_{\alpha_3+2,\ 3} & h_{\alpha_3+3,\ 3} & h_{\alpha_3+4,\ 3} \end{pmatrix}$$

$$+ \operatorname{rowdet}\begin{pmatrix} h_{\alpha_1+1,\ 1} & h_{\alpha_1+2,\ 1} & h_{\alpha_1+3,\ 1} \\ h_{\alpha_2+2,\ 2} & h_{\alpha_2+3,\ 2} & h_{\alpha_2+4,\ 2} \\ h_{\alpha_3+2,\ 3} & h_{\alpha_3+3,\ 3} & h_{\alpha_3+4,\ 3} \end{pmatrix}.$$

Therefore, (83) rewrites as

$$\begin{aligned}
&\operatorname{rowdet}\begin{pmatrix} h_{\alpha_1+2,\ 1} & h_{\alpha_1+3,\ 1} & h_{\alpha_1+4,\ 1} \\ h_{\alpha_2+2,\ 2} & h_{\alpha_2+3,\ 2} & h_{\alpha_2+4,\ 2} \\ h_{\alpha_3+1,\ 3} & h_{\alpha_3+2,\ 3} & h_{\alpha_3+3,\ 3} \end{pmatrix} \\
&\quad + \operatorname{rowdet}\begin{pmatrix} h_{\alpha_1+2,\ 1} & h_{\alpha_1+3,\ 1} & h_{\alpha_1+4,\ 1} \\ h_{\alpha_2+1,\ 2} & h_{\alpha_2+2,\ 2} & h_{\alpha_2+3,\ 2} \\ h_{\alpha_3+2,\ 3} & h_{\alpha_3+3,\ 3} & h_{\alpha_3+4,\ 3} \end{pmatrix} \\
&\quad + \operatorname{rowdet}\begin{pmatrix} h_{\alpha_1+1,\ 1} & h_{\alpha_1+2,\ 1} & h_{\alpha_1+3,\ 1} \\ h_{\alpha_2+2,\ 2} & h_{\alpha_2+3,\ 2} & h_{\alpha_2+4,\ 2} \\ h_{\alpha_3+2,\ 3} & h_{\alpha_3+3,\ 3} & h_{\alpha_3+4,\ 3} \end{pmatrix} \\
&= \operatorname{rowdet}\begin{pmatrix} h_{\alpha_1+1,\ 1} & h_{\alpha_1+3,\ 1} & h_{\alpha_1+4,\ 1} \\ h_{\alpha_2+1,\ 2} & h_{\alpha_2+3,\ 2} & h_{\alpha_2+4,\ 2} \\ h_{\alpha_3+1,\ 2} & h_{\alpha_3+3,\ 2} & h_{\alpha_3+4,\ 2} \end{pmatrix}.
\end{aligned}$$

This is easy to check directly by expanding all four row-determinants.

4.2. The proof

Our proof of the second pre-Pieri rule will be similar to that of the first.

Again, we will use Definition 2.4 and Definition 2.7. Again, several lemmas will be used. The first is an analogue of Lemma 2.10:

Lemma 4.3. Let $n \in \mathbb{N}$. Let $q \in \mathbb{Z}$ and $\nu \in \mathbb{Z}^n$ and $\xi \in \mathbb{Z}^n$ satisfy $|\nu| = |\xi|$ and

$$\{\nu_1, \nu_2, \ldots, \nu_n\} \subseteq \{\xi_1, \xi_2, \ldots, \xi_n\} \cup \{q\} \qquad \text{and} \qquad |\{\nu_1, \nu_2, \ldots, \nu_n\}| = n.$$

Then, there exists some permutation $\pi \in S_n$ satisfying $\nu = \xi \circ \pi$.

Proof of Lemma 4.3. The definition of $[n]$ yields $[n] = \{1, 2, \ldots, n\}$. The definition of $[n+1]$ yields $[n+1] = \{1, 2, \ldots, n+1\}$. Hence,

$$[n+1] \setminus \{n+1\} = \{1, 2, \ldots, n+1\} \setminus \{n+1\} = \{1, 2, \ldots, n\} = [n].$$

Thus, $[n] = [n+1] \setminus \{n+1\} \subseteq [n+1]$.

The definition of $|\xi|$ yields

$$|\xi| = \xi_1 + \xi_2 + \cdots + \xi_n = \sum_{i=1}^{n} \xi_i. \tag{84}$$

The definition of $|\nu|$ yields

$$|\nu| = \nu_1 + \nu_2 + \cdots + \nu_n = \sum_{i=1}^{n} \nu_i. \tag{85}$$

We extend the n-tuple $\xi = (\xi_1, \xi_2, \ldots, \xi_n)$ to an $(n+1)$-tuple $(\xi_1, \xi_2, \ldots, \xi_{n+1}) \in \mathbb{Z}^{n+1}$ by setting $\xi_{n+1} := q$. Then,

$$\begin{aligned}\{\xi_1, \xi_2, \ldots, \xi_{n+1}\} &= \{\xi_1, \xi_2, \ldots, \xi_n, \xi_{n+1}\} = \{\xi_1, \xi_2, \ldots, \xi_n\} \cup \left\{\underbrace{\xi_{n+1}}_{=q}\right\} \\ &= \{\xi_1, \xi_2, \ldots, \xi_n\} \cup \{q\}. \end{aligned} \tag{86}$$

Now,

$$\begin{aligned}\{\nu_1, \nu_2, \ldots, \nu_n\} &\subseteq \{\xi_1, \xi_2, \ldots, \xi_n\} \cup \{q\} \\ &= \{\xi_1, \xi_2, \ldots, \xi_{n+1}\} \qquad \text{(by (86))}. \end{aligned} \tag{87}$$

Now, recall that $n \in \mathbb{N}$ and $|\{\nu_1, \nu_2, \ldots, \nu_n\}| = n$. Hence, Lemma 2.9 (applied to $m = n$ and $u_i = \nu_i$) shows that the n objects $\nu_1, \nu_2, \ldots, \nu_n$ are distinct. In other words, if i and j are two distinct elements of $[n]$, then

$$\nu_i \neq \nu_j. \tag{88}$$

We define a map $f : [n] \to [n+1]$ as follows: For each $i \in [n]$, we choose some $j \in [n+1]$ satisfying $\nu_i = \xi_j$ (indeed, such a j exists, because

$$\begin{aligned}\nu_i &\in \{\nu_1, \nu_2, \ldots, \nu_n\} \qquad (\text{since } i \in [n] = \{1, 2, \ldots, n\}) \\ &\subseteq \{\xi_1, \xi_2, \ldots, \xi_{n+1}\} \qquad (\text{by (87)}) \\ &= \left\{\xi_j \mid j \in \underbrace{\{1, 2, \ldots, n+1\}}_{=[n+1]}\right\} = \{\xi_j \mid j \in [n+1]\}\end{aligned}$$

), and we define $f(i)$ to be this j. Thus, we have defined a map $f : [n] \to [n+1]$.

For each $i \in [n]$, the image $f(i)$ of i is an element of $[n+1]$ and satisfies

$$\nu_i = \xi_{f(i)} \tag{89}$$

(since $f(i)$ is defined to be a $j \in [n+1]$ satisfying $\nu_i = \xi_j$).

The n elements $f(1), f(2), \ldots, f(n)$ are distinct[31]. Hence,

$$|\{f(1), f(2), \ldots, f(n)\}| = n.$$

[31] *Proof.* Let i and j be two distinct elements of $[n]$. We shall show that $f(i) \neq f(j)$.

Indeed, assume the contrary. Thus, $f(i) = f(j)$. However, (89) yields $\nu_i = \xi_{f(i)} = \xi_{f(j)}$ (since $f(i) = f(j)$). On the other hand, (89) (applied to j instead of i) yields $\nu_j = \xi_{f(j)}$. Comparing these two equalities, we obtain $\nu_i = \nu_j$. However, (88) yields $\nu_i \neq \nu_j$. This contradicts $\nu_i = \nu_j$. This contradiction shows that our assumption was false. Hence, $f(i) \neq f(j)$.

Forget that we fixed i and j. We thus have shown that $f(i) \neq f(j)$ whenever i and j are two distinct elements of $[n]$. In other words, the n elements $f(1), f(2), \ldots, f(n)$ are distinct.

Clearly, $\{f(1), f(2), \ldots, f(n)\}$ is a subset of $[n+1]$ (since f is a map from $[n]$ to $[n+1]$). Thus,

$$\begin{aligned}\left|[n+1] \setminus \{f(1), f(2), \ldots, f(n)\}\right| &= \underbrace{\left|[n+1]\right|}_{\substack{=n+1 \\ \text{(since } [n+1]=\{1,2,\ldots,n+1\}\text{)}}} - \underbrace{\left|\{f(1), f(2), \ldots, f(n)\}\right|}_{=n} \\ &= (n+1) - n = 1.\end{aligned}$$

In other words, $[n+1] \setminus \{f(1), f(2), \ldots, f(n)\}$ is a 1-element set. In other words,

$$[n+1] \setminus \{f(1), f(2), \ldots, f(n)\} = \{p\} \qquad \text{for some element } p.$$

Consider this p. We have $p \in \{p\} = [n+1] \setminus \{f(1), f(2), \ldots, f(n)\}$. In other words,

$$p \in [n+1] \qquad \text{and} \qquad p \notin \{f(1), f(2), \ldots, f(n)\}.$$

As we have seen, the n elements $f(1), f(2), \ldots, f(n)$ are distinct. Hence, the n-tuple $(f(1), f(2), \ldots, f(n))$ is a list of all n elements of the set $\{f(1), f(2), \ldots, f(n)\}$ with no repetition. Therefore,

$$\begin{aligned}\sum_{i \in \{f(1), f(2), \ldots, f(n)\}} \xi_i &= \xi_{f(1)} + \xi_{f(2)} + \cdots + \xi_{f(n)} = \sum_{i=1}^{n} \underbrace{\xi_{f(i)}}_{\substack{=\nu_i \\ \text{(by (89))}}} = \sum_{i=1}^{n} \nu_i \\ &= |\nu| \qquad \text{(by (85))}. \end{aligned} \tag{90}$$

However,

$$\begin{aligned}\sum_{i \in [n+1]} \xi_i &= \underbrace{\sum_{\substack{i \in [n+1]; \\ i \in \{f(1), f(2), \ldots, f(n)\}}} \xi_i}_{\substack{= \sum\limits_{i \in \{f(1), f(2), \ldots, f(n)\}} \\ \text{(since } \{f(1), f(2), \ldots, f(n)\} \\ \text{is a subset of } [n+1]\text{)}}} + \underbrace{\sum_{\substack{i \in [n+1]; \\ i \notin \{f(1), f(2), \ldots, f(n)\}}} \xi_i}_{\substack{= \sum\limits_{i \in [n+1] \setminus \{f(1), f(2), \ldots, f(n)\}} \\ = \sum\limits_{i \in \{p\}} \\ \text{(since } [n+1] \setminus \{f(1), f(2), \ldots, f(n)\} = \{p\}\text{)}}} \\ &\qquad \left(\begin{array}{c} \text{since each } i \in [n+1] \text{ satisfies} \\ \text{either } i \in \{f(1), f(2), \ldots, f(n)\} \\ \text{or } i \notin \{f(1), f(2), \ldots, f(n)\} \text{ (but not both)} \end{array} \right) \\ &= \underbrace{\sum_{i \in \{f(1), f(2), \ldots, f(n)\}} \xi_i}_{\substack{=|\nu| \\ \text{(by (90))}}} + \underbrace{\sum_{i \in \{p\}} \xi_i}_{=\xi_p} = \underbrace{|\nu|}_{=|\xi|} + \xi_p = |\xi| + \xi_p.\end{aligned}$$

Therefore,

$$\begin{aligned}|\xi| + \xi_p &= \underbrace{\sum_{i \in [n+1]}}_{= \sum_{i=1}^{n+1}} \xi_i = \sum_{i=1}^{n+1} \xi_i = \xi_1 + \xi_2 + \cdots + \xi_{n+1} \\ &= \underbrace{\left(\xi_1 + \xi_2 + \cdots + \xi_n\right)}_{\substack{= \sum_{i=1}^{n} \xi_i = |\xi| \\ \text{(by (84))}}} + \xi_{n+1} = |\xi| + \xi_{n+1}.\end{aligned}$$

Subtracting $|\xi|$ from both sides of this equality, we obtain

$$\xi_p = \xi_{n+1}. \tag{91}$$

Using this equality, it is now easy to see that

$$\{\nu_1, \nu_2, \ldots, \nu_n\} \subseteq \{\xi_1, \xi_2, \ldots, \xi_n\}. \tag{92}$$

[*Proof of (92):* Let $i \in \{1, 2, \ldots, n\}$. We shall show that $\nu_i \in \{\xi_1, \xi_2, \ldots, \xi_n\}$.

We have $i \in \{1, 2, \ldots, n\} = [n]$, so that $\nu_i = \xi_{f(i)}$ (by (89)). We are in one of the following two cases:

Case 1: We have $f(i) = n + 1$.

Case 2: We have $f(i) \neq n + 1$.

Let us first consider Case 1. In this case, we have $f(i) = n + 1$. Hence, $\xi_{f(i)} = \xi_{n+1}$. However, from $f(i) = n + 1$, we obtain $n + 1 = f(i) \in \{f(1), f(2), \ldots, f(n)\}$ (since $i \in \{1, 2, \ldots, n\}$). Thus, we cannot have $p = n + 1$ (because if we had $p = n + 1$, then we would have $p = n + 1 \in \{f(1), f(2), \ldots, f(n)\}$, which would contradict $p \notin \{f(1), f(2), \ldots, f(n)\}$). In other words, we have $p \neq n + 1$. Combining $p \in [n + 1]$ with $p \neq n + 1$, we obtain $p \in [n + 1] \setminus \{n + 1\} = [n] = \{1, 2, \ldots, n\}$. Hence, $\xi_p \in \{\xi_1, \xi_2, \ldots, \xi_n\}$. Now,

$$\begin{aligned}\nu_i = \xi_{f(i)} = \xi_{n+1} = \xi_p \qquad &\text{(by (91))} \\ \in \{\xi_1, \xi_2, \ldots, \xi_n\}.&\end{aligned}$$

Thus, $\nu_i \in \{\xi_1, \xi_2, \ldots, \xi_n\}$ is proved in Case 1.

Next, let us consider Case 2. In this case, we have $f(i) \neq n + 1$. Combining $f(i) \in [n + 1]$ with $f(i) \neq n + 1$, we obtain $f(i) \in [n + 1] \setminus \{n + 1\} = [n] = \{1, 2, \ldots, n\}$. Hence, $\xi_{f(i)} \in \{\xi_1, \xi_2, \ldots, \xi_n\}$. Now,

$$\nu_i = \xi_{f(i)} \in \{\xi_1, \xi_2, \ldots, \xi_n\}.$$

Thus, $\nu_i \in \{\xi_1, \xi_2, \ldots, \xi_n\}$ is proved in Case 2.

We have now proved $\nu_i \in \{\xi_1, \xi_2, \ldots, \xi_n\}$ in both Cases 1 and 2. Hence, $\nu_i \in \{\xi_1, \xi_2, \ldots, \xi_n\}$ always holds.

Now, forget that we fixed i. We thus have shown that $\nu_i \in \{\xi_1, \xi_2, \ldots, \xi_n\}$ for each $i \in \{1, 2, \ldots, n\}$. In other words, all n numbers $\nu_1, \nu_2, \ldots, \nu_n$ are elements of $\{\xi_1, \xi_2, \ldots, \xi_n\}$. In other words, $\{\nu_1, \nu_2, \ldots, \nu_n\} \subseteq \{\xi_1, \xi_2, \ldots, \xi_n\}$. This proves (92).]

Now, we define a map $g : [n] \to [n]$ as follows: For each $i \in [n]$, we choose some $j \in [n]$ satisfying $\nu_i = \xi_j$ (indeed, such a j exists, because

$$\begin{aligned} \nu_i &\in \{\nu_1, \nu_2, \ldots, \nu_n\} && (\text{since } i \in [n] = \{1, 2, \ldots, n\}) \\ &\subseteq \{\xi_1, \xi_2, \ldots, \xi_n\} && (\text{by (92)}) \\ &= \left\{ \xi_j \mid j \in \underbrace{\{1, 2, \ldots, n\}}_{=[n]} \right\} = \{\xi_j \mid j \in [n]\} \end{aligned}$$

), and we define $g(i)$ to be this j. Thus, we have defined a map $g : [n] \to [n]$.

For each $i \in [n]$, the image $g(i)$ of i is an element of $[n]$ and satisfies

$$\nu_i = \xi_{g(i)} \tag{93}$$

(since $g(i)$ is defined to be a $j \in [n]$ satisfying $\nu_i = \xi_j$).

The map g is injective[32]. However, a well-known fact from set theory (actually one of the versions of the pigeonhole principle) says that any injective map ϕ from a finite set X to X must be a permutation of X. Applying this to $X = [n]$ and $\phi = g$, we conclude that g must be a permutation of $[n]$ (since g is an injective map from the finite set $[n]$ to $[n]$). In other words, $g \in S_n$ (since S_n is the set of all permutations of $[n]$).

From (93), we know that $\nu_i = \xi_{g(i)}$ for each $i \in [n]$. In other words,

$$(\nu_1, \nu_2, \ldots, \nu_n) = \left(\xi_{g(1)}, \xi_{g(2)}, \ldots, \xi_{g(n)}\right).$$

However, Definition 2.4 yields

$$\xi \circ g = \left(\xi_{g(1)}, \xi_{g(2)}, \ldots, \xi_{g(n)}\right).$$

Comparing these two equalities, we obtain

$$\xi \circ g = (\nu_1, \nu_2, \ldots, \nu_n) = \nu.$$

In other words, $\nu = \xi \circ g$. Hence, there exists some permutation $\pi \in S_n$ satisfying $\nu = \xi \circ \pi$ (namely, $\pi = g$). This proves Lemma 4.3. □

We shall furthermore use the following notation:

[32] *Proof.* Let i and j be two distinct elements of $[n]$. We shall show that $g(i) \neq g(j)$.

Indeed, assume the contrary. Thus, $g(i) = g(j)$. However, (93) yields $\nu_i = \xi_{g(i)} = \xi_{g(j)}$ (since $g(i) = g(j)$). On the other hand, (93) (applied to j instead of i) yields $\nu_j = \xi_{g(j)}$. Comparing these two equalities, we obtain $\nu_i = \nu_j$. However, (88) yields $\nu_i \neq \nu_j$. This contradicts $\nu_i = \nu_j$. This contradiction shows that our assumption was false. Hence, $g(i) \neq g(j)$.

Forget that we fixed i and j. We thus have shown that $g(i) \neq g(j)$ whenever i and j are two distinct elements of $[n]$. In other words, the map g is injective.

Definition 4.4. Let u and v be two integers. We write "$u \trianglerighteq v$" if and only if $u - v \in \{0, 1\}$.

Thus, for example, $2 \trianglerighteq 2$ and $2 \trianglerighteq 1$, but we don't have $2 \trianglerighteq 0$.
We need the following simple lemma:

Lemma 4.5. Let $n \in \mathbb{N}$ and $p \in \{0, 1, \ldots, n\}$. Let ξ be the n-tuple

$$(1, 2, \ldots, n) + \left(\underbrace{0, 0, \ldots, 0}_{n-p \text{ zeroes}}, \underbrace{1, 1, \ldots, 1}_{p \text{ ones}} \right)$$
$$= (1, 2, \ldots, n-p, n-p+2, n-p+3, \ldots, n+1) \in \mathbb{Z}^n.$$

Let $\sigma \in S_n$ be a permutation such that $\sigma \neq \mathrm{id}$ (where id denotes the identity map $[n] \to [n]$). Then,

$$\prod_{i=1}^{n} [\xi_i \trianglerighteq \sigma(i)] = 0.$$

Proof of Lemma 4.5. The definition of ξ yields

$$\xi = (1, 2, \ldots, n) + \left(\underbrace{0, 0, \ldots, 0}_{n-p \text{ zeroes}}, \underbrace{1, 1, \ldots, 1}_{p \text{ ones}} \right)$$
$$= (1, 2, \ldots, n-p, n-p+2, n-p+3, \ldots, n+1).$$

Thus,

$$(\xi_1, \xi_2, \ldots, \xi_n)$$
$$= \xi = (1, 2, \ldots, n-p, n-p+2, n-p+3, \ldots, n+1). \tag{94}$$

In other words, we have

$$\xi_i = i \qquad \text{for each } i \in \{1, 2, \ldots, n-p\} \tag{95}$$

and

$$\xi_i = i + 1 \qquad \text{for each } i \in \{n-p+1, n-p+2, \ldots, n\}. \tag{96}$$

We define a subset Z of $[n]$ by

$$Z = \{i \in [n] \mid \sigma(i) \neq i\}.$$

This set Z is nonempty[33] and finite[34]. Hence, this set Z has a smallest element and a largest element. Let us denote these two elements by a and b, respectively. Thus, a is

[33] *Proof.* Assume the contrary. Thus, Z is empty. In other words, $Z = \varnothing$. In other words, $\{i \in [n] \mid \sigma(i) \neq i\} = \varnothing$ (since $Z = \{i \in [n] \mid \sigma(i) \neq i\}$). In other words, there exists no $i \in [n]$ satisfying $\sigma(i) \neq i$. In other words, each $i \in [n]$ satisfies $\sigma(i) = i$. Hence, each $i \in [n]$ satisfies $\sigma(i) = i = \mathrm{id}(i)$. In other words, we have $\sigma = \mathrm{id}$ (since both σ and id are maps from $[n]$ to $[n]$). This contradicts $\sigma \neq \mathrm{id}$. This contradiction shows that our assumption was false, qed.

[34] since it is a subset of the finite set $[n]$ (by its definition)

the smallest element of Z. In other words, $a \in Z$ and

$$a \leq z \qquad \text{for each } z \in Z. \tag{97}$$

Furthermore, b is the largest element of Z. In other words, $b \in Z$ and

$$b \geq z \qquad \text{for each } z \in Z. \tag{98}$$

Applying (98) to $z = a$, we obtain $b \geq a$ (since $a \in Z$). Now, we are in one of the following two cases:

Case 1: We have $a \leq n - p$.

Case 2: We have $a > n - p$.

Let us first consider Case 1. In this case, we have $a \leq n - p$. Also, $a \in Z \subseteq [n] = \{1, 2, \ldots, n\}$ (by the definition of $[n]$), so that $a \geq 1$. Combining this with $a \leq n - p$, we obtain $a \in \{1, 2, \ldots, n - p\}$. Hence, (95) (applied to $i = a$) yields $\xi_a = a$. However, it is easy to see that $\sigma(a) > a$ [35]. Hence, $\xi_a = a < \sigma(a)$ (since $\sigma(a) > a$), so that $\xi_a - \sigma(a) < 0$. Therefore, we do not have $\xi_a - \sigma(a) \in \{0, 1\}$ (because if we had $\xi_a - \sigma(a) \in \{0, 1\}$, then we would have $\xi_a - \sigma(a) \in \{0, 1\} \subseteq \mathbb{N}$ and therefore $\xi_a - \sigma(a) \geq 0$, which would contradict $\xi_a - \sigma(a) < 0$). However, the statement "$\xi_a \trianglerighteq \sigma(a)$" is equivalent to "$\xi_a - \sigma(a) \in \{0, 1\}$" (by Definition 4.4). Hence, we do not have $\xi_a \trianglerighteq \sigma(a)$ (since we do not have $\xi_a - \sigma(a) \in \{0, 1\}$). In other words, we have $[\xi_a \trianglerighteq \sigma(a)] = 0$.

Now, recall that $a \in \{1, 2, \ldots, n\}$. Thus, the term $[\xi_a \trianglerighteq \sigma(a)]$ is one of the factors of the product $\prod_{i=1}^{n} [\xi_i \trianglerighteq \sigma(i)]$ (namely, the factor for $i = a$). Since this term is 0 (because we have just shown that $[\xi_a \trianglerighteq \sigma(a)] = 0$), we thus conclude that one of the factors of the product $\prod_{i=1}^{n} [\xi_i \trianglerighteq \sigma(i)]$ is 0. Therefore, the entire product is 0. In other words, we have $\prod_{i=1}^{n} [\xi_i \trianglerighteq \sigma(i)] = 0$. Thus, Lemma 4.5 is proved in Case 1.

Let us next consider Case 2. In this case, we have $a > n - p$. Thus, $b \geq a > n - p$, so that $b \geq n - p + 1$ (since b and $n - p$ are integers). Also, $b \in Z \subseteq [n] = \{1, 2, \ldots, n\}$ (by the definition of $[n]$), so that $b \leq n$. Combining this with $b \geq n - p + 1$, we obtain $b \in \{n - p + 1, n - p + 2, \ldots, n\}$. Hence, (96) (applied to $i = b$) yields $\xi_b = b + 1$. However, it is easy to see that $\sigma(b) < b$ [36]. Thus, $\underbrace{\xi_b}_{=b+1} - \underbrace{\sigma(b)}_{<b} > (b + 1) - b = 1$.

[35] *Proof.* We have $a \in Z = \{i \in [n] \mid \sigma(i) \neq i\}$. In other words, a is an $i \in [n]$ satisfying $\sigma(i) \neq i$. In other words, a is an element of $[n]$ and satisfies $\sigma(a) \neq a$.

The map σ belongs to S_n, and thus is a permutation of $[n]$ (since S_n is the set of all permutations of $[n]$). Hence, this map is bijective, therefore injective. In other words, if i and j are two elements of $[n]$ satisfying $i \neq j$, then $\sigma(i) \neq \sigma(j)$. Applying this to $i = \sigma(a)$ and $j = a$, we obtain $\sigma(\sigma(a)) \neq \sigma(a)$ (since $\sigma(a) \neq a$). Thus, $\sigma(a)$ is an $i \in [n]$ satisfying $\sigma(i) \neq i$. In other words, $\sigma(a) \in \{i \in [n] \mid \sigma(i) \neq i\}$. In other words, $\sigma(a) \in Z$ (since $Z = \{i \in [n] \mid \sigma(i) \neq i\}$). Hence, (97) (applied to $z = \sigma(a)$) yields $a \leq \sigma(a)$. In other words, $\sigma(a) \geq a$. Combining this with $\sigma(a) \neq a$, we obtain $\sigma(a) > a$.

[36] *Proof.* We have $b \in Z = \{i \in [n] \mid \sigma(i) \neq i\}$. In other words, b is an $i \in [n]$ satisfying $\sigma(i) \neq i$. In other words, b is an element of $[n]$ and satisfies $\sigma(b) \neq b$.

Therefore, we do not have $\xi_b - \sigma(b) \in \{0,1\}$ (because if we had $\xi_b - \sigma(b) \in \{0,1\}$, then we would have $\xi_b - \sigma(b) \leq 1$ (because any element of the set $\{0,1\}$ is ≤ 1), which would contradict $\xi_b - \sigma(b) > 1$). However, the statement "$\xi_b \trianglerighteq \sigma(b)$" is equivalent to "$\xi_b - \sigma(b) \in \{0,1\}$" (by Definition 4.4). Hence, we do not have $\xi_b \trianglerighteq \sigma(b)$ (since we do not have $\xi_b - \sigma(b) \in \{0,1\}$). In other words, we have $[\xi_b \trianglerighteq \sigma(b)] = 0$.

Now, recall that $b \in \{1,2,\ldots,n\}$. Thus, the term $[\xi_b \trianglerighteq \sigma(b)]$ is one of the factors of the product $\prod_{i=1}^{n} [\xi_i \trianglerighteq \sigma(i)]$ (namely, the factor for $i = b$). Since this term is 0 (because we have just shown that $[\xi_b \trianglerighteq \sigma(b)] = 0$), we thus conclude that one of the factors of the product $\prod_{i=1}^{n} [\xi_i \trianglerighteq \sigma(i)]$ is 0. Therefore, the entire product is 0. In other words, we have $\prod_{i=1}^{n} [\xi_i \trianglerighteq \sigma(i)] = 0$. Thus, Lemma 4.5 is proved in Case 2.

We have now proved Lemma 4.5 in both Cases 1 and 2. Hence, the proof of Lemma 4.5 is complete. □

The following lemma is an analogue of Lemma 2.11:

Lemma 4.6. Let $n \in \mathbb{N}$ and $p \in \{0,1,\ldots,n\}$. Let ξ be the n-tuple

$$(1,2,\ldots,n) + \left(\underbrace{0,0,\ldots,0}_{n-p \text{ zeroes}}, \underbrace{1,1,\ldots,1}_{p \text{ ones}}\right)$$
$$= (1,2,\ldots,n-p,n-p+2,n-p+3,\ldots,n+1) \in \mathbb{Z}^n.$$

Let $\nu \in \mathbb{Z}^n$ be an n-tuple satisfying $|\nu| = |\xi|$. Then,

$$\det\left(([\nu_i \trianglerighteq j])_{i,j\in[n]}\right) = \sum_{\substack{\sigma \in S_n; \\ \nu = \xi \circ \sigma}} (-1)^{\sigma}. \tag{99}$$

Note that the matrix $([\nu_i \trianglerighteq j])_{i,j\in[n]}$ in (99) is a matrix with integer entries; thus, its determinant is a well-defined integer.

Before we prove Lemma 4.6, a remark is in order:

Remark 4.7. The sum $\sum_{\substack{\sigma \in S_n; \\ \nu = \xi \circ \sigma}} (-1)^{\sigma}$ on the right hand side of (99) always has either no addends or only one addend. (Indeed, it is easy to see that the n-tuples $\xi \circ \sigma$ for

The map σ belongs to S_n, and thus is a permutation of $[n]$ (since S_n is the set of all permutations of $[n]$). Hence, this map is bijective, therefore injective. In other words, if i and j are two elements of $[n]$ satisfying $i \neq j$, then $\sigma(i) \neq \sigma(j)$. Applying this to $i = \sigma(b)$ and $j = b$, we obtain $\sigma(\sigma(b)) \neq \sigma(b)$ (since $\sigma(b) \neq b$). Thus, $\sigma(b)$ is an $i \in [n]$ satisfying $\sigma(i) \neq i$. In other words, $\sigma(b) \in \{i \in [n] \mid \sigma(i) \neq i\}$. In other words, $\sigma(b) \in Z$ (since $Z = \{i \in [n] \mid \sigma(i) \neq i\}$). Hence, (98) (applied to $z = \sigma(b)$) yields $b \geq \sigma(b)$. In other words, $\sigma(b) \leq b$. Combining this with $\sigma(b) \neq b$, we obtain $\sigma(b) < b$.

different $\sigma \in S_n$ are distinct; thus, no more than one of these n-tuples can equal ν.) Thus, this sum can be rewritten as

$$\begin{cases} (-1)^{\sigma}, & \text{if } \nu = \xi \circ \sigma \text{ for some } \sigma \in S_n; \\ 0, & \text{otherwise.} \end{cases}$$

Proof of Lemma 4.6. The definition of ξ yields

$$\begin{aligned} \xi &= (1, 2, \ldots, n) + \left(\underbrace{0, 0, \ldots, 0}_{n-p \text{ zeroes}}, \underbrace{1, 1, \ldots, 1}_{p \text{ ones}} \right) \\ &= (1, 2, \ldots, n-p, n-p+2, n-p+3, \ldots, n+1) . \end{aligned}$$

Thus,

$$\begin{aligned} &(\xi_1, \xi_2, \ldots, \xi_n) \\ &= \xi = (1, 2, \ldots, n-p, n-p+2, n-p+3, \ldots, n+1) . \end{aligned} \tag{100}$$

In other words, we have

$$\xi_i = i \qquad \text{for each } i \in \{1, 2, \ldots, n-p\} \tag{101}$$

and

$$\xi_i = i+1 \qquad \text{for each } i \in \{n-p+1, n-p+2, \ldots, n\} . \tag{102}$$

Clearly, the following two chains of inequalities hold:

$$1 < 2 < \cdots < n-p \qquad \text{and} \qquad n-p+2 < n-p+3 < \cdots < n+1.$$

Since $n-p < n-p+2$, we can splice these two chains together to obtain a single chain of inequalities:

$$1 < 2 < \cdots < n-p < n-p+2 < n-p+3 < \cdots < n+1.$$

In view of (100), we can rewrite this chain of inequalities as follows:

$$\xi_1 < \xi_2 < \cdots < \xi_n.$$

Hence, the n numbers $\xi_1, \xi_2, \ldots, \xi_n$ are distinct. Therefore, $|\{\xi_1, \xi_2, \ldots, \xi_n\}| = n$.

We have $[n] = \{1, 2, \ldots, n\}$ (by the definition of $[n]$) and $[n+1] = \{1, 2, \ldots, n+1\}$ (by the definition of $[n+1]$). Hence, $[n] = \{1, 2, \ldots, n\} \subseteq \{1, 2, \ldots, n+1\} = [n+1]$.

We are in one of the following three cases:
Case 1: We have $\{\nu_1, \nu_2, \ldots, \nu_n\} \not\subseteq [n+1]$.
Case 2: We have $|\{\nu_1, \nu_2, \ldots, \nu_n\}| \neq n$.
Case 3: We have neither $\{\nu_1, \nu_2, \ldots, \nu_n\} \not\subseteq [n+1]$ nor $|\{\nu_1, \nu_2, \ldots, \nu_n\}| \neq n$.

Let us first consider Case 1. In this case, we have $\{\nu_1, \nu_2, \ldots, \nu_n\} \not\subseteq [n+1]$. In other words, there exists some $w \in \{\nu_1, \nu_2, \ldots, \nu_n\}$ such that $w \notin [n+1]$. Consider this w. We have $w \in \{\nu_1, \nu_2, \ldots, \nu_n\}$; in other words, there exists some $k \in \{1, 2, \ldots, n\}$ such that $w = \nu_k$. Consider this k. Hence, $\nu_k = w \notin [n+1]$.

Now, it is easy to see that

$$[\nu_k \trianglerighteq j] = 0 \qquad \text{for each } j \in [n]\,. \tag{103}$$

[*Proof of (103):* Let $j \in [n]$. Then, $j \in [n] = \{1, 2, \ldots, n\}$, so that $j+1 \in \{2, 3, \ldots, n+1\} \subseteq \{1, 2, \ldots, n+1\} = [n+1]$. It is easy to see that we do not have $\nu_k - j \in \{0, 1\}$ [37]. However, the statement "$\nu_k \trianglerighteq j$" is equivalent to "$\nu_k - j \in \{0, 1\}$" (by Definition 4.4). Hence, we do not have $\nu_k \trianglerighteq j$ (since we do not have $\nu_k - j \in \{0, 1\}$). In other words, we have $[\nu_k \trianglerighteq j] = 0$. This proves (103).]

The equality (103) shows that each entry in the k-th row of the matrix $([\nu_i \trianglerighteq j])_{i,j\in[n]}$ equals 0 (because the j-th entry in the k-th row of this matrix is $[\nu_k \trianglerighteq j]$). In other words, the k-th row of the matrix $([\nu_i \trianglerighteq j])_{i,j\in[n]}$ is zero. Hence, this matrix $([\nu_i \trianglerighteq j])_{i,j\in[n]}$ has a zero row. However, it is well-known that if a square matrix[38] has a zero row, then its determinant is 0. Hence, the determinant of the matrix $([\nu_i \trianglerighteq j])_{i,j\in[n]}$ is 0 (since this matrix has a zero row). In other words,

$$\det\left(([\nu_i \trianglerighteq j])_{i,j\in[n]}\right) = 0. \tag{104}$$

On the other hand, there exists no $\sigma \in S_n$ satisfying $\nu = \xi \circ \sigma$ [39]. Hence, the sum

[37] *Proof.* Assume the contrary. Thus, $\nu_k - j \in \{0, 1\}$. In other words, $\nu_k - j = 0$ or $\nu_k - j = 1$. Hence, we are in one of the following two cases:

Case 1: We have $\nu_k - j = 0$.

Case 2: We have $\nu_k - j = 1$.

Let us consider Case 1 first. In this case, we have $\nu_k - j = 0$. Thus, $\nu_k = j \in [n] \subseteq [n+1]$. This contradicts $\nu_k \notin [n+1]$. Thus, we have obtained a contradiction in Case 1.

Let us now consider Case 2. In this case, we have $\nu_k - j = 1$. Thus, $\nu_k = j + 1 \in [n+1]$. This contradicts $\nu_k \notin [n+1]$. Thus, we have obtained a contradiction in Case 2.

We have now found a contradiction in each of the two Cases 1 and 2. Thus, we always have a contradiction. This shows that our assumption was false, qed.

[38] In this proof, the word "matrix" always means a matrix with integer entries.

[39] *Proof.* Let $\sigma \in S_n$ satisfy $\nu = \xi \circ \sigma$. We shall obtain a contradiction.

Indeed, we have $\sigma \in S_n$. In other words, σ is a permutation of $[n]$ (since S_n was defined as the set of all permutations of $[n]$). Hence,

$$\begin{aligned}
\left\{\xi_{\sigma(1)}, \xi_{\sigma(2)}, \ldots, \xi_{\sigma(n)}\right\} &= \{\xi_1, \xi_2, \ldots, \xi_n\} \\
&= \{1, 2, \ldots, n-p, n-p+2, n-p+3, \ldots, n+1\} \qquad \text{(by (100))} \\
&= \underbrace{\{1, 2, \ldots, n+1\}}_{=[n+1]} \setminus \{n-p+1\} = [n+1] \setminus \{n-p+1\} \subseteq [n+1]\,.
\end{aligned}$$

However, $(\nu_1, \nu_2, \ldots, \nu_n) = \nu = \xi \circ \sigma = \left(\xi_{\sigma(1)}, \xi_{\sigma(2)}, \ldots, \xi_{\sigma(n)}\right)$ (by Definition 2.4). Hence,

$$\{\nu_1, \nu_2, \ldots, \nu_n\} = \left\{\xi_{\sigma(1)}, \xi_{\sigma(2)}, \ldots, \xi_{\sigma(n)}\right\} \subseteq [n+1]\,.$$

But this contradicts $\{\nu_1, \nu_2, \ldots, \nu_n\} \not\subseteq [n+1]$.

$\sum\limits_{\substack{\sigma \in S_n; \\ \nu = \xi \circ \sigma}} (-1)^\sigma$ is empty. Thus,

$$\sum_{\substack{\sigma \in S_n; \\ \nu = \xi \circ \sigma}} (-1)^\sigma = (\text{empty sum}) = 0.$$

Comparing this with (104), we obtain $\det\left(\left([\nu_i \trianglerighteq j]\right)_{i,j \in [n]}\right) = \sum\limits_{\substack{\sigma \in S_n; \\ \nu = \xi \circ \sigma}} (-1)^\sigma$. Thus, Lemma 4.6 is proved in Case 1.

Let us next consider Case 2. In this case, we have $|\{\nu_1, \nu_2, \ldots, \nu_n\}| \neq n$. Hence, the n numbers $\nu_1, \nu_2, \ldots, \nu_n$ cannot be distinct (because if they were distinct, then we would have $|\{\nu_1, \nu_2, \ldots, \nu_n\}| = n$, which would contradict $|\{\nu_1, \nu_2, \ldots, \nu_n\}| \neq n$). In other words, two of these n numbers are equal. In other words, there exist two distinct elements r and s of $[n]$ such that $\nu_r = \nu_s$. Consider these r and s. Now, for each $j \in [n]$, we have

$$[\nu_r \trianglerighteq j] = [\nu_s \trianglerighteq j] \qquad (\text{since } \nu_r = \nu_s).$$

This shows that each entry in the r-th row of the matrix $\left([\nu_i \trianglerighteq j]\right)_{i,j \in [n]}$ equals the corresponding entry in the s-th row of this matrix (because the j-th entry in the r-th row of this matrix is $[\nu_r \trianglerighteq j]$, whereas the corresponding entry in the s-th row is $[\nu_s \trianglerighteq j]$). In other words, the r-th row of the matrix $\left([\nu_i \trianglerighteq j]\right)_{i,j \in [n]}$ equals the s-th row of this matrix. Hence, this matrix $\left([\nu_i \trianglerighteq j]\right)_{i,j \in [n]}$ has two equal rows (since r and s are distinct). However, it is well-known that if a square matrix[40] has two equal rows, then its determinant is 0. Hence, the determinant of the matrix $\left([\nu_i \trianglerighteq j]\right)_{i,j \in [n]}$ is 0 (since this matrix has two equal rows). In other words,

$$\det\left(\left([\nu_i \trianglerighteq j]\right)_{i,j \in [n]}\right) = 0. \tag{105}$$

On the other hand, there exists no $\sigma \in S_n$ satisfying $\nu = \xi \circ \sigma$ [41]. Hence, the sum $\sum\limits_{\substack{\sigma \in S_n; \\ \nu = \xi \circ \sigma}} (-1)^\sigma$ is empty. Thus,

$$\sum_{\substack{\sigma \in S_n; \\ \nu = \xi \circ \sigma}} (-1)^\sigma = (\text{empty sum}) = 0.$$

Forget that we fixed σ. We thus have found a contradiction for each $\sigma \in S_n$ satisfying $\nu = \xi \circ \sigma$. Hence, there exists no such σ. Qed.

[40] In this proof, the word "matrix" always means a matrix with integer entries.

[41] *Proof.* Let $\sigma \in S_n$ satisfy $\nu = \xi \circ \sigma$. We shall obtain a contradiction.

Indeed, we have $\sigma \in S_n$. In other words, σ is a permutation of $[n]$ (since S_n was defined as the set of all permutations of $[n]$). Hence, $\left\{\xi_{\sigma(1)}, \xi_{\sigma(2)}, \ldots, \xi_{\sigma(n)}\right\} = \{\xi_1, \xi_2, \ldots, \xi_n\}$.

However, $(\nu_1, \nu_2, \ldots, \nu_n) = \nu = \xi \circ \sigma = \left(\xi_{\sigma(1)}, \xi_{\sigma(2)}, \ldots, \xi_{\sigma(n)}\right)$ (by Definition 2.4). Hence,

$$\{\nu_1, \nu_2, \ldots, \nu_n\} = \left\{\xi_{\sigma(1)}, \xi_{\sigma(2)}, \ldots, \xi_{\sigma(n)}\right\} = \{\xi_1, \xi_2, \ldots, \xi_n\}.$$

Thus, $|\{\nu_1, \nu_2, \ldots, \nu_n\}| = |\{\xi_1, \xi_2, \ldots, \xi_n\}| = n$. This contradicts $|\{\nu_1, \nu_2, \ldots, \nu_n\}| \neq n$.

Forget that we fixed σ. We thus have found a contradiction for each $\sigma \in S_n$ satisfying $\nu = \xi \circ \sigma$. Hence, there exists no such σ. Qed.

Comparing this with (105), we obtain $\det\left(\left([\nu_i \trianglerighteq j]\right)_{i,j\in[n]}\right) = \sum\limits_{\substack{\sigma\in S_n;\\ \nu=\xi\circ\sigma}} (-1)^\sigma$. Thus, Lemma 4.6 is proved in Case 2.

Finally, let us consider Case 3. In this case, we have neither $\{\nu_1,\nu_2,\ldots,\nu_n\} \not\subseteq [n+1]$ nor $|\{\nu_1,\nu_2,\ldots,\nu_n\}| \neq n$. In other words, we have $\{\nu_1,\nu_2,\ldots,\nu_n\} \subseteq [n+1]$ and $|\{\nu_1,\nu_2,\ldots,\nu_n\}| = n$.

Set $q := n-p+1$. Thus, $q \in \mathbb{Z}$. From (100), we obtain

$$\begin{aligned}\{\xi_1,\xi_2,\ldots,\xi_n\} &= \{1,2,\ldots,n-p,n-p+2,n-p+3,\ldots,n+1\}\\ &= \underbrace{\{1,2,\ldots,n+1\}}_{=[n+1]} \setminus \left\{\underbrace{n-p+1}_{=q}\right\} = [n+1]\setminus\{q\}.\end{aligned}$$

Hence,

$$\begin{aligned}\underbrace{\{\xi_1,\xi_2,\ldots,\xi_n\}}_{=[n+1]\setminus\{q\}} \cup \{q\} &= ([n+1]\setminus\{q\})\cup\{q\}\\ &= [n+1]\cup\{q\}. \qquad (106)\end{aligned}$$

Now,

$$\{\nu_1,\nu_2,\ldots,\nu_n\} \subseteq [n+1] \subseteq [n+1]\cup\{q\} = \{\xi_1,\xi_2,\ldots,\xi_n\}\cup\{q\}$$

(by (106)). Thus, Lemma 4.3 yields that there exists some permutation $\pi \in S_n$ satisfying $\nu = \xi\circ\pi$. Consider this π.

It is easy to see that π is the **only** permutation $\sigma \in S_n$ satisfying $\nu = \xi\circ\sigma$ [42]. Hence, the sum $\sum\limits_{\substack{\sigma\in S_n;\\ \nu=\xi\circ\sigma}} (-1)^\sigma$ has only one addend, namely the addend for $\sigma=\pi$. Thus,

$$\sum_{\substack{\sigma\in S_n;\\ \nu=\xi\circ\sigma}} (-1)^\sigma = (-1)^\pi. \qquad (107)$$

We have $\nu = \xi\circ\pi = \left(\xi_{\pi(1)},\xi_{\pi(2)},\ldots,\xi_{\pi(n)}\right)$ (by Definition 2.4). Thus, for each $i\in[n]$, we have

$$\nu_i = \left(\xi_{\pi(1)},\xi_{\pi(2)},\ldots,\xi_{\pi(n)}\right)_i = \xi_{\pi(i)}. \qquad (108)$$

[42] *Proof.* It is clear that π is a permutation $\sigma\in S_n$ satisfying $\nu=\xi\circ\sigma$ (because $\pi\in S_n$ is a permutation satisfying $\nu=\xi\circ\pi$). It thus remains to show that π is the **only** such permutation. In other words, it remains to show that if $\sigma\in S_n$ is a permutation satisfying $\nu=\xi\circ\sigma$, then $\sigma=\pi$. So let us show this.

Let $\sigma\in S_n$ be a permutation satisfying $\nu=\xi\circ\sigma$. Then, $\xi\circ\sigma=\nu=\xi\circ\pi$.

Recall that the n numbers $\xi_1,\xi_2,\ldots,\xi_n$ are distinct. Thus, Proposition 2.6 (applied to $\eta=\xi$) shows that if the permutations σ and π were distinct, then we would have $\xi\circ\sigma\neq\xi\circ\pi$; but this would contradict $\xi\circ\sigma=\xi\circ\pi$. Hence, σ and π cannot be distinct. In other words, we must have $\sigma=\pi$.

Forget that we fixed σ. We thus have shown that if $\sigma\in S_n$ is a permutation satisfying $\nu=\xi\circ\sigma$, then $\sigma=\pi$. In other words, π is the **only** permutation $\sigma\in S_n$ satisfying $\nu=\xi\circ\sigma$ (since we already know that π is such a permutation). Qed.

On the other hand, it is well-known (see, e.g., [Grinbe21, Corollary 6.4.15] or [Grinbe15, Lemma 6.17 **(a)**]) that when the rows of a matrix are permuted, then the determinant of this matrix gets multiplied by $(-1)^\tau$, where τ is the permutation used to permute the rows. In other words: If $(a_{i,j})_{i,j\in[n]}$ is a square matrix (with integer entries), and if $\tau \in S_n$ is a permutation, then

$$\det\left(\left(a_{\tau(i),j}\right)_{i,j\in[n]}\right) = (-1)^\tau \cdot \det\left(\left(a_{i,j}\right)_{i,j\in[n]}\right).$$

Applying this to $a_{i,j} = [\xi_i \trianglerighteq j]$ and $\tau = \pi$, we obtain

$$\det\left(\left(\left[\xi_{\pi(i)} \trianglerighteq j\right]\right)_{i,j\in[n]}\right) = (-1)^\pi \cdot \det\left(\left([\xi_i \trianglerighteq j]\right)_{i,j\in[n]}\right).$$

In view of (108), we can rewrite this as

$$\det\left(\left([\nu_i \trianglerighteq j]\right)_{i,j\in[n]}\right) = (-1)^\pi \cdot \det\left(\left([\xi_i \trianglerighteq j]\right)_{i,j\in[n]}\right). \tag{109}$$

However, we have

$$[\xi_i \trianglerighteq i] = 1 \qquad \text{for each } i \in \{1,2,\ldots,n\} \tag{110}$$

[43].

Now, consider the identity permutation $\mathrm{id} : [n] \to [n]$ of the set $[n]$. The definition

[43] *Proof of (110):* Let $i \in \{1,2,\ldots,n\}$. We must prove that $[\xi_i \trianglerighteq i] = 1$. We are in one of the following two cases:

Case 1: We have $i \leq n-p$.

Case 2: We have $i > n-p$.

Let us first consider Case 1. In this case, we have $i \leq n-p$. Combining this with $i \geq 1$ (which follows from $i \in \{1,2,\ldots,n\}$), we obtain $i \in \{1,2,\ldots,n-p\}$. Thus, (101) yields $\xi_i = i$. Hence, $\xi_i - i = 0 \in \{0,1\}$. In other words, $\xi_i \trianglerighteq i$ (because Definition 4.4 shows that the statement "$\xi_i \trianglerighteq i$" is equivalent to "$\xi_i - i \in \{0,1\}$"). Therefore, $[\xi_i \trianglerighteq i] = 1$. Thus, (110) is proved in Case 1.

Let us next consider Case 2. In this case, we have $i > n-p$. Hence, $i \geq n-p+1$ (since i and $n-p$ are integers). Combining this with $i \leq n$ (which follows from $i \in \{1,2,\ldots,n\}$), we obtain $i \in \{n-p+1, n-p+2, \ldots, n\}$. Thus, (102) yields $\xi_i = i+1$. Hence, $\xi_i - i = 1 \in \{0,1\}$. In other words, $\xi_i \trianglerighteq i$ (because Definition 4.4 shows that the statement "$\xi_i \trianglerighteq i$" is equivalent to "$\xi_i - i \in \{0,1\}$"). Therefore, $[\xi_i \trianglerighteq i] = 1$. Thus, (110) is proved in Case 2.

We have now proved (110) in both Cases 1 and 2. Thus, the proof of (110) is complete.

of the determinant of a matrix yields

$$\begin{aligned}
\det\left(\left(\left[\xi_i \trianglerighteq j\right]\right)_{i,j\in[n]}\right) &= \sum_{\sigma\in S_n} (-1)^{\sigma} \prod_{i=1}^{n} \left[\xi_i \trianglerighteq \sigma(i)\right] \\
&= \underbrace{(-1)^{\mathrm{id}}}_{=1} \prod_{i=1}^{n} \left[\xi_i \trianglerighteq \underbrace{\mathrm{id}(i)}_{=i}\right] + \sum_{\substack{\sigma\in S_n;\\ \sigma\neq\mathrm{id}}} (-1)^{\sigma} \underbrace{\prod_{i=1}^{n} \left[\xi_i \trianglerighteq \sigma(i)\right]}_{\substack{=0\\ \text{(by Lemma 4.5)}}} \\
&\qquad \left(\begin{array}{c}\text{here, we have split off the addend}\\ \text{for } \sigma = \mathrm{id} \text{ from the sum (since } \mathrm{id}\in S_n)\end{array}\right) \\
&= \prod_{i=1}^{n} \underbrace{\left[\xi_i \trianglerighteq i\right]}_{\substack{=1\\ \text{(by (110))}}} + \underbrace{\sum_{\substack{\sigma\in S_n;\\ \sigma\neq\mathrm{id}}} (-1)^{\sigma} 0}_{=0} = \prod_{i=1}^{n} 1 = 1.
\end{aligned}$$

Thus, (109) becomes

$$\det\left(\left(\left[\nu_i \trianglerighteq j\right]\right)_{i,j\in[n]}\right) = (-1)^{\pi} \cdot \underbrace{\det\left(\left(\left[\xi_i \trianglerighteq j\right]\right)_{i,j\in[n]}\right)}_{=1} = (-1)^{\pi} = \sum_{\substack{\sigma\in S_n;\\ \nu=\xi\circ\sigma}} (-1)^{\sigma}$$

(by (107)). Thus, Lemma 4.6 is proved in Case 3.

We have now proved Lemma 4.6 in all three Cases 1, 2 and 3. This completes the proof of Lemma 4.6. □

Our next lemma is an analogue to Lemma 2.13:

Lemma 4.8. Let $n \in \mathbb{N}$. For any permutation $\sigma \in S_n$, we let $\overline{\sigma}$ denote the n-tuple $(\sigma(1), \sigma(2), \ldots, \sigma(n)) \in \mathbb{Z}^n$.

Let $\nu \in \mathbb{Z}^n$ be an n-tuple. Then,

$$\det\left(\left(\left[\nu_i \trianglerighteq j\right]\right)_{i,j\in[n]}\right) = \sum_{\substack{\sigma\in S_n;\\ \nu-\overline{\sigma}\in\{0,1\}^n}} (-1)^{\sigma}.$$

The following proof of Lemma 4.8 is strongly similar to the above proof of Lemma 2.13; we nevertheless provide it here in full for the sake of completeness.

Proof of Lemma 4.8. We first observe that each $\sigma \in S_n$ satisfies

$$\left[\nu - \overline{\sigma} \in \{0,1\}^n\right] = \prod_{i=1}^{n} \left[\nu_i \trianglerighteq \sigma(i)\right]. \tag{111}$$

[*Proof of (111):* Let $\sigma \in S_n$ be arbitrary. We must prove the equality (111).

The definition of $\overline{\sigma}$ yields $\overline{\sigma} = (\sigma(1), \sigma(2), \ldots, \sigma(n))$. Hence, for each $i \in [n]$, we have

$$\overline{\sigma}_i = \sigma(i). \tag{112}$$

We are in one of the following two cases:
Case 1: We have $\nu - \overline{\sigma} \in \{0,1\}^n$.
Case 2: We have $\nu - \overline{\sigma} \notin \{0,1\}^n$.

Let us first consider Case 1. In this case, we have $\nu - \overline{\sigma} \in \{0,1\}^n$. Now, let $i \in [n]$. Then, $(\nu - \overline{\sigma})_i \in \{0,1\}$ (since $\nu - \overline{\sigma} \in \{0,1\}^n$). In view of

$$(\nu - \overline{\sigma})_i = \nu_i - \underbrace{\overline{\sigma}_i}_{\substack{=\sigma(i) \\ \text{(by (112))}}} = \nu_i - \sigma(i),$$

we can rewrite this as $\nu_i - \sigma(i) \in \{0,1\}$. In other words, $\nu_i \trianglerighteq \sigma(i)$ (since Definition 4.4 says that the statement "$\nu_i \trianglerighteq \sigma(i)$" is equivalent to "$\nu_i - \sigma(i) \in \{0,1\}$"). Thus, $[\nu_i \trianglerighteq \sigma(i)] = 1$.

Forget that we fixed i. Thus, we have shown that $[\nu_i \trianglerighteq \sigma(i)] = 1$ for each $i \in [n]$. Multiplying these equalities over all $i \in [n]$, we obtain $\prod\limits_{i \in [n]} [\nu_i \trianglerighteq \sigma(i)] = \prod\limits_{i \in [n]} 1 = 1$. On the other hand, $[\nu - \overline{\sigma} \in \{0,1\}^n] = 1$ (since $\nu - \overline{\sigma} \in \{0,1\}^n$). Comparing these two equalities, we obtain

$$[\nu - \overline{\sigma} \in \{0,1\}^n] = \underbrace{\prod_{i \in [n]} [\nu_i \trianglerighteq \sigma(i)]}_{=\prod\limits_{i=1}^{n}} = \prod_{i=1}^{n} [\nu_i \trianglerighteq \sigma(i)].$$

Thus, (111) is proved in Case 1.

Let us now consider Case 2. In this case, we have $\nu - \overline{\sigma} \notin \{0,1\}^n$. Hence, there exists some $j \in [n]$ such that $(\nu - \overline{\sigma})_j \notin \{0,1\}$. Consider this j. We have $\overline{\sigma}_j = \sigma(j)$ (by (112), applied to $i = j$). Now,

$$(\nu - \overline{\sigma})_j = \nu_j - \underbrace{\overline{\sigma}_j}_{=\sigma(j)} = \nu_j - \sigma(j),$$

so that $\nu_j - \sigma(j) = (\nu - \overline{\sigma})_j \notin \{0,1\}$. In other words, we don't have $\nu_j - \sigma(j) \in \{0,1\}$. In other words, we don't have $\nu_j \trianglerighteq \sigma(j)$ (since Definition 4.4 says that the statement "$\nu_j \trianglerighteq \sigma(j)$" is equivalent to "$\nu_j - \sigma(j) \in \{0,1\}$"). Thus, $[\nu_j \trianglerighteq \sigma(j)] = 0$.

Now, consider the product $\prod\limits_{i=1}^{n} [\nu_i \trianglerighteq \sigma(i)]$. The term $[\nu_j \trianglerighteq \sigma(j)]$ is one of the factors of this product (namely, the factor for $i = j$). Since this term is 0 (because we have just shown that $[\nu_j \trianglerighteq \sigma(j)] = 0$), we thus conclude that one of the factors of the product $\prod\limits_{i=1}^{n} [\nu_i \trianglerighteq \sigma(i)]$ is 0. Therefore, the entire product is 0. In other words,

$$\prod_{i=1}^{n} [\nu_i \trianglerighteq \sigma(i)] = 0. \tag{113}$$

On the other hand, we do not have $\nu - \overline{\sigma} \in \{0,1\}^n$ (since $\nu - \overline{\sigma} \notin \{0,1\}^n$). Hence, $[\nu - \overline{\sigma} \in \{0,1\}^n] = 0$. Comparing this with (113), we obtain

$$[\nu - \overline{\sigma} \in \{0,1\}^n] = \prod_{i=1}^{n} [\nu_i \trianglerighteq \sigma(i)] .$$

Thus, (111) is proved in Case 2.

We have now proved (111) in both Cases 1 and 2. Thus, the proof of (111) is complete.]

Now, the definition of the determinant of a matrix yields

$$\begin{aligned}
&\det\left(([\nu_i \trianglerighteq j])_{i,j\in[n]}\right) \\
&= \sum_{\sigma\in S_n} (-1)^\sigma \underbrace{\prod_{i=1}^{n} [\nu_i \trianglerighteq \sigma(i)]}_{\substack{=[\nu-\overline{\sigma}\in\{0,1\}^n] \\ \text{(by (111))}}} = \sum_{\sigma\in S_n} (-1)^\sigma [\nu - \overline{\sigma} \in \{0,1\}^n] \\
&= \sum_{\substack{\sigma\in S_n; \\ \nu-\overline{\sigma}\in\{0,1\}^n}} (-1)^\sigma \underbrace{[\nu - \overline{\sigma} \in \{0,1\}^n]}_{\substack{=1 \\ \text{(since } \nu-\overline{\sigma}\in\{0,1\}^n\text{)}}} + \sum_{\substack{\sigma\in S_n; \\ \nu-\overline{\sigma}\notin\{0,1\}^n}} (-1)^\sigma \underbrace{[\nu - \overline{\sigma} \in \{0,1\}^n]}_{\substack{=0 \\ \text{(since we don't have } \nu-\overline{\sigma}\in\{0,1\}^n \\ \text{(because } \nu-\overline{\sigma}\notin\{0,1\}^n\text{))}}} \\
&\qquad \left(\begin{array}{c} \text{since each } \sigma\in S_n \text{ satisfies either } \nu-\overline{\sigma}\in\{0,1\}^n \\ \text{or } \nu-\overline{\sigma}\notin\{0,1\}^n \text{ (but not both at the same time)} \end{array}\right) \\
&= \sum_{\substack{\sigma\in S_n; \\ \nu-\overline{\sigma}\in\{0,1\}^n}} (-1)^\sigma + \underbrace{\sum_{\substack{\sigma\in S_n; \\ \nu-\overline{\sigma}\notin\{0,1\}^n}} (-1)^\sigma 0}_{=0} = \sum_{\substack{\sigma\in S_n; \\ \nu-\overline{\sigma}\in\{0,1\}^n}} (-1)^\sigma .
\end{aligned}$$

This proves Lemma 4.8. □

We can now prove Theorem 4.1:

Proof of Theorem 4.1. The definition of ξ yields

$$\xi = (1,2,\ldots,n) + \left(\underbrace{0,0,\ldots,0}_{n-p \text{ zeroes}}, \underbrace{1,1,\ldots,1}_{p \text{ ones}}\right). \tag{114}$$

Thus,

$$\begin{aligned}
|\xi| &= \left|(1,2,\ldots,n) + \left(\underbrace{0,0,\ldots,0}_{n-p \text{ zeroes}}, \underbrace{1,1,\ldots,1}_{p \text{ ones}}\right)\right| \\
&= |(1,2,\ldots,n)| + \left|\left(\underbrace{0,0,\ldots,0}_{n-p \text{ zeroes}}, \underbrace{1,1,\ldots,1}_{p \text{ ones}}\right)\right|
\end{aligned}$$

(by Lemma 2.14, applied to $\alpha = (1, 2, \ldots, n)$ and $\beta = \left(\underbrace{0, 0, \ldots, 0}_{n-p \text{ zeroes}}, \underbrace{1, 1, \ldots, 1}_{p \text{ ones}}\right)$). In view of

$$|(1, 2, \ldots, n)| = 1 + 2 + \cdots + n \qquad \text{(by the definition of } |(1, 2, \ldots, n)|)$$

and

$$\begin{aligned}
&\left|\left(\underbrace{0, 0, \ldots, 0}_{n-p \text{ zeroes}}, \underbrace{1, 1, \ldots, 1}_{p \text{ ones}}\right)\right| \\
&= \underbrace{0 + 0 + \cdots + 0}_{n-p \text{ zeroes}} + \underbrace{1 + 1 + \cdots + 1}_{p \text{ ones}} \\
&\qquad \left(\text{by the definition of } \left|\left(\underbrace{0, 0, \ldots, 0}_{n-p \text{ zeroes}}, \underbrace{1, 1, \ldots, 1}_{p \text{ ones}}\right)\right|\right) \\
&= (n - p) \cdot 0 + p \cdot 1 = p,
\end{aligned}$$

this rewrites as

$$|\xi| = (1 + 2 + \cdots + n) + p. \tag{115}$$

For any permutation $\sigma \in S_n$, we let $\overline{\sigma}$ denote the n-tuple $(\sigma(1), \sigma(2), \ldots, \sigma(n)) \in \mathbb{Z}^n$. This n-tuple $\overline{\sigma}$ satisfies $|\overline{\sigma}| = 1 + 2 + \cdots + n$ [44] and therefore

$$\underbrace{|\overline{\sigma}|}_{=1+2+\cdots+n} + p = (1 + 2 + \cdots + n) + p = |\xi| \tag{116}$$

(by (115)). Moreover, for each $\sigma \in S_n$ and each $i \in [n]$, we have

$$\overline{\sigma}_i = \sigma(i) \tag{117}$$

(since $\overline{\sigma} = (\sigma(1), \sigma(2), \ldots, \sigma(n))$).

Now, we claim the following:

> *Claim 1:* Let $\beta \in \{0, 1\}^n$ and $\sigma \in S_n$ be arbitrary. Then, the statement "$|\beta| = p$" is equivalent to "$|\beta + \overline{\sigma}| = |\xi|$".

[44] *Proof.* Let $\sigma \in S_n$. Thus, σ is a permutation of $[n]$ (since S_n was defined as the set of all permutations of $[n]$). In other words, σ is a bijection from $[n]$ to $[n]$. However, the definition of $\overline{\sigma}$ yields $\overline{\sigma} = (\sigma(1), \sigma(2), \ldots, \sigma(n))$. Thus,

$$\begin{aligned}
|\overline{\sigma}| &= |(\sigma(1), \sigma(2), \ldots, \sigma(n))| = \sigma(1) + \sigma(2) + \cdots + \sigma(n) \\
&\qquad \text{(by the definition of } |(\sigma(1), \sigma(2), \ldots, \sigma(n))|) \\
&= \sum_{i \in [n]} \sigma(i) = \sum_{i \in [n]} i \qquad \left(\begin{array}{c}\text{here, we have substituted } i \text{ for } \sigma(i) \text{ in the sum,} \\ \text{since the map } \sigma : [n] \to [n] \text{ is a bijection}\end{array}\right) \\
&= 1 + 2 + \cdots + n,
\end{aligned}$$

qed.

[*Proof of Claim 1:* We have $\beta + \overline{\sigma} = \overline{\sigma} + \beta$ (since $\mathbb{Z}^n$ is an abelian group). Thus, $|\beta + \overline{\sigma}| = |\overline{\sigma} + \beta| = |\overline{\sigma}| + |\beta|$ (by Lemma 2.14, applied to $\alpha = \overline{\sigma}$). Therefore,

$$\begin{aligned} \underbrace{|\beta + \overline{\sigma}|}_{=|\overline{\sigma}|+|\beta|} - \underbrace{|\xi|}_{\substack{=|\overline{\sigma}|+p \\ \text{(by (116))}}} &= (|\overline{\sigma}| + |\beta|) - (|\overline{\sigma}| + p) \\ &= |\beta| - p. \end{aligned} \tag{118}$$

Hence, we have the following chain of logical equivalences:

$$\begin{aligned} (|\beta| = p) &\iff \left(\underbrace{|\beta| - p}_{\substack{=|\beta+\overline{\sigma}|-|\xi| \\ \text{(by (118))}}} = 0 \right) \iff (|\beta + \overline{\sigma}| - |\xi| = 0) \\ &\iff (|\beta + \overline{\sigma}| = |\xi|) . \end{aligned}$$

In other words, the statement "$|\beta| = p$" is equivalent to "$|\beta + \overline{\sigma}| = |\xi|$". This proves Claim 1.]

Let us now set

$$\prod_{i=1}^{n} a_i := a_1 a_2 \cdots a_n \tag{119}$$

for any $a_1, a_2, \ldots, a_n \in R$. (This is, of course, the usual meaning of the notation $\prod_{i=1}^{n} a_i$ when the ring R is commutative; however, we are now extending it to the case of arbitrary R by using the formula (119).)

Thus, if $(a_{i,j})_{i,j\in[n]}$ is any $n \times n$-matrix over R, then the definition of rowdet $\left((a_{i,j})_{i,j\in[n]}\right)$ yields

$$\begin{aligned} \operatorname{rowdet}\left((a_{i,j})_{i,j\in[n]}\right) &= \sum_{\sigma\in S_n} (-1)^\sigma \underbrace{a_{1,\sigma(1)} a_{2,\sigma(2)} \cdots a_{n,\sigma(n)}}_{=\prod_{i=1}^{n} a_{i,\sigma(i)}} \\ &= \sum_{\sigma\in S_n} (-1)^\sigma \prod_{i=1}^{n} a_{i,\sigma(i)}. \end{aligned} \tag{120}$$

For each $\beta \in \{0,1\}^n$, we have

$$\begin{aligned} t_{\alpha+\beta} &= \operatorname{rowdet}\left(\left(h_{(\alpha+\beta)_i+j,\ i}\right)_{i,j\in[n]}\right) \qquad \text{(by the definition of } t_{\alpha+\beta}) \\ &= \operatorname{rowdet}\left(\left(h_{\alpha_i+\beta_i+j,\ i}\right)_{i,j\in[n]}\right) \\ &\qquad \text{(since } (\alpha+\beta)_i = \alpha_i + \beta_i \text{ for each } i \in [n]) \\ &= \sum_{\sigma\in S_n} (-1)^\sigma \prod_{i=1}^{n} h_{\alpha_i+\beta_i+\sigma(i),\ i} \\ &\qquad \text{(by (120), applied to } a_{i,j} = h_{\alpha_i+\beta_i+j,\ i}) . \end{aligned} \tag{121}$$

However, for each $\beta \in \{0,1\}^n$ and each $\sigma \in S_n$, we have

$$\beta_i + \underbrace{\sigma(i)}_{\substack{=\overline{\sigma}_i \\ \text{(by (117))}}} = \beta_i + \overline{\sigma}_i = (\beta + \overline{\sigma})_i$$

and therefore

$$h_{\alpha_i+\beta_i+\sigma(i),\ i} = h_{\alpha_i+(\beta+\overline{\sigma})_i,\ i}. \tag{122}$$

Hence, for each $\beta \in \{0,1\}^n$, we have the equality

$$\begin{aligned} t_{\alpha+\beta} &= \sum_{\sigma \in S_n} (-1)^\sigma \prod_{i=1}^{n} \underbrace{h_{\alpha_i+\beta_i+\sigma(i),\ i}}_{\substack{=h_{\alpha_i+(\beta+\overline{\sigma})_i,\ i} \\ \text{(by (122))}}} \qquad \text{(by (121))} \\ &= \sum_{\sigma \in S_n} (-1)^\sigma \prod_{i=1}^{n} h_{\alpha_i+(\beta+\overline{\sigma})_i,\ i}. \end{aligned}$$

Summing these equalities over all $\beta \in \{0,1\}^n$ satisfying $|\beta| = p$, we obtain

$$\begin{aligned} \sum_{\substack{\beta \in \{0,1\}^n; \\ |\beta|=p}} t_{\alpha+\beta} &= \underbrace{\sum_{\substack{\beta \in \{0,1\}^n; \\ |\beta|=p}} \sum_{\sigma \in S_n}}_{= \sum_{\sigma \in S_n} \sum_{\substack{\beta \in \{0,1\}^n; \\ |\beta|=p}}} (-1)^\sigma \prod_{i=1}^{n} h_{\alpha_i+(\beta+\overline{\sigma})_i,\ i} \\ &= \sum_{\sigma \in S_n} \sum_{\substack{\beta \in \{0,1\}^n; \\ |\beta|=p}} (-1)^\sigma \prod_{i=1}^{n} h_{\alpha_i+(\beta+\overline{\sigma})_i,\ i} \\ &= \sum_{\sigma \in S_n} (-1)^\sigma \sum_{\substack{\beta \in \{0,1\}^n; \\ |\beta|=p}} \prod_{i=1}^{n} h_{\alpha_i+(\beta+\overline{\sigma})_i,\ i}. \end{aligned} \tag{123}$$

Now, fix a permutation $\sigma \in S_n$. We shall rewrite the sum $\sum\limits_{\substack{\beta \in \{0,1\}^n; \\ |\beta|=p}} \prod\limits_{i=1}^{n} h_{\alpha_i+(\beta+\overline{\sigma})_i,\ i}$ in terms of $\beta + \overline{\sigma}$. Indeed, Claim 1 shows that for any n-tuple $\beta \in \{0,1\}^n$, the statement "$|\beta| = p$" is equivalent to "$|\beta + \overline{\sigma}| = |\xi|$". Hence, we have the following equality of summation signs:

$$\begin{aligned} \sum_{\substack{\beta \in \{0,1\}^n; \\ |\beta|=p}} &= \sum_{\substack{\beta \in \{0,1\}^n; \\ |\beta+\overline{\sigma}|=|\xi|}} = \sum_{\substack{\beta \in \mathbb{Z}^n; \\ \beta \in \{0,1\}^n; \\ |\beta+\overline{\sigma}|=|\xi|}} \qquad (\text{since } \{0,1\}^n \subseteq \mathbb{Z}^n) \\ &= \sum_{\substack{\beta \in \mathbb{Z}^n; \\ (\beta+\overline{\sigma})-\overline{\sigma} \in \{0,1\}^n; \\ |\beta+\overline{\sigma}|=|\xi|}} \qquad (\text{since } \beta = (\beta+\overline{\sigma})-\overline{\sigma}). \end{aligned}$$

Thus,

$$\underbrace{\sum_{\substack{\beta\in\{0,1\}^n;\\ |\beta|=p}}}_{=\sum\limits_{\substack{\beta\in\mathbb{Z}^n;\\ (\beta+\overline{\sigma})-\overline{\sigma}\in\{0,1\}^n;\\ |\beta+\overline{\sigma}|=|\xi|}}} \prod_{i=1}^{n} h_{\alpha_i+(\beta+\overline{\sigma})_i,\ i} = \sum_{\substack{\beta\in\mathbb{Z}^n;\\ (\beta+\overline{\sigma})-\overline{\sigma}\in\{0,1\}^n;\\ |\beta+\overline{\sigma}|=|\xi|}} \prod_{i=1}^{n} h_{\alpha_i+(\beta+\overline{\sigma})_i,\ i}.$$

However, $\mathbb{Z}^n$ is a group (under addition). Hence, for any $\rho \in \mathbb{Z}^n$, the map

$$\begin{aligned} \mathbb{Z}^n &\to \mathbb{Z}^n, \\ \beta &\mapsto \beta + \rho \end{aligned}$$

is a bijection (whose inverse is the map $\mathbb{Z}^n \to \mathbb{Z}^n,\ \nu \mapsto \nu - \rho$). Applying this to $\rho = \overline{\sigma}$, we conclude that the map

$$\begin{aligned} \mathbb{Z}^n &\to \mathbb{Z}^n, \\ \beta &\mapsto \beta + \overline{\sigma} \end{aligned}$$

is a bijection (whose inverse is the map $\mathbb{Z}^n \to \mathbb{Z}^n,\ \nu \mapsto \nu - \overline{\sigma}$). Therefore, we can substitute ν for $\beta + \overline{\sigma}$ in the sum $\sum\limits_{\substack{\beta\in\mathbb{Z}^n;\\ (\beta+\overline{\sigma})-\overline{\sigma}\in\{0,1\}^n;\\ |\beta+\overline{\sigma}|=|\xi|}} \prod\limits_{i=1}^{n} h_{\alpha_i+(\beta+\overline{\sigma})_i,\ i}$. We thus obtain

$$\sum_{\substack{\beta\in\mathbb{Z}^n;\\ (\beta+\overline{\sigma})-\overline{\sigma}\in\{0,1\}^n;\\ |\beta+\overline{\sigma}|=|\xi|}} \prod_{i=1}^{n} h_{\alpha_i+(\beta+\overline{\sigma})_i,\ i} = \sum_{\substack{\nu\in\mathbb{Z}^n;\\ \nu-\overline{\sigma}\in\{0,1\}^n;\\ |\nu|=|\xi|}} \prod_{i=1}^{n} h_{\alpha_i+\nu_i,\ i}.$$

Hence, our above computation becomes

$$\begin{aligned} \sum_{\substack{\beta\in\{0,1\}^n;\\ |\beta|=p}} \prod_{i=1}^{n} h_{\alpha_i+(\beta+\overline{\sigma})_i,\ i} &= \sum_{\substack{\beta\in\mathbb{Z}^n;\\ (\beta+\overline{\sigma})-\overline{\sigma}\in\{0,1\}^n;\\ |\beta+\overline{\sigma}|=|\xi|}} \prod_{i=1}^{n} h_{\alpha_i+(\beta+\overline{\sigma})_i,\ i} \\ &= \sum_{\substack{\nu\in\mathbb{Z}^n;\\ \nu-\overline{\sigma}\in\{0,1\}^n;\\ |\nu|=|\xi|}} \prod_{i=1}^{n} h_{\alpha_i+\nu_i,\ i}. \end{aligned} \tag{124}$$

Forget that we fixed σ. We thus have proved (124) for each permutation $\sigma \in S_n$.

Now, (123) becomes

$$\sum_{\substack{\beta\in\{0,1\}^n;\\ |\beta|=p}} t_{\alpha+\beta} = \sum_{\sigma\in S_n} (-1)^\sigma \underbrace{\sum_{\substack{\beta\in\{0,1\}^n;\\ |\beta|=p}} \prod_{i=1}^{n} h_{\alpha_i+(\beta+\overline{\sigma})_i,\ i}}_{\substack{= \sum\limits_{\substack{\nu\in\mathbb{Z}^n;\\ \nu-\overline{\sigma}\in\{0,1\}^n;\\ |\nu|=|\xi|}} \prod\limits_{i=1}^{n} h_{\alpha_i+\nu_i,\ i} \\ \text{(by (124))}}}$$

$$= \sum_{\sigma\in S_n} (-1)^\sigma \sum_{\substack{\nu\in\mathbb{Z}^n;\\ \nu-\overline{\sigma}\in\{0,1\}^n;\\ |\nu|=|\xi|}} \prod_{i=1}^{n} h_{\alpha_i+\nu_i,\ i}$$

$$= \underbrace{\sum_{\sigma\in S_n} \sum_{\substack{\nu\in\mathbb{Z}^n;\\ \nu-\overline{\sigma}\in\{0,1\}^n;\\ |\nu|=|\xi|}}}_{= \sum\limits_{\substack{\nu\in\mathbb{Z}^n;\\ |\nu|=|\xi|}} \sum\limits_{\substack{\sigma\in S_n;\\ \nu-\overline{\sigma}\in\{0,1\}^n}}} (-1)^\sigma \prod_{i=1}^{n} h_{\alpha_i+\nu_i,\ i}$$

$$= \sum_{\substack{\nu\in\mathbb{Z}^n;\\ |\nu|=|\xi|}} \sum_{\substack{\sigma\in S_n;\\ \nu-\overline{\sigma}\in\{0,1\}^n}} (-1)^\sigma \prod_{i=1}^{n} h_{\alpha_i+\nu_i,\ i}$$

$$= \sum_{\substack{\nu\in\mathbb{Z}^n;\\ |\nu|=|\xi|}} \left(\sum_{\substack{\sigma\in S_n;\\ \nu-\overline{\sigma}\in\{0,1\}^n}} (-1)^\sigma \right) \prod_{i=1}^{n} h_{\alpha_i+\nu_i,\ i}. \tag{125}$$

Now, if $\nu \in \mathbb{Z}^n$ is an n-tuple satisfying $|\nu| = |\xi|$, then

$$\sum_{\substack{\sigma\in S_n;\\ \nu-\overline{\sigma}\in\{0,1\}^n}} (-1)^\sigma = \det\left(([\nu_i \trianglerighteq j])_{i,j\in[n]}\right) \qquad \text{(by Lemma 4.8)}$$

$$= \sum_{\substack{\sigma\in S_n;\\ \nu=\xi\circ\sigma}} (-1)^\sigma \qquad \text{(by Lemma 4.6)}\,. \tag{126}$$

Thus, (125) becomes

$$\sum_{\substack{\beta\in\{0,1\}^n;\\ |\beta|=p}} t_{\alpha+\beta} = \sum_{\substack{\nu\in\mathbb{Z}^n;\\ |\nu|=|\xi|}} \underbrace{\left(\sum_{\substack{\sigma\in S_n;\\ \nu-\overline{\sigma}\in\{0,1\}^n}} (-1)^\sigma \right)}_{\substack{= \sum\limits_{\substack{\sigma\in S_n;\\ \nu=\xi\circ\sigma}} (-1)^\sigma \\ \text{(by (126))}}} \prod_{i=1}^{n} h_{\alpha_i+\nu_i,\ i}$$

$$= \sum_{\substack{\nu\in\mathbb{Z}^n;\\ |\nu|=|\xi|}} \left(\sum_{\substack{\sigma\in S_n;\\ \nu=\xi\circ\sigma}} (-1)^\sigma \right) \prod_{i=1}^{n} h_{\alpha_i+\nu_i,\ i}$$

$$= \sum_{\substack{\nu\in\mathbb{Z}^n;\\ |\nu|=|\xi|}} \sum_{\substack{\sigma\in S_n;\\ \nu=\xi\circ\sigma}} (-1)^\sigma \prod_{i=1}^{n} h_{\alpha_i+\nu_i,\ i}. \tag{127}$$

On the other hand,

$$\operatorname{rowdet}\left(\left(h_{\alpha_i+\xi_j,\ i} \right)_{i,j\in[n]} \right) = \sum_{\sigma\in S_n} (-1)^\sigma \prod_{i=1}^{n} h_{\alpha_i+\xi_{\sigma(i)},\ i} \tag{128}$$

(by (120), applied to $a_{i,j} = h_{\alpha_i+\xi_j,\ i}$). However, for each $\sigma \in S_n$, the n-tuple $\xi \circ \sigma$ belongs to $\mathbb{Z}^n$ and satisfies $|\xi \circ \sigma| = |\xi|$ (by Proposition 2.5). In other words, for each $\sigma \in S_n$, the n-tuple $\xi \circ \sigma$ is a $\nu \in \mathbb{Z}^n$ satisfying $|\nu| = |\xi|$. Hence, any sum ranging over all $\sigma \in S_n$ can be split according to the value of $\xi \circ \sigma$. In other words, we have the following equality of summation signs:

$$\sum_{\sigma\in S_n} = \sum_{\substack{\nu\in\mathbb{Z}^n;\\ |\nu|=|\xi|}} \underbrace{\sum_{\substack{\sigma\in S_n;\\ \xi\circ\sigma=\nu}}}_{= \sum\limits_{\substack{\sigma\in S_n;\\ \nu=\xi\circ\sigma}}} = \sum_{\substack{\nu\in\mathbb{Z}^n;\\ |\nu|=|\xi|}} \sum_{\substack{\sigma\in S_n;\\ \nu=\xi\circ\sigma}}.$$

Thus, (128) becomes

$$\begin{aligned}\operatorname{rowdet}\left(\left(h_{\alpha_i+\xi_j,\ i}\right)_{i,j\in[n]}\right) &= \underbrace{\sum_{\sigma\in S_n}}_{} (-1)^\sigma \prod_{i=1}^{n} h_{\alpha_i+\xi_{\sigma(i)},\ i} \\ &= \sum_{\substack{\nu\in\mathbb{Z}^n;\\ |\nu|=|\xi|}} \sum_{\substack{\sigma\in S_n;\\ \nu=\xi\circ\sigma}} \\ &= \sum_{\substack{\nu\in\mathbb{Z}^n;\\ |\nu|=|\xi|}} \sum_{\substack{\sigma\in S_n;\\ \nu=\xi\circ\sigma}} (-1)^\sigma \prod_{i=1}^{n} \underbrace{h_{\alpha_i+\xi_{\sigma(i)},\ i}}_{\substack{=h_{\alpha_i+\nu_i,\ i}\\ \text{(since } \xi_{\sigma(i)}=\nu_i \\ \text{(because } \nu=\xi\circ\sigma=\left(\xi_{\sigma(1)},\xi_{\sigma(2)},\ldots,\xi_{\sigma(n)}\right) \\ \text{(by Definition 2.4) and therefore } \nu_i=\xi_{\sigma(i)}))}} \\ &= \sum_{\substack{\nu\in\mathbb{Z}^n;\\ |\nu|=|\xi|}} \sum_{\substack{\sigma\in S_n;\\ \nu=\xi\circ\sigma}} (-1)^\sigma \prod_{i=1}^{n} h_{\alpha_i+\nu_i,\ i}.\end{aligned}$$

Comparing this with (127), we obtain

$$\sum_{\substack{\beta\in\{0,1\}^n;\\ |\beta|=p}} t_{\alpha+\beta} = \operatorname{rowdet}\left(\left(h_{\alpha_i+\xi_j,\ i}\right)_{i,j\in[n]}\right).$$

This proves Theorem 4.1. □

4.3. Corollaries

Several corollaries can be obtained from Theorem 4.1 in the same way as we did above with Theorem 2.1. Here is an analogue of Corollary 3.1:

Corollary 4.9. Let $n \in \mathbb{N}$ and $p \in \{0, 1, \ldots, n\}$. Let $q = n - p$. Let $h_{k,\ i}$ be an element of R for all $k \in \mathbb{Z}$ and $i \in [n]$. Assume that

$$h_{k,\ i} = 0 \qquad \text{for all } k < 0 \text{ and } i > q. \tag{129}$$

For any $\alpha \in \mathbb{Z}^n$, we define

$$t_\alpha := \operatorname{rowdet}\left(\left(h_{\alpha_i+j,\ i}\right)_{i,j\in[n]}\right) \in R.$$

Let $\alpha \in \mathbb{Z}^n$. Assume that

$$\alpha_i < -q \qquad \text{for each } i > q. \tag{130}$$

Then,

$$\sum_{\substack{\beta\in\{0,1\}^n;\\ |\beta|=p}} t_{\alpha+\beta} = \operatorname{rowdet}\left(\left(h_{\alpha_i+j,\ i}\right)_{i,j\in[q]}\right) \cdot \operatorname{rowdet}\left(\left(h_{\alpha_{q+i}+q+j+1,\ q+i}\right)_{i,j\in[p]}\right).$$

This can be derived from Theorem 4.1 using the following lemma (which generalizes both our Lemma 3.2 and [Grinbe15, Exercise 6.29], although it is stated in a rather different way):

Lemma 4.10. Let $n \in \mathbb{N}$. Let $A = (a_{i,j})_{i,j\in[n]} \in R^{n\times n}$ be an $n \times n$-matrix. Let $k \in \{0, 1, \ldots, n\}$. Assume that

$$a_{i,j} = 0 \qquad \text{for every } i \in \{k+1, k+2, \ldots, n\} \text{ and } j \in \{1, 2, \ldots, k\}.$$

Then,

$$\operatorname{rowdet} A = \operatorname{rowdet}\left((a_{i,j})_{i,j\in[k]}\right) \cdot \operatorname{rowdet}\left((a_{k+i,k+j})_{i,j\in[n-k]}\right).$$

Example 4.11. For $n = 4$ and $k = 2$, Lemma 4.10 is saying that

$$\operatorname{rowdet}\begin{pmatrix} a_{1,1} & a_{1,2} & a_{1,3} & a_{1,4} \\ a_{2,1} & a_{2,2} & a_{2,3} & a_{2,4} \\ 0 & 0 & a_{3,3} & a_{3,4} \\ 0 & 0 & a_{4,3} & a_{4,4} \end{pmatrix} = \operatorname{rowdet}\begin{pmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{pmatrix} \cdot \operatorname{rowdet}\begin{pmatrix} a_{3,3} & a_{3,4} \\ a_{4,3} & a_{4,4} \end{pmatrix}.$$

Lemma 4.10 is a straightforward generalization of [Grinbe15, Exercise 6.29] to the case of an arbitrary ring R (that is not necessarily commutative)[45]. Unsurprisingly, it can be proved in more or less the same way as [Grinbe15, Exercise 6.29], as long as the requisite attention is paid to the order of factors in products[46]. A reader could easily perform the necessary modifications to obtain a proof of Lemma 4.10. We shall,

[45] This might not be immediately obvious from the statements of Lemma 4.10 and [Grinbe15, Exercise 6.29], since the notations used are rather different. However, when the ring R is commutative, Lemma 4.10 and [Grinbe15, Exercise 6.29] are the same fact viewed from different angles. Indeed, if n, A, $a_{i,j}$ and k are as in Lemma 4.10, then $A = \begin{pmatrix} U & V \\ 0_{(n-k)\times k} & W \end{pmatrix}$ for the three matrices

$$U = (a_{i,j})_{i,j\in[k]}, \qquad V = (a_{i,k+j})_{i\in[k],\ j\in[n-k]} \qquad \text{and} \qquad W = (a_{k+i,k+j})_{i,j\in[n-k]}.$$

Thus, applying [Grinbe15, Exercise 6.29] to k, $n-k$, U, V and W instead of n, m, A, B and D yields that

$$\det A = \det U \cdot \det W = \det\left((a_{i,j})_{i,j\in[k]}\right) \cdot \det\left((a_{k+i,k+j})_{i,j\in[n-k]}\right)$$

in this case. Since the determinant of a matrix over a commutative ring R equals its row-determinant, we can rewrite this as

$$\operatorname{rowdet} A = \operatorname{rowdet}\left((a_{i,j})_{i,j\in[k]}\right) \cdot \operatorname{rowdet}\left((a_{k+i,k+j})_{i,j\in[n-k]}\right).$$

Thus, our Lemma 4.10 follows from [Grinbe15, Exercise 6.29] in the case when R is commutative. The converse implication is equally easy to see.

[46] This means, in particular, that some products in [Grinbe15, solution to Exercise 6.29] need to be reordered (and the finite product notation $\prod_{i\in I} a_i$ needs to be understood as the product of the a_i

however, give a different proof of Lemma 4.10, which proceeds from scratch using a combinatorial lemma. (This will obviously yield a new solution to [Grinbe15, Exercise 6.29].)

Before we state this lemma, we need to recall two notations concerning permutations:

- If $n \in \mathbb{N}$ and if $\sigma \in S_n$, then an *inversion* of σ means a pair (i, j) of integers satisfying $1 \leq i < j \leq n$ and $\sigma(i) > \sigma(j)$.
- If $n \in \mathbb{N}$ and if $\sigma \in S_n$, then the *length* of σ is defined to be the number of inversions of σ. This length is denoted by $\ell(\sigma)$.

Recall that the sign $(-1)^\sigma$ of a permutation $\sigma \in S_n$ is defined to be $(-1)^{\ell(\sigma)}$.
Now, we can state our combinatorial lemma:

Lemma 4.12. Let $n \in \mathbb{N}$. Let I be a subset of $\{1, 2, \ldots, n\}$. Let $k = |I|$. Let $(a_1, a_2, \ldots, a_k)$ be the list of all elements of I in increasing order (with no repetitions). Let $(b_1, b_2, \ldots, b_{n-k})$ be the list of all elements of $\{1, 2, \ldots, n\} \setminus I$ in increasing order (with no repetitions). Let $\alpha \in S_k$ and $\beta \in S_{n-k}$. Then:

(a) There exists a unique $\sigma \in S_n$ satisfying

$$(\sigma(1), \sigma(2), \ldots, \sigma(n)) = \left(a_{\alpha(1)}, a_{\alpha(2)}, \ldots, a_{\alpha(k)}, b_{\beta(1)}, b_{\beta(2)}, \ldots, b_{\beta(n-k)}\right).$$

Denote this σ by $\sigma_{I,\alpha,\beta}$.

(b) Let $\sum I$ denote the sum of all elements of I. (Thus, $\sum I = \sum\limits_{i \in I} i$.) We have

$$\ell(\sigma_{I,\alpha,\beta}) = \ell(\alpha) + \ell(\beta) + \sum I - (1 + 2 + \cdots + k)$$

and

$$(-1)^{\sigma_{I,\alpha,\beta}} = (-1)^\alpha \cdot (-1)^\beta \cdot (-1)^{\sum I - (1+2+\cdots+k)}.$$

(c) Forget that we fixed α and β. We thus have defined an element $\sigma_{I,\alpha,\beta} \in S_n$ for every $\alpha \in S_k$ and every $\beta \in S_{n-k}$. The map

$$\begin{aligned} S_k \times S_{n-k} &\to \{\tau \in S_n \mid \tau(\{1, 2, \ldots, k\}) = I\}, \\ (\alpha, \beta) &\mapsto \sigma_{I,\alpha,\beta} \end{aligned}$$

is well-defined and a bijection.

in the order of increasing i). Furthermore, instead of using [Grinbe15, Theorem 6.82 **(a)**], we need to use the fact that any $n \times n$-matrix $A = (a_{i,j})_{i,j \in [n]}$ with $n > 0$ satisfies

$$\operatorname{rowdet} A = \sum_{q=1}^{n} (-1)^{n+q} \operatorname{rowdet}(A_{\sim n, \sim q}) \cdot a_{n,q}.$$

(This generalizes the $p = n$ case of [Grinbe15, Theorem 6.82 **(a)**] to the case of noncommutative R. The general case of [Grinbe15, Theorem 6.82 **(a)**] cannot be generalized to noncommutative R, but fortunately we only need the $p = n$ case in our proof.)

Proof of Lemma 4.12. Lemma 4.12 is precisely [Grinbe15, Exercise 5.14]. □

We will use a particular case of Lemma 4.12, which has a slightly simpler form:

Lemma 4.13. Let $n \in \mathbb{N}$. Let $k \in \{0, 1, \ldots, n\}$. Let $\alpha \in S_k$ and $\beta \in S_{n-k}$. Then:

(a) There exists a unique $\sigma \in S_n$ satisfying

$$\begin{aligned}&(\sigma(1), \sigma(2), \ldots, \sigma(n))\\ &= (\alpha(1),\ \alpha(2),\ \ldots,\ \alpha(k),\ k+\beta(1),\ k+\beta(2),\ \ldots,\ k+\beta(n-k)).\end{aligned}$$

Denote this σ by $\alpha \oplus \beta$.

(b) We have

$$(-1)^{\alpha\oplus\beta} = (-1)^{\alpha} \cdot (-1)^{\beta}.$$

(c) Forget that we fixed α and β. We thus have defined an element $\alpha \oplus \beta \in S_n$ for every $\alpha \in S_k$ and every $\beta \in S_{n-k}$. The map

$$\begin{aligned}S_k \times S_{n-k} &\to \{\tau \in S_n \mid \tau([k]) = [k]\},\\ (\alpha, \beta) &\mapsto \alpha \oplus \beta\end{aligned}$$

is well-defined and a bijection.

Proof of Lemma 4.13. From $k \in \{0, 1, \ldots, n\}$, we obtain $0 \leq k \leq n$. The definition of $[n]$ yields $[n] = \{1, 2, \ldots, n\}$. The definition of $[k]$ yields $[k] = \{1, 2, \ldots, k\}$.

Let I be the set $[k]$. Thus, $I = [k] = \{1, 2, \ldots, k\} \subseteq \{1, 2, \ldots, n\}$ (since $k \leq n$). In other words, I is a subset of $\{1, 2, \ldots, n\}$. Moreover, from $I = [k]$, we obtain $|I| = |[k]| = k$ (since $[k]$ is a k-element set), so that $k = |I|$.

Define a k-tuple $(a_1, a_2, \ldots, a_k) \in \mathbb{Z}^k$ by $(a_1, a_2, \ldots, a_k) = (1, 2, \ldots, k)$. Thus,

$$a_i = i \qquad \text{for each } i \in [k]. \tag{131}$$

Define an $(n-k)$-tuple $(b_1, b_2, \ldots, b_{n-k}) \in \mathbb{Z}^{n-k}$ by $(b_1, b_2, \ldots, b_{n-k}) = (k+1, k+2, \ldots, n)$. Thus,

$$(b_1, b_2, \ldots, b_{n-k}) = (k+1, k+2, \ldots, n) = (k+1, k+2, \ldots, k+(n-k))$$

(since $n = k + (n-k)$). Hence,

$$b_i = k + i \qquad \text{for each } i \in [n-k]. \tag{132}$$

We have $I = \{1, 2, \ldots, k\}$ and $1 < 2 < \cdots < k$. Hence, $(1, 2, \ldots, k)$ is the list of all elements of I in increasing order (with no repetitions). In other words, $(a_1, a_2, \ldots, a_k)$ is the list of all elements of I in increasing order (with no repetitions) (since $(a_1, a_2, \ldots, a_k) = (1, 2, \ldots, k)$).

Moreover, we have

$$\begin{aligned}\{1, 2, \ldots, n\} \setminus \underbrace{I}_{=\{1,2,\ldots,k\}} &= \{1, 2, \ldots, n\} \setminus \{1, 2, \ldots, k\}\\ &= \{k+1, k+2, \ldots, n\} \qquad (\text{since } 0 \leq k \leq n)\end{aligned}$$

and $k+1 < k+2 < \cdots < n$. Hence, $(k+1, k+2, \ldots, n)$ is the list of all elements of $\{1, 2, \ldots, n\} \setminus I$ in increasing order (with no repetitions). In other words, $(b_1, b_2, \ldots, b_{n-k})$ is the list of all elements of $\{1, 2, \ldots, n\} \setminus I$ in increasing order (with no repetitions) (since $(b_1, b_2, \ldots, b_{n-k}) = (k+1, k+2, \ldots, n)$).

(a) Let $\alpha \in S_k$ and $\beta \in S_{n-k}$.

We have $\alpha \in S_k$; in other words, α is a permutation of $[k]$ (since S_k is the set of all permutations of $[k]$). Hence, for each $j \in [k]$, we have $\alpha(j) \in [k]$ and thus $a_{\alpha(j)} = \alpha(j)$ (by (131), applied to $i = \alpha(j)$). In other words, we have

$$\left(a_{\alpha(1)}, a_{\alpha(2)}, \ldots, a_{\alpha(k)}\right) = (\alpha(1),\ \alpha(2),\ \ldots,\ \alpha(k)).$$

We have $\beta \in S_{n-k}$; in other words, β is a permutation of $[n-k]$ (since S_{n-k} is the set of all permutations of $[n-k]$). Hence, for each $j \in [n-k]$, we have $\beta(j) \in [n-k]$ and thus $b_{\beta(j)} = k + \beta(j)$ (by (132), applied to $i = \beta(j)$). In other words, we have

$$\left(b_{\beta(1)}, b_{\beta(2)}, \ldots, b_{\beta(n-k)}\right) = (k+\beta(1),\ k+\beta(2),\ \ldots,\ k+\beta(n-k)).$$

Now, let us use the notation $u*v$ for the concatenation of two finite lists u and v. This is defined as follows: If $u = (u_1, u_2, \ldots, u_p)$ and $v = (v_1, v_2, \ldots, v_q)$ are two finite lists, then their *concatenation* $u*v$ is defined to be the finite list $(u_1, u_2, \ldots, u_p, v_1, v_2, \ldots, v_q)$.

Clearly,

$$\begin{aligned}
&\left(a_{\alpha(1)}, a_{\alpha(2)}, \ldots, a_{\alpha(k)}, b_{\beta(1)}, b_{\beta(2)}, \ldots, b_{\beta(n-k)}\right) \\
&= \underbrace{\left(a_{\alpha(1)}, a_{\alpha(2)}, \ldots, a_{\alpha(k)}\right)}_{=(\alpha(1),\ \alpha(2),\ \ldots,\ \alpha(k))} * \underbrace{\left(b_{\beta(1)}, b_{\beta(2)}, \ldots, b_{\beta(n-k)}\right)}_{=(k+\beta(1),\ k+\beta(2),\ \ldots,\ k+\beta(n-k))} \\
&= (\alpha(1),\ \alpha(2),\ \ldots,\ \alpha(k)) * (k+\beta(1),\ k+\beta(2),\ \ldots,\ k+\beta(n-k)) \\
&= (\alpha(1),\ \alpha(2),\ \ldots,\ \alpha(k),\ k+\beta(1),\ k+\beta(2),\ \ldots,\ k+\beta(n-k)). \qquad (133)
\end{aligned}$$

Now, recall that $(a_1, a_2, \ldots, a_k)$ is the list of all elements of I in increasing order (with no repetitions), whereas $(b_1, b_2, \ldots, b_{n-k})$ is the list of all elements of $\{1, 2, \ldots, n\} \setminus I$ in increasing order (with no repetitions). Thus, we can apply Lemma 4.12 **(a)**. We thus conclude that there exists a unique $\sigma \in S_n$ satisfying

$$(\sigma(1), \sigma(2), \ldots, \sigma(n)) = \left(a_{\alpha(1)}, a_{\alpha(2)}, \ldots, a_{\alpha(k)}, b_{\beta(1)}, b_{\beta(2)}, \ldots, b_{\beta(n-k)}\right).$$

In view of (133), we can rewrite this as follows: There exists a unique $\sigma \in S_n$ satisfying

$$\begin{aligned}
&(\sigma(1), \sigma(2), \ldots, \sigma(n)) \\
&= (\alpha(1),\ \alpha(2),\ \ldots,\ \alpha(k),\ k+\beta(1),\ k+\beta(2),\ \ldots,\ k+\beta(n-k)).
\end{aligned}$$

This proves Lemma 4.13 **(a)**.

(b) Let $\alpha \in S_k$ and $\beta \in S_{n-k}$. Then, we have

$$\begin{aligned}
&\left(a_{\alpha(1)}, a_{\alpha(2)}, \ldots, a_{\alpha(k)}, b_{\beta(1)}, b_{\beta(2)}, \ldots, b_{\beta(n-k)}\right) \\
&= (\alpha(1),\ \alpha(2),\ \ldots,\ \alpha(k),\ k+\beta(1),\ k+\beta(2),\ \ldots,\ k+\beta(n-k)). \qquad (134)
\end{aligned}$$

(Indeed, this was already shown in our above proof of Lemma 4.13 **(a)**.)

The permutation $\alpha \oplus \beta$ is the unique $\sigma \in S_n$ satisfying

$$\begin{aligned}&(\sigma(1), \sigma(2), \ldots, \sigma(n)) \\ &= (\alpha(1),\ \alpha(2),\ \ldots,\ \alpha(k),\ k+\beta(1),\ k+\beta(2),\ \ldots,\ k+\beta(n-k))\end{aligned}$$

(by the definition of $\alpha \oplus \beta$). In view of (134), we can rewrite this as follows: The permutation $\alpha \oplus \beta$ is the unique $\sigma \in S_n$ satisfying

$$(\sigma(1), \sigma(2), \ldots, \sigma(n)) = \left(a_{\alpha(1)}, a_{\alpha(2)}, \ldots, a_{\alpha(k)}, b_{\beta(1)}, b_{\beta(2)}, \ldots, b_{\beta(n-k)}\right).$$

However, recall that $(a_1, a_2, \ldots, a_k)$ is the list of all elements of I in increasing order (with no repetitions), whereas $(b_1, b_2, \ldots, b_{n-k})$ is the list of all elements of $\{1, 2, \ldots, n\} \setminus I$ in increasing order (with no repetitions). Thus, we can apply Lemma 4.12. In particular, the permutation $\sigma_{I,\alpha,\beta}$ from Lemma 4.12 **(a)** is well-defined. Consider this permutation $\sigma_{I,\alpha,\beta}$. According to its definition, this permutation $\sigma_{I,\alpha,\beta}$ is the unique $\sigma \in S_n$ satisfying

$$(\sigma(1), \sigma(2), \ldots, \sigma(n)) = \left(a_{\alpha(1)}, a_{\alpha(2)}, \ldots, a_{\alpha(k)}, b_{\beta(1)}, b_{\beta(2)}, \ldots, b_{\beta(n-k)}\right).$$

However, we have already seen that the permutation $\alpha \oplus \beta$ is the unique $\sigma \in S_n$ satisfying

$$(\sigma(1), \sigma(2), \ldots, \sigma(n)) = \left(a_{\alpha(1)}, a_{\alpha(2)}, \ldots, a_{\alpha(k)}, b_{\beta(1)}, b_{\beta(2)}, \ldots, b_{\beta(n-k)}\right).$$

Comparing the preceding two sentences, we thus conclude that $\sigma_{I,\alpha,\beta}$ and $\alpha \oplus \beta$ are equal (since each of them is the unique $\sigma \in S_n$ satisfying

$$(\sigma(1), \sigma(2), \ldots, \sigma(n)) = \left(a_{\alpha(1)}, a_{\alpha(2)}, \ldots, a_{\alpha(k)}, b_{\beta(1)}, b_{\beta(2)}, \ldots, b_{\beta(n-k)}\right)$$

). In other words,

$$\sigma_{I,\alpha,\beta} = \alpha \oplus \beta. \tag{135}$$

Now, let $\sum I$ denote the sum of all elements of the set I. Thus,

$$\begin{aligned}\sum I &= \sum_{i \in I} i = \sum_{i \in \{1,2,\ldots,k\}} i \qquad \left(\text{since } I = \{1, 2, \ldots, k\}\right) \\ &= 1 + 2 + \cdots + k,\end{aligned}$$

so that $\sum I - (1 + 2 + \cdots + k) = 0$. Thus,

$$(-1)^{\sum I - (1+2+\cdots+k)} = (-1)^0 = 1. \tag{136}$$

However, Lemma 4.12 **(b)** yields that we have

$$\ell(\sigma_{I,\alpha,\beta}) = \ell(\alpha) + \ell(\beta) + \sum I - (1 + 2 + \cdots + k)$$

and

$$(-1)^{\sigma_{I,\alpha,\beta}} = (-1)^{\alpha} \cdot (-1)^{\beta} \cdot (-1)^{\sum I - (1+2+\cdots+k)}.$$

The second of these two equalities becomes

$$(-1)^{\sigma_{I,\alpha,\beta}} = (-1)^{\alpha} \cdot (-1)^{\beta} \cdot \underbrace{(-1)^{\sum I - (1+2+\cdots+k)}}_{\substack{=1 \\ \text{(by (136))}}} = (-1)^{\alpha} \cdot (-1)^{\beta}.$$

In view of $\sigma_{I,\alpha,\beta} = \alpha \oplus \beta$, we can rewrite this as $(-1)^{\alpha\oplus\beta} = (-1)^{\alpha} \cdot (-1)^{\beta}$. This proves Lemma 4.13 **(b)**.

(c) Recall that $(a_1, a_2, \ldots, a_k)$ is the list of all elements of I in increasing order (with no repetitions), whereas $(b_1, b_2, \ldots, b_{n-k})$ is the list of all elements of $\{1, 2, \ldots, n\} \setminus I$ in increasing order (with no repetitions).

For every $\alpha \in S_k$ and $\beta \in S_{n-k}$, we define a permutation $\sigma_{I,\alpha,\beta} \in S_n$ as in Lemma 4.12 **(a)**. In our above proof of Lemma 4.13 **(b)**, we have shown that this permutation $\sigma_{I,\alpha,\beta}$ satisfies (135). In other words, we have $\sigma_{I,\alpha,\beta} = \alpha \oplus \beta$ for every $\alpha \in S_k$ and $\beta \in S_{n-k}$. In other words, we have $\sigma_{I,\alpha,\beta} = \alpha \oplus \beta$ for every $(\alpha, \beta) \in S_k \times S_{n-k}$.

Lemma 4.12 **(c)** yields that the map

$$\begin{aligned} S_k \times S_{n-k} &\to \{\tau \in S_n \mid \tau(\{1, 2, \ldots, k\}) = I\}, \\ (\alpha, \beta) &\mapsto \sigma_{I,\alpha,\beta} \end{aligned}$$

is well-defined and a bijection. In other words, the map

$$\begin{aligned} S_k \times S_{n-k} &\to \{\tau \in S_n \mid \tau(\{1, 2, \ldots, k\}) = I\}, \\ (\alpha, \beta) &\mapsto \alpha \oplus \beta \end{aligned}$$

is well-defined and a bijection (because we have $\sigma_{I,\alpha,\beta} = \alpha \oplus \beta$ for every $(\alpha, \beta) \in S_k \times S_{n-k}$). In view of $\{1, 2, \ldots, k\} = [k]$ and $I = [k]$, we can rewrite this as follows: The map

$$\begin{aligned} S_k \times S_{n-k} &\to \{\tau \in S_n \mid \tau([k]) = [k]\}, \\ (\alpha, \beta) &\mapsto \alpha \oplus \beta \end{aligned}$$

is well-defined and a bijection. This proves Lemma 4.13 **(c)**. □

Our proof of Lemma 4.10 will furthermore use a piece of notation that we have already employed in some of our previous proofs:

Convention 4.14. Let us set $\prod_{i=1}^{n} a_i := a_1 a_2 \cdots a_n$ for any $a_1, a_2, \ldots, a_n \in R$. (This is, of course, the usual meaning of the notation $\prod_{i=1}^{n} a_i$ when the ring R is commutative; however, we are now extending it to the case of arbitrary R.)

Using this notation, we can rewrite the definition of a row-determinant in a slicker form: If $(a_{i,j})_{i,j\in[n]} \in R^{n\times n}$ is any $n \times n$-matrix over R (where n is any nonnegative integer), then

$$\begin{aligned}\operatorname{rowdet}\left((a_{i,j})_{i,j\in[n]}\right) &= \sum_{\sigma\in S_n} (-1)^{\sigma} \underbrace{a_{1,\sigma(1)}a_{2,\sigma(2)}\cdots a_{n,\sigma(n)}}_{=\prod_{i=1}^{n} a_{i,\sigma(i)}} \\ &\qquad \left(\text{since } \prod_{i=1}^{n} a_{i,\sigma(i)} \text{ is defined to be } a_{1,\sigma(1)}a_{2,\sigma(2)}\cdots a_{n,\sigma(n)}\right) \\ &\qquad \text{(by the definition of a row-determinant)} \\ &= \sum_{\sigma\in S_n} (-1)^{\sigma} \prod_{i=1}^{n} a_{i,\sigma(i)}. \end{aligned} \tag{137}$$

Proof of Lemma 4.10. From $k \in \{0, 1, \ldots, n\}$, we obtain $0 \leq k \leq n$. The definition of $[k]$ yields $[k] = \{1, 2, \ldots, k\} \subseteq \{1, 2, \ldots, n\}$ (since $k \leq n$). In other words, $[k] \subseteq [n]$ (since $[n] = \{1, 2, \ldots, n\}$). In other words, $[k]$ is a subset of $[n]$.

Lemma 4.13 **(a)** defines a permutation $\alpha \oplus \beta \in S_n$ for each $\alpha \in S_k$ and $\beta \in S_{n-k}$. Lemma 4.13 **(c)** shows that the map

$$\begin{aligned} S_k \times S_{n-k} &\to \{\tau \in S_n \mid \tau([k]) = [k]\}, \\ (\alpha, \beta) &\mapsto \alpha \oplus \beta \end{aligned}$$

is well-defined and a bijection.

Now, we shall show the following:

Claim 1: For each $\alpha \in S_k$ and each $\beta \in S_{n-k}$, we have

$$\prod_{i=1}^{n} a_{i,(\alpha\oplus\beta)(i)} = \left(\prod_{i=1}^{k} a_{i,\alpha(i)}\right)\left(\prod_{i=1}^{n-k} a_{k+i,k+\beta(i)}\right). \tag{138}$$

[*Proof of Claim 1:* Let $\alpha \in S_k$ and $\beta \in S_{n-k}$. Let $\gamma = \alpha \oplus \beta$.

Recall that $\alpha \oplus \beta$ is the unique $\sigma \in S_n$ satisfying

$$\begin{aligned}&(\sigma(1), \sigma(2), \ldots, \sigma(n)) \\ &= (\alpha(1),\ \alpha(2),\ \ldots,\ \alpha(k),\ k+\beta(1),\ k+\beta(2),\ \ldots,\ k+\beta(n-k))\end{aligned}$$

(because this is how $\alpha \oplus \beta$ was defined in Lemma 4.13 **(a)**). In other words, γ is the unique $\sigma \in S_n$ satisfying

$$\begin{aligned}&(\sigma(1), \sigma(2), \ldots, \sigma(n)) \\ &= (\alpha(1),\ \alpha(2),\ \ldots,\ \alpha(k),\ k+\beta(1),\ k+\beta(2),\ \ldots,\ k+\beta(n-k))\end{aligned}$$

(since $\gamma = \alpha \oplus \beta$). Hence, γ is an element of S_n and satisfies

$$\begin{aligned}&(\gamma(1),\ \gamma(2),\ \ldots,\ \gamma(n)) \\ &= (\alpha(1),\ \alpha(2),\ \ldots,\ \alpha(k),\ k+\beta(1),\ k+\beta(2),\ \ldots,\ k+\beta(n-k)).\end{aligned}$$

Thus, in particular, we have

$$\begin{aligned}&(\gamma(1),\ \gamma(2),\ \ldots,\ \gamma(n))\\&=(\alpha(1),\ \alpha(2),\ \ldots,\ \alpha(k),\ k+\beta(1),\ k+\beta(2),\ \ldots,\ k+\beta(n-k)).\end{aligned}$$

In other words, we have

$$(\gamma(i)=\alpha(i) \qquad \text{for each } i\in\{1,2,\ldots,k\}) \tag{139}$$

and

$$(\gamma(i)=k+\beta(i-k) \qquad \text{for each } i\in\{k+1,k+2,\ldots,n\}). \tag{140}$$

Now, let $i\in\{1,2,\ldots,n-k\}$. Thus, $k+i\in\{k+1,k+2,\ldots,k+(n-k)\}=\{k+1,k+2,\ldots,n\}$ (since $k+(n-k)=n$). Hence, (140) (applied to $k+i$ instead of i) yields

$$\gamma(k+i)=k+\beta\left(\underbrace{(k+i)-k}_{=i}\right)=k+\beta(i). \tag{141}$$

Forget that we fixed i. We thus have proved that (141) holds for each $i\in\{1,2,\ldots,n-k\}$.

Now, Convention 4.14 yields

$$\begin{aligned}\prod_{i=1}^{n}a_{i,\gamma(i)}&=a_{1,\gamma(1)}a_{2,\gamma(2)}\cdots a_{n,\gamma(n)}\\&=\left(a_{1,\gamma(1)}a_{2,\gamma(2)}\cdots a_{k,\gamma(k)}\right)\left(a_{k+1,\gamma(k+1)}a_{k+2,\gamma(k+2)}\cdots a_{n,\gamma(n)}\right)\end{aligned}$$

(since $0\le k\le n$). In view of

$$\begin{aligned}a_{1,\gamma(1)}a_{2,\gamma(2)}\cdots a_{k,\gamma(k)}&=\prod_{i=1}^{k}\underbrace{a_{i,\gamma(i)}}_{\substack{=a_{i,\alpha(i)}\\ \text{(since (139) yields } \gamma(i)=\alpha(i))}} \qquad \text{(by Convention 4.14)}\\&=\prod_{i=1}^{k}a_{i,\alpha(i)}\end{aligned}$$

and

$$\begin{aligned}a_{k+1,\gamma(k+1)}a_{k+2,\gamma(k+2)}\cdots a_{n,\gamma(n)}&=a_{k+1,\gamma(k+1)}a_{k+2,\gamma(k+2)}\cdots a_{k+(n-k),\gamma(k+(n-k))}\\&\qquad\qquad\text{(since } n=k+(n-k))\\&=\prod_{i=1}^{n-k}\underbrace{a_{k+i,\gamma(k+i)}}_{\substack{=a_{k+i,k+\beta(i)}\\ \text{(since (141) yields } \gamma(k+i)=k+\beta(i))}}\\&\qquad\qquad\text{(by Convention 4.14)}\\&=\prod_{i=1}^{n-k}a_{k+i,k+\beta(i)},\end{aligned}$$

we can rewrite this as

$$\prod_{i=1}^{n} a_{i,\gamma(i)} = \left(\prod_{i=1}^{k} a_{i,\alpha(i)}\right)\left(\prod_{i=1}^{n-k} a_{k+i,k+\beta(i)}\right).$$

In other words,

$$\prod_{i=1}^{n} a_{i,(\alpha\oplus\beta)(i)} = \left(\prod_{i=1}^{k} a_{i,\alpha(i)}\right)\left(\prod_{i=1}^{n-k} a_{k+i,k+\beta(i)}\right)$$

(since $\gamma = \alpha \oplus \beta$). This proves Claim 1.]

Next, we observe the following:

Claim 2: Let $\sigma \in S_n$ satisfy $\sigma([k]) \neq [k]$. Then,

$$\prod_{i=1}^{n} a_{i,\sigma(i)} = 0.$$

[*Proof of Claim 2:* We have $\sigma \in S_n$. In other words, σ is a permutation of $[n]$ (since S_n is the set of all permutations of $[n]$). In other words, σ is a bijective map from $[n]$ to $[n]$. Thus, the map σ is bijective and therefore injective.

Hence, we have $|\sigma(T)| = |T|$ for any subset T of $[n]$ [47]. Applying this to $T = [k]$, we obtain $|\sigma([k])| = |[k]|$ (since $[k]$ is a subset of $[n]$). In other words, $|[k]| = |\sigma([k])|$. Also, $[k] \neq \sigma([k])$ (since $\sigma([k]) \neq [k]$). Moreover, the sets $[k]$ and $\sigma([k])$ are subsets of $[n]$, and thus are finite (since $[n]$ is finite).

We recall the following elementary fact from set theory: If X and Y are two finite sets satisfying $|X| = |Y|$ and $X \neq Y$, then $X \not\subseteq Y$ [48]. We can apply this to $X = [k]$ and $Y = \sigma([k])$ (since $|[k]| = |\sigma([k])|$ and $[k] \neq \sigma([k])$), and conclude that $[k] \not\subseteq \sigma([k])$. Hence, there exists some $u \in [k]$ such that $u \notin \sigma([k])$. Consider this u.

We have $u \in [k] \subseteq [n]$. Thus, $\sigma^{-1}(u)$ is well-defined (since σ is a bijective map from $[n]$ to $[n]$). Let $v = \sigma^{-1}(u)$. Then, $v = \sigma^{-1}(u) \in [n]$ and $\sigma(v) = u$ (since $v = \sigma^{-1}(u)$). If we had $v \in [k]$, then we would thus have $u = \sigma\left(\underbrace{v}_{\in [k]}\right) \in \sigma([k])$, which would contradict $u \notin \sigma([k])$. Hence, we cannot have $v \in [k]$. Thus, we have $v \notin [k]$. Combining $v \in [n]$ with $v \notin [k]$, we obtain

$$v \in \underbrace{[n]}_{=\{1,2,\ldots,n\}} \setminus \underbrace{[k]}_{=\{1,2,\ldots,k\}} = \{1,2,\ldots,n\} \setminus \{1,2,\ldots,k\} = \{k+1, k+2, \ldots, n\}.$$

[47] *Proof.* Recall the following general fact from set theory: If X and Y are two finite sets, and if $f : X \to Y$ is an injective map, then we have $|f(T)| = |T|$ for any subset T of X. Applying this to $X = [n]$ and $Y = [n]$ and $f = \sigma$, we conclude that $|\sigma(T)| = |T|$ for any subset T of $[n]$ (since σ is injective).

[48] *Proof.* Let X and Y be two finite sets satisfying $|X| = |Y|$ and $X \neq Y$. We must prove that $X \not\subseteq Y$.

Assume the contrary. Thus, $X \subseteq Y$. Hence, $|Y \setminus X| = |Y| - \underbrace{|X|}_{=|Y|} = |Y| - |Y| = 0$, so that $Y \setminus X = \varnothing$. In other words, $Y \subseteq X$. Combining this with $X \subseteq Y$, we obtain $X = Y$. However, this contradicts $X \neq Y$.

This contradiction shows that our assumption was false. Hence, we have $X \not\subseteq Y$, qed.

Moreover, $u \in [k] = \{1, 2, \ldots, k\}$.

However, one of the assumptions of Lemma 4.10 says that

$$a_{i,j} = 0 \qquad \text{for every } i \in \{k+1, k+2, \ldots, n\} \text{ and } j \in \{1, 2, \ldots, k\}.$$

Applying this to $i = v$ and $j = u$, we obtain $a_{v,u} = 0$ (since $v \in \{k+1, k+2, \ldots, n\}$ and $u \in \{1, 2, \ldots, k\}$). In view of $\sigma(v) = u$, we can rewrite this as $a_{v,\sigma(v)} = 0$.

However, $v \in [n] = \{1, 2, \ldots, n\}$. Thus, $a_{v,\sigma(v)}$ is one of the factors of the product $\prod_{i=1}^{n} a_{i,\sigma(i)}$ (namely, the factor for $i = v$). Since we have just shown that $a_{v,\sigma(v)} = 0$, we thus conclude that one of the factors of this product is 0. Thus, the whole product must be 0. In other words, we have $\prod_{i=1}^{n} a_{i,\sigma(i)} = 0$. This proves Claim 2.]

Now, from $A = (a_{i,j})_{i,j\in[n]}$, we obtain

$$\begin{aligned}
\operatorname{rowdet} A &= \operatorname{rowdet}\left((a_{i,j})_{i,j\in[n]}\right) = \sum_{\sigma\in S_n} (-1)^\sigma \prod_{i=1}^{n} a_{i,\sigma(i)} \qquad \text{(by (137))} \\
&= \underbrace{\sum_{\substack{\sigma\in S_n;\\ \sigma([k])=[k]}}}_{= \sum\limits_{\sigma\in\{\tau\in S_n \mid \tau([k])=[k]\}}} (-1)^\sigma \prod_{i=1}^{n} a_{i,\sigma(i)} + \sum_{\substack{\sigma\in S_n;\\ \sigma([k])\neq[k]}} (-1)^\sigma \underbrace{\prod_{i=1}^{n} a_{i,\sigma(i)}}_{\substack{=0\\ \text{(by Claim 2)}}} \\
&\qquad\qquad \left(\begin{array}{c}\text{since each } \sigma\in S_n \text{ satisfies either } \sigma([k]) = [k] \\ \text{or } \sigma([k]) \neq [k] \text{ (but not both at the same time)}\end{array}\right) \\
&= \sum_{\sigma\in\{\tau\in S_n \mid \tau([k])=[k]\}} (-1)^\sigma \prod_{i=1}^{n} a_{i,\sigma(i)} + \underbrace{\sum_{\substack{\sigma\in S_n;\\ \sigma([k])\neq[k]}} (-1)^\sigma \cdot 0}_{=0} \\
&= \sum_{\sigma\in\{\tau\in S_n \mid \tau([k])=[k]\}} (-1)^\sigma \prod_{i=1}^{n} a_{i,\sigma(i)} \\
&= \underbrace{\sum_{(\alpha,\beta)\in S_k\times S_{n-k}}}_{= \sum\limits_{\alpha\in S_k}\sum\limits_{\beta\in S_{n-k}}} (-1)^{\alpha\oplus\beta} \prod_{i=1}^{n} a_{i,(\alpha\oplus\beta)(i)} \\
&\qquad\qquad \left(\begin{array}{c}\text{here, we have substituted } \alpha\oplus\beta \text{ for } \sigma \text{ in the sum,} \\ \text{since the map } S_k\times S_{n-k} \to \{\tau\in S_n \mid \tau([k]) = [k]\}, \\ (\alpha,\beta) \mapsto \alpha\oplus\beta \text{ is a bijection}\end{array}\right) \\
&= \sum_{\alpha\in S_k} \sum_{\beta\in S_{n-k}} \underbrace{(-1)^{\alpha\oplus\beta}}_{\substack{=(-1)^\alpha\cdot(-1)^\beta\\ \text{(by Lemma 4.13 \textbf{(b)})}}} \underbrace{\prod_{i=1}^{n} a_{i,(\alpha\oplus\beta)(i)}}_{\substack{=\left(\prod\limits_{i=1}^{k} a_{i,\alpha(i)}\right)\left(\prod\limits_{i=1}^{n-k} a_{k+i,k+\beta(i)}\right)\\ \text{(by Claim 1)}}} \\
&= \sum_{\alpha\in S_k} \sum_{\beta\in S_{n-k}} (-1)^\alpha\cdot(-1)^\beta \left(\prod_{i=1}^{k} a_{i,\alpha(i)}\right)\left(\prod_{i=1}^{n-k} a_{k+i,k+\beta(i)}\right).
\end{aligned}$$

Comparing this with

$$\begin{aligned}
&\underbrace{\operatorname{rowdet}\left(\left(a_{i,j}\right)_{i,j\in[k]}\right)}_{=\sum_{\sigma\in S_k}(-1)^{\sigma}\prod_{i=1}^{k}a_{i,\sigma(i)}\ \text{(by (137), applied to } k \text{ instead of } n)}\cdot\underbrace{\operatorname{rowdet}\left(\left(a_{k+i,k+j}\right)_{i,j\in[n-k]}\right)}_{=\sum_{\sigma\in S_{n-k}}(-1)^{\sigma}\prod_{i=1}^{n-k}a_{k+i,k+\sigma(i)}\ \text{(by (137), applied to } n-k \text{ and } a_{k+i,k+j} \text{ instead of } n \text{ and } a_{i,j})}\\
&=\underbrace{\left(\sum_{\sigma\in S_k}(-1)^{\sigma}\prod_{i=1}^{k}a_{i,\sigma(i)}\right)}_{=\sum_{\alpha\in S_k}(-1)^{\alpha}\prod_{i=1}^{k}a_{i,\alpha(i)}\ \text{(here, we have renamed the summation index } \sigma \text{ as } \alpha)}\cdot\underbrace{\left(\sum_{\sigma\in S_{n-k}}(-1)^{\sigma}\prod_{i=1}^{n-k}a_{k+i,k+\sigma(i)}\right)}_{=\sum_{\beta\in S_{n-k}}(-1)^{\beta}\prod_{i=1}^{n-k}a_{k+i,k+\beta(i)}\ \text{(here, we have renamed the summation index } \sigma \text{ as } \beta)}\\
&=\left(\sum_{\alpha\in S_k}(-1)^{\alpha}\prod_{i=1}^{k}a_{i,\alpha(i)}\right)\cdot\left(\sum_{\beta\in S_{n-k}}(-1)^{\beta}\prod_{i=1}^{n-k}a_{k+i,k+\beta(i)}\right)\\
&=\sum_{\alpha\in S_k}\sum_{\beta\in S_{n-k}}\underbrace{\left((-1)^{\alpha}\prod_{i=1}^{k}a_{i,\alpha(i)}\right)\cdot\left((-1)^{\beta}\prod_{i=1}^{n-k}a_{k+i,k+\beta(i)}\right)}_{=(-1)^{\alpha}\cdot(-1)^{\beta}\left(\prod_{i=1}^{k}a_{i,\alpha(i)}\right)\left(\prod_{i=1}^{n-k}a_{k+i,k+\beta(i)}\right)\ \text{(because } (-1)^{\beta} \text{ belongs to } \mathbb{Z} \text{ and thus commutes with every element of } R)}\\
&=\sum_{\alpha\in S_k}\sum_{\beta\in S_{n-k}}(-1)^{\alpha}\cdot(-1)^{\beta}\left(\prod_{i=1}^{k}a_{i,\alpha(i)}\right)\left(\prod_{i=1}^{n-k}a_{k+i,k+\beta(i)}\right),
\end{aligned}$$

we obtain

$$\operatorname{rowdet}A=\operatorname{rowdet}\left(\left(a_{i,j}\right)_{i,j\in[k]}\right)\cdot\operatorname{rowdet}\left(\left(a_{k+i,k+j}\right)_{i,j\in[n-k]}\right).$$

This proves Lemma 4.10. □

We can now easily derive Corollary 4.9 from Theorem 4.1:

Proof of Corollary 4.9. We have $q=n-p\in\{0,1,\ldots,n\}$ (since $p\in\{0,1,\ldots,n\}$). From $q=n-p$, we obtain $n-q=p$ and $q+p=n$.

Define an n-tuple $\xi\in\mathbb{Z}^n$ as in Theorem 4.1. Then, Theorem 4.1 yields

$$\sum_{\substack{\beta\in\{0,1\}^n;\\ |\beta|=p}}t_{\alpha+\beta}=\operatorname{rowdet}\left(\left(h_{\alpha_i+\xi_j,\ i}\right)_{i,j\in[n]}\right). \tag{142}$$

However, the definition of ξ yields

$$\begin{aligned}
\xi&=(1,2,\ldots,n)+\left(\underbrace{0,0,\ldots,0}_{n-p\text{ zeroes}},\underbrace{1,1,\ldots,1}_{p\text{ ones}}\right)\\
&=(1,2,\ldots,n-p,n-p+2,n-p+3,\ldots,n+1).
\end{aligned}$$

Thus,

$$\begin{aligned}(\xi_1, \xi_2, \ldots, \xi_n) = \xi &= (1, 2, \ldots, n-p, n-p+2, n-p+3, \ldots, n+1) \\ &= (1, 2, \ldots, q, q+2, q+3, \ldots, n+1) \qquad (\text{since } n-p = q)\,.\end{aligned}$$

In other words, we have

$$\xi_k = k \qquad \text{for each } k \in \{1, 2, \ldots, q\} \tag{143}$$

and

$$\xi_k = k+1 \qquad \text{for each } k \in \{q+1, q+2, \ldots, n\}\,. \tag{144}$$

Now, it is easy to see that

$$h_{\alpha_i+\xi_j,\ i} = 0 \qquad \text{for every } i \in \{q+1, q+2, \ldots, n\} \text{ and } j \in \{1, 2, \ldots, q\}$$

[49]. Therefore, Lemma 4.10 (applied to $a_{i,j} = h_{\alpha_i+\xi_j,\ i}$ and $A = \left(h_{\alpha_i+\xi_j,\ i}\right)_{i,j\in[n]}$ and $k = q$) yields

$$\begin{aligned}&\operatorname{rowdet}\left(\left(h_{\alpha_i+\xi_j,\ i}\right)_{i,j\in[n]}\right) \\ &= \operatorname{rowdet}\left(\left(h_{\alpha_i+\xi_j,\ i}\right)_{i,j\in[q]}\right) \cdot \operatorname{rowdet}\left(\left(h_{\alpha_{q+i}+\xi_{q+j},\ q+i}\right)_{i,j\in[n-q]}\right) \\ &= \operatorname{rowdet}\left(\left(h_{\alpha_i+\xi_j,\ i}\right)_{i,j\in[q]}\right) \cdot \operatorname{rowdet}\left(\left(h_{\alpha_{q+i}+\xi_{q+j},\ q+i}\right)_{i,j\in[p]}\right)\end{aligned} \tag{145}$$

(since $n - q = p$).

However, for each $i \in [q]$ and $j \in [q]$, we have $h_{\alpha_i+\xi_j,\ i} = h_{\alpha_i+j,\ i}$ [50]. Hence,

$$\left(h_{\alpha_i+\xi_j,\ i}\right)_{i,j\in[q]} = \left(h_{\alpha_i+j,\ i}\right)_{i,j\in[q]}\,. \tag{146}$$

Furthermore, for each $i \in [p]$ and $j \in [p]$, we have $h_{\alpha_{q+i}+\xi_{q+j},\ q+i} = h_{\alpha_{q+i}+q+j+1,\ q+i}$ [51]. Hence,

$$\left(h_{\alpha_{q+i}+\xi_{q+j},\ q+i}\right)_{i,j\in[p]} = \left(h_{\alpha_{q+i}+q+j+1,\ q+i}\right)_{i,j\in[p]}\,. \tag{147}$$

In light of (146) and (147), we can rewrite (145) as

$$\begin{aligned}&\operatorname{rowdet}\left(\left(h_{\alpha_i+\xi_j,\ i}\right)_{i,j\in[n]}\right) \\ &= \operatorname{rowdet}\left(\left(h_{\alpha_i+j,\ i}\right)_{i,j\in[q]}\right) \cdot \operatorname{rowdet}\left(\left(h_{\alpha_{q+i}+q+j+1,\ q+i}\right)_{i,j\in[p]}\right).\end{aligned}$$

[49] *Proof.* Let $i \in \{q+1, q+2, \ldots, n\}$ and $j \in \{1, 2, \ldots, q\}$. From $i \in \{q+1, q+2, \ldots, n\}$, we obtain $i \geq q+1 > q$. Hence, (130) yields $\alpha_i < -q$. On the other hand, $j \in \{1, 2, \ldots, q\}$. Hence, applying (143) to $k = j$, we obtain $\xi_j = j \leq q$ (since $j \in \{1, 2, \ldots, q\}$). Thus, $\underbrace{\alpha_i}_{<-q} + \underbrace{\xi_j}_{\leq q} < -q + q = 0$.

Therefore, (129) (applied to $k = \alpha_i + \xi_j$) yields $h_{\alpha_i+\xi_j,\ i} = 0$ (since $i > q$). Qed.

[50] *Proof.* Let $i \in [q]$ and $j \in [q]$. Then, $j \in [q] = \{1, 2, \ldots, q\}$. Hence, (143) (applied to $k = j$) yields $\xi_j = j$. Thus, $h_{\alpha_i+\xi_j,\ i} = h_{\alpha_i+j,\ i}$. Qed.

[51] *Proof.* Let $i \in [p]$ and $j \in [p]$. Then, $j \in [p] = \{1, 2, \ldots, p\}$, so that $q+j \in \{q+1, q+2, \ldots, q+p\} = \{q+1, q+2, \ldots, n\}$ (since $q + p = n$). Thus, (144) (applied to $k = q + j$) yields $\xi_{q+j} = q + j + 1$. Hence, $h_{\alpha_{q+i}+\xi_{q+j},\ q+i} = h_{\alpha_{q+i}+q+j+1,\ q+i}$. Qed.

Hence, (142) yields

$$\sum_{\substack{\beta\in\{0,1\}^n;\\ |\beta|=p}} t_{\alpha+\beta} = \operatorname{rowdet}\left(\left(h_{\alpha_i+\xi_j,\ i}\right)_{i,j\in[n]}\right)$$
$$= \operatorname{rowdet}\left(\left(h_{\alpha_i+j,\ i}\right)_{i,j\in[q]}\right)\cdot \operatorname{rowdet}\left(\left(h_{\alpha_{q+i}+q+j+1,\ q+i}\right)_{i,j\in[p]}\right).$$

This proves Corollary 4.9. □

Next, let us state a counterpart to Corollary 3.5:

Corollary 4.15. Let $n \in \mathbb{N}$ and $p \in \{0,1,\ldots,n\}$. Let $q = n-p$. Let $h_{k,\ i}$ be an element of R for all $k \in \mathbb{Z}$ and $i \in [n]$. Assume that

$$h_{k,\ i} = 0 \qquad \text{for all } k < 0 \text{ and } i > q. \tag{148}$$

For any $m \in \{0,1,\ldots,n\}$ and any $\lambda \in \mathbb{Z}^m$, we define

$$s_\lambda := \operatorname{rowdet}\left(\left(h_{\lambda_i+j-i,\ i}\right)_{i,j\in[m]}\right) \in R.$$

Set

$$e_{p,\ q} := \operatorname{rowdet}\left(\left(h_{1+j-i,\ q+i}\right)_{i,j\in[p]}\right) \in R.$$

Fix an n-tuple $\mu \in \mathbb{Z}^n$. Assume that

$$\mu_i = 0 \qquad \text{for all } i > q. \tag{149}$$

Let $\overline{\mu} = (\mu_1, \mu_2, \ldots, \mu_q)$. Then,

$$s_{\overline{\mu}} \cdot e_{p,\ q} = \sum_{\substack{\beta\in\{0,1\}^n;\\ |\beta|=p}} s_{\mu+\beta}.$$

Proof of Corollary 4.15. Define $t_\alpha \in R$ for each $\alpha \in \mathbb{Z}^n$ as in Corollary 4.9.

Define an n-tuple $\alpha \in \mathbb{Z}^n$ by

$$\alpha = (\mu_1 - 1,\ \mu_2 - 2,\ \ldots,\ \mu_n - n).$$

Thus,

$$\alpha_i = \mu_i - i \qquad \text{for each } i \in [n]. \tag{150}$$

Hence, for each $i \in [n]$ satisfying $i > q$, we have

$$\alpha_i = \underbrace{\mu_i}_{\substack{=0\\ \text{(by (149))}}} - i = 0 - i = -i < -q \qquad (\text{since } i > q).$$

Hence, Corollary 4.9 yields

$$\sum_{\substack{\beta\in\{0,1\}^n;\\ |\beta|=p}} t_{\alpha+\beta} = \operatorname{rowdet}\left(\left(h_{\alpha_i+j,\ i}\right)_{i,j\in[q]}\right)\cdot \operatorname{rowdet}\left(\left(h_{\alpha_{q+i}+q+j+1,\ q+i}\right)_{i,j\in[p]}\right). \tag{151}$$

However, each $\beta\in\mathbb{N}^n$ satisfies

$$t_{\alpha+\beta} = s_{\mu+\beta} \tag{152}$$

[52] Furthermore, we have

$$\operatorname{rowdet}\left(\left(h_{\alpha_i+j,\ i}\right)_{i,j\in[q]}\right) = s_{\overline{\mu}} \tag{153}$$

[52] *Proof.* Let $\beta\in\mathbb{N}^n$. For each $i,j\in[n]$, we have

$$\underbrace{(\alpha+\beta)_i}_{=\alpha_i+\beta_i}+j = \underbrace{\alpha_i}_{\substack{=\mu_i-i\\ \text{(by (150))}}}+\beta_i+j = \mu_i-i+\beta_i+j = \underbrace{\mu_i+\beta_i}_{=(\mu+\beta)_i}+j-i = (\mu+\beta)_i+j-i$$

and therefore

$$h_{(\alpha+\beta)_i+j,\ i} = h_{(\mu+\beta)_i+j-i,\ i}.$$

In other words, we have

$$\left(h_{(\alpha+\beta)_i+j,\ i}\right)_{i,j\in[n]} = \left(h_{(\mu+\beta)_i+j-i,\ i}\right)_{i,j\in[n]}.$$

The definition of $t_{\alpha+\beta}$ yields

$$t_{\alpha+\beta} = \operatorname{rowdet}\left(\underbrace{\left(h_{(\alpha+\beta)_i+j,\ i}\right)_{i,j\in[n]}}_{=\left(h_{(\mu+\beta)_i+j-i,\ i}\right)_{i,j\in[n]}}\right) = \operatorname{rowdet}\left(\left(h_{(\mu+\beta)_i+j-i,\ i}\right)_{i,j\in[n]}\right) = s_{\mu+\beta}$$

(since the definition of $s_{\mu+\beta}$ yields $s_{\mu+\beta} = \operatorname{rowdet}\left(\left(h_{(\mu+\beta)_i+j-i,\ i}\right)_{i,j\in[n]}\right)$). Qed.

[53]. Finally, we have

$$\operatorname{rowdet}\left(\left(h_{\alpha_{q+i}+q+j+1,\ q+i}\right)_{i,j\in[p]}\right) = e_{p,\ q} \tag{156}$$

[54].

---

[53] *Proof.* We have $\overline{\mu} = (\mu_1, \mu_2, \ldots, \mu_q)$; thus,

$$\overline{\mu}_i = \mu_i \qquad \text{for each } i \in [q]\,. \tag{154}$$

The definition of $s_{\overline{\mu}}$ yields

$$s_{\overline{\mu}} = \operatorname{rowdet}\left(\left(h_{\overline{\mu}_i+j-i,\ i}\right)_{i,j\in[q]}\right). \tag{155}$$

We have $q = n - p \in \{0, 1, \ldots, n\}$ (since $p \in \{0, 1, \ldots, n\}$). Thus, $q \leq n$. Therefore, $\{1, 2, \ldots, q\} \subseteq \{1, 2, \ldots, n\}$. In other words, $[q] \subseteq [n]$ (since $[q] = \{1, 2, \ldots, q\}$ and $[n] = \{1, 2, \ldots, n\}$).

For each $i, j \in [q]$, we have $i \in [q] \subseteq [n]$ and thus

$$\underbrace{\alpha_i}_{\substack{=\mu_i - i \\ \text{(by (150))}}} + j = \underbrace{\mu_i}_{\substack{=\overline{\mu}_i \\ \text{(by (154))}}} - i + j = \overline{\mu}_i - i + j = \overline{\mu}_i + j - i$$

and therefore $h_{\alpha_i+j,\ i} = h_{\overline{\mu}_i+j-i,\ i}$. In other words, we have

$$\left(h_{\alpha_i+j,\ i}\right)_{i,j\in[q]} = \left(h_{\overline{\mu}_i+j-i,\ i}\right)_{i,j\in[q]}.$$

Hence,

$$\operatorname{rowdet}\left(\underbrace{\left(h_{\alpha_i+j,\ i}\right)_{i,j\in[q]}}_{=\left(h_{\overline{\mu}_i+j-i,\ i}\right)_{i,j\in[q]}}\right) = \operatorname{rowdet}\left(\left(h_{\overline{\mu}_i+j-i,\ i}\right)_{i,j\in[q]}\right) = s_{\overline{\mu}}$$

(by (155)).

[54] *Proof of (156):* Let $i \in [p]$ and $j \in [p]$. Then, $i \in [p] = \{1, 2, \ldots, p\}$, so that $q + i \in \{q + 1, q + 2, \ldots, q + p\} = \{q + 1, q + 2, \ldots, n\}$ (since $q + p = n$ (because $q = n - p$)). Thus, $q + i \geq q + 1 > q$. Also, $q + i \in \{q + 1, q + 2, \ldots, n\} \subseteq \{1, 2, \ldots, n\} = [n]$. Now, (149) (applied to $q + i$ instead of i) yields $\mu_{q+i} = 0$ (since $q + i > q$). However, (150) (applied to $q + i$ instead of i) yields $\alpha_{q+i} = \underbrace{\mu_{q+i}}_{=0} - (q + i) = -(q + i)$. Thus,

$$\underbrace{\alpha_{q+i}}_{=-(q+i)} + q + j + 1 = -(q + i) + q + j + 1 = 1 + j - i.$$

Thus, $h_{\alpha_{q+i}+q+j+1,\ q+i} = h_{1+j-i,\ q+i}$.

Forget that we fixed i and j. We thus have shown that $h_{\alpha_{q+i}+q+j+1,\ q+i} = h_{1+j-i,\ q+i}$ for all $i \in [p]$ and $j \in [p]$. Thus,

$$\left(h_{\alpha_{q+i}+q+j+1,\ q+i}\right)_{i,j\in[p]} = \left(h_{1+j-i,\ q+i}\right)_{i,j\in[p]}.$$

Hence,

$$\operatorname{rowdet}\left(\left(h_{\alpha_{q+i}+q+j+1,\ q+i}\right)_{i,j\in[p]}\right) = \operatorname{rowdet}\left(\left(h_{1+j-i,\ q+i}\right)_{i,j\in[p]}\right).$$

Using (152), (153) and (156), we can rewrite (151) as

$$\sum_{\substack{\beta\in\{0,1\}^n;\\ |\beta|=p}} s_{\mu+\beta} = s_{\overline{\mu}} \cdot e_{p,\ q}.$$

This proves Corollary 4.15. □

Counterparts to Corollary 3.4 and Corollary 3.6 can be stated as well, but we omit them to save space. (They are trivial consequences of Corollary 4.9 and Corollary 4.15.)

We note that (6) is the particular case of Corollary 4.15 for $R = \Lambda$, $h_{k,\ i} = h_k$ and $\mu = \lambda$.

5. A pre-LR rule?

We have now proved two fairly general determinantal identities – Theorem 2.1 and Theorem 4.1 – and seen some of their consequences. Anyone familiar with symmetric functions will likely view these two identities as two "antipodal" statements, in the sense in which the complete homogeneous symmetric functions are "antipodal" to the elementary symmetric functions.[55] The latter "antipodality" can be understood particularly well by viewing both families of symmetric functions as corner cases of *Schur functions* (see, e.g., [Stanle24, Chapter 7] or [Macdon95, §I.3]). It is thus natural to ask whether our determinantal identities can be viewed as corner cases of something more general, too:

Question 5.1. Is there a common generalization of Theorem 2.1 and Theorem 4.1?

Such a generalization might resemble (perhaps even generalize) the "immaculate Littlewood–Richardson rule" of Berg, Bergeron, Saliola, Serrano and Zabrocki ([BBSSZ15, Theorem 7.3]). Indeed, as we have seen above (in our proof of Proposition 3.8), the "right-Pieri rule" [BBSSZ13, Theorem 3.5] is a particular case of our Theorem 2.1; one can likewise derive a "second right-Pieri rule" (with $E_s = \mathfrak{S}_{(1^s)}$ taking the place of H_s) from our Theorem 4.1. Both of these "right-Pieri rules" are particular cases of the "immaculate Littlewood–Richardson rule". Thus, it is not too outlandish to suspect

On the other hand,

$$e_{p,\ q} = \operatorname{rowdet}\left(\left(h_{1+j-i,\ q+i}\right)_{i,j\in[p]}\right)$$

(by the definition of $e_{p,\ q}$). Comparing these two equalities, we obtain $\operatorname{rowdet}\left(\left(h_{\alpha_{q+i}+q+j+1,\ q+i}\right)_{i,j\in[p]}\right) = e_{p,\ q}$. This proves (156).

[55] The analogy strikes the eye from many directions: The summation signs $\sum_{\substack{\beta\in\mathbb{N}^n;\\ |\beta|=p}}$ and $\sum_{\substack{\beta\in\{0,1\}^n;\\ |\beta|=p}}$ in Theorem 2.1 and Theorem 4.1 are precisely the ones that appear in the definitions of the respective symmetric functions; the determinant $e_{p,\ q}$ in Corollary 4.15 is a rather straightforward generalization of the Jacobi–Trudi determinant for the p-th elementary symmetric function; and so on.

that the latter rule can, too, be viewed as a particular case of a (noncommutative) determinantal identity.

A step in the general direction of such a generalization appears to be the following proposition:

Proposition 5.2. Let $n \in \mathbb{N}$. Let $h_{k,\ i}$ be an element of R for all $k \in \mathbb{Z}$ and $i \in [n]$.

Let B be a finite set of n-tuples $\beta \in \mathbb{Z}^n$. Assume that this set B is invariant under the right S_n-action on $\mathbb{Z}^n$. (This S_n-action was introduced in Definition 2.4.)

For any $\alpha \in \mathbb{Z}^n$, we define

$$t_\alpha := \operatorname{rowdet}\left(\left(h_{\alpha_i+j,\ i}\right)_{i,j\in[n]}\right) \in R.$$

Let $\alpha \in \mathbb{Z}^n$. Then, there exists a family $(\lambda_\gamma)_{\gamma\in\mathbb{Z}^n}$ of coefficients $\lambda_\gamma \in \mathbb{Z}$ such that all but finitely many $\gamma \in \mathbb{Z}^n$ satisfy $\lambda_\gamma = 0$, and such that

$$\sum_{\beta\in B} t_{\alpha+\beta} = \sum_{\gamma\in\mathbb{Z}^n} \lambda_\gamma \operatorname{rowdet}\left(\left(h_{\alpha_i+\gamma_j,\ i}\right)_{i,j\in[n]}\right). \tag{157}$$

Proof of Proposition 5.2 (sketched). Define the ring

$$N := \mathbb{Z}\left\langle X_{k,\ i} \ \mid \ k \in \mathbb{Z} \text{ and } i \in [n]\right\rangle.$$

This is the ring of noncommutative polynomials over $\mathbb{Z}$ in the variables $X_{k,\ i}$ for all $k \in \mathbb{Z}$ and $i \in [n]$.

A noncommutative monomial in N will be called *multilinear* if it has the form $X_{p_1,\ 1}X_{p_2,\ 2}\cdots X_{p_n,\ n}$ for some $p \in \mathbb{Z}^n$. Let N^{mult} denote the $\mathbb{Z}$-linear span of all multilinear monomials in N. The symmetric group S_n acts $\mathbb{Z}$-linearly on this $\mathbb{Z}$-submodule N^{mult} from the right according to the rule

$$\left(X_{p_1,\ 1}X_{p_2,\ 2}\cdots X_{p_n,\ n}\right)\cdot\tau = X_{p_{\tau(1)},\ 1}X_{p_{\tau(2)},\ 2}\cdots X_{p_{\tau(n)},\ n} \tag{158}$$

(for all multilinear monomials $X_{p_1,\ 1}X_{p_2,\ 2}\cdots X_{p_n,\ n}$ and all $\tau \in S_n$).

An element $p \in N^{\mathrm{mult}}$ will be called *antisymmetric* if each $\tau \in S_n$ satisfies $p\cdot\tau = (-1)^\tau p$. Let N^{sign} denote the set of all antisymmetric elements $p \in N^{\mathrm{mult}}$; this is a $\mathbb{Z}$-submodule of N^{mult}.

For each $\alpha \in \mathbb{Z}^n$, we define

$$T_\alpha := \operatorname{rowdet}\left(\left(X_{\alpha_i+j,\ i}\right)_{i,j\in[n]}\right) \in N.$$

It is easy to see that $T_\alpha \in N^{\mathrm{mult}}$ for each $\alpha \in \mathbb{Z}^n$. Hence, $\sum\limits_{\beta\in B} T_\beta \in N^{\mathrm{mult}}$. Set

$$T_B := \sum_{\beta\in B} T_\beta. \tag{159}$$

We shall now show that $T_B \in N^{\mathrm{sign}}$.

[*Proof:* We have $T_B = \sum\limits_{\beta \in B} T_\beta \in N^{\mathrm{mult}}$. It thus remains to show that T_B is antisymmetric, i.e., that each $\tau \in S_n$ satisfies $T_B \cdot \tau = (-1)^\tau T_B$.

Let $\tau \in S_n$ be arbitrary. For each $\beta \in \mathbb{Z}^n$, we have

$$\begin{aligned} T_\beta &= \operatorname{rowdet}\left(\left(X_{\beta_i + j,\ i} \right)_{i,j \in [n]} \right) \qquad \text{(by the definition of } T_\beta\text{)} \\ &= \sum_{\sigma \in S_n} (-1)^\sigma X_{\beta_1 + \sigma(1),\ 1} X_{\beta_2 + \sigma(2),\ 2} \cdots X_{\beta_n + \sigma(n),\ n} \end{aligned} \tag{160}$$

and thus

$$\begin{aligned} & T_\beta \cdot \tau \\ &= \left(\sum_{\sigma \in S_n} (-1)^\sigma X_{\beta_1 + \sigma(1),\ 1} X_{\beta_2 + \sigma(2),\ 2} \cdots X_{\beta_n + \sigma(n),\ n} \right) \cdot \tau \\ &= \sum_{\sigma \in S_n} (-1)^\sigma \underbrace{\left(X_{\beta_1 + \sigma(1),\ 1} X_{\beta_2 + \sigma(2),\ 2} \cdots X_{\beta_n + \sigma(n),\ n} \right) \cdot \tau}_{\substack{= X_{\beta_{\tau(1)} + \sigma(\tau(1)),\ 1} X_{\beta_{\tau(2)} + \sigma(\tau(2)),\ 2} \cdots X_{\beta_{\tau(n)} + \sigma(\tau(n)),\ n} \\ \text{(by (158))}}} \\ &= \sum_{\sigma \in S_n} (-1)^\sigma X_{\beta_{\tau(1)} + \sigma(\tau(1)),\ 1} X_{\beta_{\tau(2)} + \sigma(\tau(2)),\ 2} \cdots X_{\beta_{\tau(n)} + \sigma(\tau(n)),\ n} \\ &= \sum_{\sigma \in S_n} (-1)^\sigma X_{(\beta \cdot \tau)_1 + (\sigma \circ \tau)(1),\ 1} X_{(\beta \cdot \tau)_2 + (\sigma \circ \tau)(2),\ 2} \cdots X_{(\beta \cdot \tau)_n + (\sigma \circ \tau)(n),\ n} \\ & \qquad \left(\text{since } \beta_{\tau(i)} = (\beta \cdot \tau)_i \text{ and } \sigma(\tau(i)) = (\sigma \circ \tau)(i) \text{ for each } i \right) \\ &= \sum_{\sigma \in S_n} \underbrace{(-1)^{\sigma \circ \tau^{-1}}}_{= (-1)^\sigma (-1)^\tau} X_{(\beta \cdot \tau)_1 + \sigma(1),\ 1} X_{(\beta \cdot \tau)_2 + \sigma(2),\ 2} \cdots X_{(\beta \cdot \tau)_n + \sigma(n),\ n} \\ & \qquad \text{(here, we have substituted } \sigma \text{ for } \sigma \circ \tau \text{ in the sum)} \\ &= (-1)^\tau \underbrace{\sum_{\sigma \in S_n} (-1)^\sigma X_{(\beta \cdot \tau)_1 + \sigma(1),\ 1} X_{(\beta \cdot \tau)_2 + \sigma(2),\ 2} \cdots X_{(\beta \cdot \tau)_n + \sigma(n),\ n}}_{\substack{= T_{\beta \cdot \tau} \\ \text{(by (160), applied to } \beta \cdot \tau \text{ instead of } \beta\text{)}}} \\ &= (-1)^\tau \cdot T_{\beta \cdot \tau}. \end{aligned} \tag{161}$$

Now, from (159), we obtain

$$\begin{aligned} T_B \cdot \tau &= \left(\sum_{\beta \in B} T_\beta \right) \cdot \tau = \sum_{\beta \in B} \underbrace{T_\beta \cdot \tau}_{\substack{= (-1)^\tau \cdot T_{\beta \cdot \tau} \\ \text{(by (161))}}} = \sum_{\beta \in B} (-1)^\tau \cdot T_{\beta \cdot \tau} = (-1)^\tau \cdot \sum_{\beta \in B} T_{\beta \cdot \tau} \\ &= (-1)^\tau \cdot \underbrace{\sum_{\beta \in B} T_\beta}_{= T_B} \qquad \left(\begin{array}{c} \text{here, we have substituted } \beta \text{ for } \beta \cdot \tau \text{ in the sum,} \\ \text{since the map } B \to B,\ \beta \mapsto \beta \cdot \tau \text{ is a bijection} \\ \text{(because } B \text{ is invariant under the } S_n\text{-action)} \end{array} \right) \\ &= (-1)^\tau T_B. \end{aligned}$$

This completes our proof of $T_B \in N^{\mathrm{sign}}$.]

On the other hand, we claim the following:

Claim 1: Each $p \in N^{\mathrm{sign}}$ is a $\mathbb{Z}$-linear combination of the row-determinants $\operatorname{rowdet}\left(\left(X_{\gamma_j,\ i}\right)_{i,j\in[n]}\right)$ with $\gamma \in \mathbb{Z}^n$.

Before we prove this, let us note that all these row-determinants $\operatorname{rowdet}\left(\left(X_{\gamma_j,\ i}\right)_{i,j\in[n]}\right)$ actually belong to N^{sign} (since each $\gamma \in \mathbb{Z}^n$ satisfies

$$\begin{aligned}\operatorname{rowdet}\left(\left(X_{\gamma_j,\ i}\right)_{i,j\in[n]}\right) &= \sum_{\sigma\in S_n} (-1)^\sigma \underbrace{X_{\gamma_{\sigma(1)},\ 1} X_{\gamma_{\sigma(2)},\ 2} \cdots X_{\gamma_{\sigma(n)},\ n}}_{\substack{=\left(X_{\gamma_1,\ 1} X_{\gamma_2,\ 2} \cdots X_{\gamma_n,\ n}\right)\cdot\sigma \\ \text{(by (158))}}} \\ &= \sum_{\sigma\in S_n} (-1)^\sigma \left(X_{\gamma_1,\ 1} X_{\gamma_2,\ 2} \cdots X_{\gamma_n,\ n}\right)\cdot\sigma,\end{aligned}$$

which easily yields that $\operatorname{rowdet}\left(\left(X_{\gamma_j,\ i}\right)_{i,j\in[n]}\right) \in N^{\mathrm{sign}}$); thus, Claim 1 shows that these row-determinants span the $\mathbb{Z}$-module N^{sign}. However, we will not need this.

[*Proof of Claim 1:* Let $p \in N^{\mathrm{sign}}$. We shall show that p is a $\mathbb{Z}$-linear combination of the row-determinants $\operatorname{rowdet}\left(\left(X_{\gamma_j,\ i}\right)_{i,j\in[n]}\right)$.

We know that p is antisymmetric (since $p \in N^{\mathrm{sign}}$). In other words, each $\sigma \in S_n$ satisfies

$$p\cdot\sigma = (-1)^\sigma p. \tag{162}$$

On the other hand, $p \in N^{\mathrm{sign}} \subseteq N^{\mathrm{mult}}$. Hence, we can write p as a $\mathbb{Z}$-linear combination of multilinear monomials (by the definition of N^{mult}). In other words,

$$p = \sum_{\alpha\in\mathbb{Z}^n} \mu_\alpha X_{\alpha_1,\ 1} X_{\alpha_2,\ 2} \cdots X_{\alpha_n,\ n} \tag{163}$$

for some scalars $\mu_\alpha \in \mathbb{Z}$ (almost all of them 0). These scalars μ_α must furthermore satisfy

$$\mu_{\alpha\cdot\sigma} = (-1)^\sigma \cdot \mu_\alpha \qquad \text{for all } \alpha \in \mathbb{Z}^n \text{ and all } \sigma \in S_n \tag{164}$$

(by comparing coefficients in (162)). Hence, we have $\mu_\alpha = 0$ for any n-tuple $\alpha \in \mathbb{Z}^n$ that has at least two equal entries (because for any such α, there exists some transposition $\sigma \in S_n$ such that $\alpha\cdot\sigma = \alpha$, so that the equality (164) simplifies to $\mu_\alpha = \underbrace{(-1)^\sigma}_{=-1}\cdot\mu_\alpha =$

$-\mu_\alpha$, and therefore we have $\mu_\alpha = 0$). Therefore, (163) simplifies to

$$\begin{aligned}
p &= \sum_{\substack{\alpha\in\mathbb{Z}^n; \\ \text{all entries of } \alpha \text{ are distinct}}} \mu_\alpha X_{\alpha_1,\ 1} X_{\alpha_2,\ 2} \cdots X_{\alpha_n,\ n} \\
&= \sum_{\substack{\gamma\in\mathbb{Z}^n; \\ \gamma_1<\gamma_2<\cdots<\gamma_n}} \sum_{\sigma\in S_n} \underbrace{\mu_{\gamma\cdot\sigma}}_{\substack{=(-1)^\sigma\mu_\gamma \\ \text{(by (164))}}} X_{\gamma_{\sigma(1)},\ 1} X_{\gamma_{\sigma(2)},\ 2} \cdots X_{\gamma_{\sigma(n)},\ n} \\
&\qquad \left(\begin{array}{c} \text{here, we have split the sum according to the } n\text{-tuple } \gamma \\ \text{obtained by sorting the } n\text{-tuple } \alpha \text{ in increasing order} \end{array} \right) \\
&= \sum_{\substack{\gamma\in\mathbb{Z}^n; \\ \gamma_1<\gamma_2<\cdots<\gamma_n}} \sum_{\sigma\in S_n} (-1)^\sigma \mu_\gamma X_{\gamma_{\sigma(1)},\ 1} X_{\gamma_{\sigma(2)},\ 2} \cdots X_{\gamma_{\sigma(n)},\ n} \\
&= \sum_{\substack{\gamma\in\mathbb{Z}^n; \\ \gamma_1<\gamma_2<\cdots<\gamma_n}} \mu_\gamma \underbrace{\sum_{\sigma\in S_n} (-1)^\sigma X_{\gamma_{\sigma(1)},\ 1} X_{\gamma_{\sigma(2)},\ 2} \cdots X_{\gamma_{\sigma(n)},\ n}}_{=\operatorname{rowdet}\left(\left(X_{\gamma_j,\ i}\right)_{i,j\in[n]}\right)} \\
&= \sum_{\substack{\gamma\in\mathbb{Z}^n; \\ \gamma_1<\gamma_2<\cdots<\gamma_n}} \mu_\gamma \operatorname{rowdet}\left(\left(X_{\gamma_j,\ i}\right)_{i,j\in[n]}\right).
\end{aligned}$$

This equality shows that p is a $\mathbb{Z}$-linear combination of the row-determinants $\operatorname{rowdet}\left(\left(X_{\gamma_j,\ i}\right)_{i,j\in[n]}\right)$. This completes our proof of Claim 1.]

Now, recall that $T_B \in N^{\text{sign}}$. Hence, Claim 1 shows that T_B is a $\mathbb{Z}$-linear combination of the row-determinants $\operatorname{rowdet}\left(\left(X_{\gamma_j,\ i}\right)_{i,j\in[n]}\right)$ with $\gamma \in \mathbb{Z}^n$. In other words, there exists a family $(\lambda_\gamma)_{\gamma\in\mathbb{Z}^n}$ of coefficients $\lambda_\gamma \in \mathbb{Z}$ such that all but finitely many $\gamma \in \mathbb{Z}^n$ satisfy $\lambda_\gamma = 0$, and such that

$$T_B = \sum_{\gamma\in\mathbb{Z}^n} \lambda_\gamma \operatorname{rowdet}\left(\left(X_{\gamma_j,\ i}\right)_{i,j\in[n]}\right). \tag{165}$$

Consider this family $(\lambda_\gamma)_{\gamma\in\mathbb{Z}^n}$. We shall now show that this family also satisfies (157).

Indeed, consider the ring homomorphism $f : N \to R$ that sends each $X_{k,\ i}$ to $h_{\alpha_i+k,\ i}$. (This clearly exists by the universal property of the free $\mathbb{Z}$-algebra N.) For each $\beta \in \mathbb{Z}^n$, we have

$$\begin{aligned}
f(T_\beta) &= f\left(\operatorname{rowdet}\left(\left(X_{\beta_i+j,\ i}\right)_{i,j\in[n]}\right)\right) && \text{(by the definition of } T_\beta) \\
&= \operatorname{rowdet}\left(\left(f\left(X_{\beta_i+j,\ i}\right)\right)_{i,j\in[n]}\right) && \text{(since } f \text{ is a ring homomorphism)} \\
&= \operatorname{rowdet}\left(\left(h_{(\alpha+\beta)_i+j,\ i}\right)_{i,j\in[n]}\right) \\
&\qquad \left(\begin{array}{c} \text{because we have } f\left(X_{\beta_i+j,\ i}\right) = h_{\alpha_i+\beta_i+j,\ i} = h_{(\alpha+\beta)_i+j,\ i} \\ \text{for all } i \in [n] \text{ and } j \in [n] \end{array} \right) \\
&= t_{\alpha+\beta}
\end{aligned} \tag{166}$$

(by the definition of $t_{\alpha+\beta}$). Now, applying the homomorphism f to both sides of the equality (159), we obtain

$$f\left(T_B\right) = f\left(\sum_{\beta\in B} T_\beta\right) = \sum_{\beta\in B} \underbrace{f\left(T_\beta\right)}_{\substack{=t_{\alpha+\beta} \\ \text{(by (166))}}} \qquad \text{(since } f \text{ is a ring homomorphism)}$$
$$= \sum_{\beta\in B} t_{\alpha+\beta}.$$

Hence, applying the homomorphism f to both sides of (165), we obtain

$$\sum_{\beta\in B} t_{\alpha+\beta} = \sum_{\gamma\in\mathbb{Z}^n} \lambda_\gamma \underbrace{f\left(\operatorname{rowdet}\left(\left(X_{\gamma_j,\ i}\right)_{i,j\in[n]}\right)\right)}_{\substack{=\operatorname{rowdet}\left(\left(f\left(X_{\gamma_j,\ i}\right)\right)_{i,j\in[n]}\right) \\ \text{(since } f \text{ is a ring homomorphism)}}}$$
$$= \sum_{\gamma\in\mathbb{Z}^n} \lambda_\gamma \operatorname{rowdet}\left(\left(\underbrace{f\left(X_{\gamma_j,\ i}\right)}_{\substack{=h_{\alpha_i+\gamma_j,\ i} \\ \text{(by the definition of } f\text{)}}}\right)_{i,j\in[n]}\right)$$
$$= \sum_{\gamma\in\mathbb{Z}^n} \lambda_\gamma \operatorname{rowdet}\left(\left(h_{\alpha_i+\gamma_j,\ i}\right)_{i,j\in[n]}\right).$$

In other words, (157) holds.

Thus, we have found a family $\left(\lambda_\gamma\right)_{\gamma\in\mathbb{Z}^n}$ of coefficients $\lambda_\gamma \in \mathbb{Z}$ such that all but finitely many $\gamma \in \mathbb{Z}^n$ satisfy $\lambda_\gamma = 0$, and such that (157) holds. The proof of Proposition 5.2 is thus complete. □

References

[BBSSZ13] Chris Berg, Nantel Bergeron, Franco Saliola, Luis Serrano, Mike Zabrocki, *A Lift of the Schur and Hall–Littlewood Bases to Non-commutative Symmetric Functions*, Canadian Journal of Mathematics **66** (2014), Issue 3, pp. 525–565.

[BBSSZ15] Chris Berg, Nantel Bergeron, Franco Saliola, Luis Serrano, Mike Zabrocki, *Multiplicative structures of the immaculate basis of non-commutative symmetric functions*, Journal of Combinatorial Theory, Series A **152** (2017), pp. 10–44.
See arXiv:1305.4700v3 for a preprint.

[Fun12] Alex Fun, *Raising operators, recurrences, and the Littlewood–Richardson polynomials*, PhD thesis, University of Sydney 2012.

[Grinbe15] Darij Grinberg, *Notes on the combinatorial fundamentals of algebra*, 7 May 2026.
`http://www.cip.ifi.lmu.de/~grinberg/primes2015/sols.pdf`
The numbering of theorems and formulas in this link might shift when the project gets updated; for a "frozen" version whose numbering is guaranteed to match that in the citations above, see arXiv:2008.09862v3 (somewhat out of date) or `https://github.com/darijgr/detnotes/releases/tag/2026-05-07` (current).

[Grinbe19] Darij Grinberg, *A basis for a quotient of symmetric polynomials (draft)*, 22 September 2021, arXiv:1910.00207v2.
`https://www.cip.ifi.lmu.de/~grinberg/algebra/basisquot.pdf`

[Grinbe21] Darij Grinberg, *An Introduction to Algebraic Combinatorics [Math 701, Spring 2021 lecture notes]*, 4 May 2026.
`https://www.cip.ifi.lmu.de/~grinberg/t/21s/lecs.pdf`

[LakTho07] Dan Laksov, Anders Thorup, *A Determinantal Formula for the Exterior Powers of the Polynomial Ring*, Indiana University Mathematics Journal **56**, No. 2 (2007), pp. 825–845.

[Macdon95] Ian G. Macdonald, *Symmetric Functions and Hall Polynomials*, Oxford Mathematical Monographs, 2nd edition, Oxford Science Publications 1995.

[Stanle24] Richard P. Stanley, *Enumerative Combinatorics, volume 2*, Second edition, Cambridge University Press 2024.
See `http://math.mit.edu/~rstan/ec/` for errata.

[Tamvak13] Harry Tamvakis, *The theory of Schur polynomials revisited*, Enseign. Math. **58** (2012), pp. 147–163, arXiv:1008.3094v2.